Functional Analysis

— *EYNTKA* Series —

Nathanael Chwojko-Srawley

July 18, 2026

Contents

Contents 3

I Functional Analysis 5

1 Metric Function Spaces 9
1.1 Basic Function Spaces 10
1.2 Application to other concepts 15
1.3 Compactness and equicontinuity 17
1.4 Uniform Apprixmation in bounded continuous functions 20
1.5 Ordinary Differential Equations 23
1.6 Nowhere Differentiable Continuous Functions 26
1.7 Baire Theory 29
1.8 Cardinality quick 29
1.9 Condition for Smooth to be Analytic 30

2 Banach and Fréchet Spaces 35
2.1 Banach Spaces 35
2.2 Baire Category Theorem 47
2.3 Dual Spaces 55
2.4 Weak Topologies and TVS's 59
2.5 Fréchet Spaces 64

3 Lebesgue Spaces **69**
3.1 Basics . . . 69
3.1.1 Relating Lebesuge spaces: Interpolation . . . 84
3.2 The dual of Lebesgue Space . . . 90
3.3 Some Useful Inequalities . . . 91
3.4 Interpolation: The Riesz-Thorin Theorem . . . 97

4 Radon Measures and Dual Theory **109**
4.1 Positive Linear Functionals on Compact Support Continuous Functions . . . 109
4.2 Dual of Continuous Decaying Functions . . . 114

5 Hilbert Spaces **117**
5.1 Basic definition and properties . . . 124
5.2 Operators on Hilbert Spaces . . . 138
5.3 Compact Operators . . . 149
5.4 Trace . . . 155
5.5 *Intuition to add . . . 155

6 Spectral Theory **157**
6.1 Banach Algebras . . . 157
6.2 Continuous Functional Calculus . . . 161

Part I

Functional Analysis

The transition from Calculus and Linear Algebra to Functional Analysis marks a fundamental shift in mathematical perspective. In classical analysis, we study individual functions, their derivatives, and their integrals. In linear algebra, we study vectors in finite dimensions. *Functional Analysis* we treat the function f itself as a point and look at transformations between spaces of these functions.

Note the term functional historically meant a function F take takes a function returns a number. We shall show that most linear functional are integrating against a function as we'll show in section 3.2 and chapter 4), and many important exceptions will be shown to still integrating against distributions.

The Historical Imperative: Taming the Infinite

Historically, the field grew out of a crisis in physics and differential equations. By the early 20th century, physicists dealing with Quantum Mechanics and classical wave equations realized that their problems could not be solved by looking at individual coordinates.

- In Quantum Mechanics, the state of a particle is not a position in $\mathbb{R}^3$, but a wave function ψ. The superposition principle—the fact that if ψ_1 and ψ_2 are states, then $\alpha\psi_1 + \beta\psi_2$ is a state—suggests that these wave functions live in a linear vector space.
- In Partial Differential Equations, viewing the heat or wave equation as an operation on an infinite-dimensional space allows us to decouple the "geometry" of the operator from the specific details of the function.

However, unlike the finite-dimensional world of $\mathbb{R}^n$, where linear algebra is rigid and predictable, infinite-dimensional spaces are wild. The intuitions of Euclidean geometry breaks down: bounded sets are no longer compact, linear maps are not necessarily continuous, and the notion of "convergence" splinters into multiple, distinct topologies.

Functional Analysis is the rigorous attempt to tame this infinite wilderness. It builds a hierarchy of structures to recover the tools of calculus and algebra in the infinite setting.

The Three Pillars of the Theory

The narrative of this course is driven by the tension between three mathematical forces:

1. The Algebraic Structure (Linearity): At its core, we are doing linear algebra. We want to solve equations like $Lx = y$. If L is a matrix, we invert it. If L is a differential operator, we want to do the same. The dream of Functional Analysis is to treat differential operators as "infinite matrices" and apply the Spectral Theorem to diagonalize them. When this works (as in Hilbert Spaces), we can solve complex PDEs with the same ease as a 2×2 matrix equation.
2. The Topological Structure (Continuity): Algebra alone is insufficient in infinite dimensions. In $\mathbb{R}^n$, all norms are equivalent, and continuity is automatic for linear maps. In infinite dimensions, this rigidity evaporates. We must introduce *Topology* via norms and metrics to distinguish between operators that are "well-behaved" (bounded) and those that are explosive (unbounded, like differentiation). The interplay between the algebraic dual space (all linear functionals) and the topological dual space (continuous functionals) becomes a central theme of the theory.

3. The Geometric Structure (Distance and Angle): This is where the "Zoo" of spaces emerges.
   (a) *Hilbert Spaces (L^2):* These are "correct" generalization of finite-dimensional euclidean space. They possess an inner product, allowing us to measure angles and define orthogonality.
   (b) *Banach Spaces ($L^p, C(K)$):* They have a notion of size (norm) but lack a notion of angle (i.e. a linear notion of relatedness). The geometry here can be strange – unit balls can be "pointy" or "flat," and the shortest path to a subspace might not be unique.
   (c) *Fréchet and Topological Vector Spaces:* necessary for distributions and generalized functions, where the concept of a "norm" is abandoned for families of semi-norms. They are equivalent to invariant metric spaces

There are more general spaces, but they are beyond an introductory text.

Ultimately, Functional Analysis is the study of *Dualities*. It is the duality between a space and the (linear) functions acting on it. We will see that by lifting concrete problems into these abstract spaces, we gain a new vantage point. The goal of this Part is to map this universe, understanding how the shape of the space dictates the physics of the operators within it.

1

Metric Function Spaces

(This entire article is so enlightening: Intuition on Function Spaces)

Recall in section **??** that the space L^1 is a vector-space. As an exercise, it should be shown that it is an infinite dimensional vector-space[1]. Since it is a real vector-space, it is isomorphic to $\mathbb{R}^N$ for appropriate cardinality N. However, N is usually uncountable, finding such N is equivalent to the axiom of choice, and the notion of measure on such a space is not well-defined. Instead, to extract information from a vector space that is an infinite dimensional function space, we assign functionals to extract information. The type of space we work with determines which type of functionals we can assign (inner products, norms, semi-norms, metrics, etc.)

The term *Function Analysis* or *Functional Analysis* is the traditional name for the study of infinite-dimensional vector-spaces over $\mathbb{R}$ or $\mathbb{C}$ and the linear maps between them. When in finite dimension, we have a lot of very nice algebraic results such as: Let V be a finite dimensional vector space,

1. If $W \subseteq V$ and $f \in W^*$, then f can be extended to be a linear functional on V
2. Bijective homomorphisms are isomorphisms
3. Others

What makes functional analysis different form linear algebra over finite dimensional vector space is that in finite dimension, there is only one "resonable" topology that makes linear functions automatically continuous. This is not the case for ininite dimensional spaces (recall the amount of interpretations we saw of how a sequence $\{f_n\}$ of function on $\mathbb{R}$ can convege $f_n \to f$). This chapter will be an introduction to this world.

(Why do we want maps to be continuous? Is it that we saw that many spaces are Banach spaces, and we want to thus apply algebraic results, but they are infinite dimensional meaning many finite-

[1] answer: the functions $\chi_{[n,n+1)}$ are linearly independent for all $n \in \mathbb{N}$

dimensional results don't hold, and so we limit our attention to "continuous" continuous functions to fix this?)

(When does a metric induce a norm: https://math.stackexchange.com/questions/166380/not-every-metric-is-induced-from-a-norm)

(I don't konw if this is the right place to put this, but it is really really really cool. It is bassically the classical duality between algebra and geoemtry. Given a topological space X, we can define an algebra by taking the set of all continuous functions $C(X)$ with the operations being pointwise. Now, given just the information $C(X)$, does there exist a X such that $C(X)$ is isomorphic to A? In other words, can we recover X? The answer is yes if X is a locally compact topological space! This is the algebra/topology duality. The same is true if X is a Riemann manifold, though you need more information, something called the *spectral triple*. Later, Grothendick generlized this and we got to create form any algebra A a topological space $\text{Spec}(A)$!!! Given the appropriate topological properties on $\text{Spec}(A)$, we can recover A!! In other words, there is this way to go back and forth between algebra and geometry. This same exists in analysis too: the dual of $C_c(X)$ for a LCH X is intimately related to Radon measure, and the dual of $C_c(X)$ with the appropriate conditions let's us generlize the notion of derivatives to locally integrable functions!!!)

One of the key problems a norm tries to fix is making sure there is still a reasonable "finiteness" to your space. Though a normed vector-space doesn't have an ordering, there is something simlar to it: in the same way the distance between any two real numbers if a real numbers (i.e. finite) the *norm* of any element in a Banach space is *finite*. Thus, though a space might be infinite like L^1, there is still some notion of finiteness allowing us to bring in many of the concepts we have worked with before and avoid the pathologies of the infinite.

(wouldn't mind mentioning how one of the key take-aways for Banach spaces is the Hahn-Banach Theorem, Open mapping theorem, and uniform boundedness principle.)

1.1 Basic Function Spaces

Example 1.1.1: pointwise convergence

1. here
2. Note that it's possible that a poinwise converging sequence of functions "miss" points. Take

$$f_n(x) : (a, b) \to \mathbb{R} \qquad f_n(x) = x^n$$

then $f_n \to 0$, the zero-function, since every point is indeed eventually convering to 0. However, it seems like there will be a point which is sort of "up there". We need a strong notion to fix that

Definition 1.1.2: Uniform Convergence

Let M and N be arbitrary metric spaces. A sequence of function $f_n : M \to N$ *converges uniformally* to the limit $f : M \to N$ if for each $\epsilon > 0$ there is an N such that for all $n \geq N$, and all $x \in M$,

$$|f_n(x) - f(x)| < \epsilon$$

The function f is the *uniform limit* of the sequence (f_n) and we write

$$f_n \rightrightarrows f \qquad \text{or} \qquad \text{unif} \lim_{n\to\infty} f_n = f$$

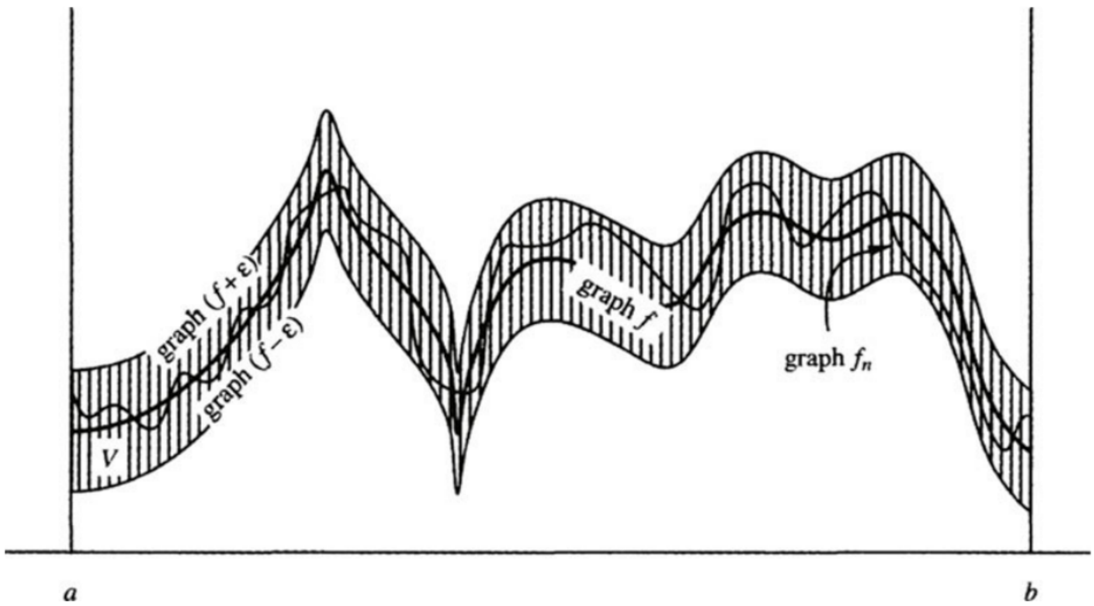


Figure 1.1: uniform convergence visually

Theorem 1.1.3: Uniform Convergence Preserve's Continuity

if $f_n \rightrightarrows f$ and eachf_n is continuous at x_0 then f is continuous at x_0

***Proof*:**
Key is to use the triangle inequality. Details are omitted to be field. Once you choose any $\epsilon > 0$, you know for some $n \geq N$,

$$|f_n(x) - f(x)| < \frac{\epsilon}{3}$$

Then you know for f_N at x_0 there is a $\delta > 0$ st

$$|f_N(x) - f_N(x_0)| < \frac{\epsilon}{3}$$

> and then you do the triangle inequality twice, to get split the first equation into 3 equations with f_N being part of it at least once, then use they're all less than $\frac{\epsilon}{3}$, and done

So, with $f_n(x) = x^n$, these function do *not* uniformally converge to 0, but to

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x = 1 \end{cases}$$

Note that the converse is not true: if at every point x_0, f_n is continuous and f is continuous at x_0, this does not imply (f_n) is uniformally convergent:

Example 1.1.4: growing steeple
This is called the growing steeple:

$$f_n(x) = \begin{cases} n^2 x & \text{if } 0 \leq x \leq \frac{1}{n} \\ 2n - n^2 x & \text{if } \frac{1}{n} \leq x \leq \frac{2}{n} \\ 0 & \text{if } \frac{2}{n} \leq x \leq 1 \end{cases}$$

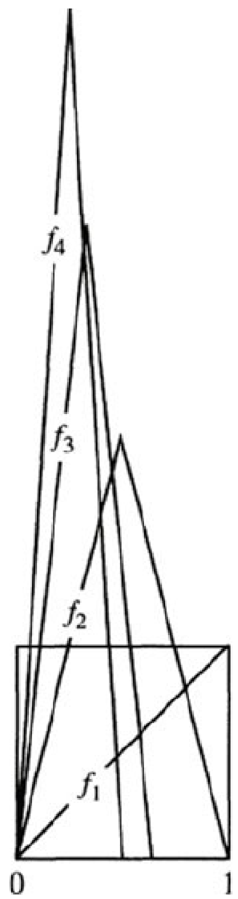


Figure 1.2: growing steeple function

and if you don't like that it goes to infinity, multiply it by $\frac{1}{n}$

Application to function spaces

Definition 1.1.5: Space Of Bounded Functions

Let $C_b([a,b],\mathbb{R})$ represent the set of all bounded functions from $f:[a,b]\to\mathbb{R}$

Then we can define the following norm on this space:

Definition 1.1.6: Sup Norm

Let $f \in C_b$. Then:

$$\|f\| = \sup\{|f(x)||x \in [a,b]\}$$

you should check that this is indeed a norm, and that it will induce a metric as well, and hence a metric topology, and hence the idea of convergence.

In this metric space under this metric, the definition is the same as uniform convergence

Theorem 1.1.7: sup norm convergence if and only if uniformally convergence

Convergence with respect to the sup metric d is equivalent to uniform convergence

***Proof*:**
The two definition's are in fact identical.

Since $d(f_n, f) \to 0$, then for every $x \in [a,b]$ the distnce is becomming smaller, so $|f(x) - f_n(x)|\epsilon$, and so $f_n \rightrightarrows f$.

Theorem 1.1.8: C_b Is A Complete Metric Space

C_b is a complete metric space

***Proof*:**
p.216. If you take any cauchy sequence of functions, then at every point, it's a cuachy sequence in $\mathbb{R}$. use the face that $\mathbb{R}$ is complete while using the sup-norm to complete the proof.

With this, we can show that the space of continuous bounded functions is complete too.

Definition 1.1.9: Continuous Bounded Funcitons

Let $C^0([a,b], \mathbb{R})$ denote the set of continuous functions from $f : [a,b] \to \mathbb{R}$. Note that since the domain is compact, the image is compact, and in $\mathbb{R}$ that implies it's bounded, so $C^0 \subset C_b$.

Corollary 1.1.10: C^0 Is A Compelte Metric Space

C^0 Is A Compelte Metric Space

***Proof*:**
By theorem 1.1.3, every convergent sequence in C^0 lies in C^0, thus C^0 contains it's limits, and thus is closed. so it's a closed subset of a compelte metric space. Thus, by HERE, it's complete

1.2 Application to other concepts

We'll now explore how uniform convergence works with derivatives, integrals, and series

Let

$$F_n = \sum_{i=1}^{n} f_i(x)$$

Then we can say that (F_n) conveges uniformally just like with (f_n) since it's also a sequence of functions.

If we have

$$F_n = \sum_{i=1}^{\infty} f_i(x)$$

Then we can say it converges uniformally if the sequence of partial sums converges uniformally.

Also, we can say the series of absolutes values $\sum |f_k(x)|$ converges then the series $\sum f_k$ *converges absolutely*

Theorem 1.2.1: Weierstrass M-Test

If $\sum M_k$ is a convergetn sereis of constants and if $f_k \in C_b$ satisfies $\|f_k\| \leq M_k$ for all k then $\sum f_k$ converges uniformly and absolutely

***Proof*:**
For n large enough, we can use the triangle enquality on the partial sums to expand out our series:

$$\begin{aligned} d(F_n, F_m) &\leq d(F_n, F_{n-1}) + \cdots + d(F_{m+1}, F_m) \\ &= \sum k = m + 1^n \|f_k\| \leq \sum k = m + 1^n M_k \end{aligned}$$

and thus, since $\sum M_k$ converges, then we can always use it's epsilon for large enough m, n. This shows that (F_n) is cauchy, and so since C_b is compelte, it converges uniformly.

Theorem 1.2.2: Uniform Limits Through Integrals

The uniform limit of Riemann integrable functiosn is Riemann integrable, and hte limit of hte integrals is the integral of hte limit

$$\lim_{n\to\infty} \int_a^b f_n(x)dx = \int_a^b \text{uniflim}_{n\to\infty} f_n(x)dx$$

***Proof*:**
p.218

Note that since all bounded contiuous function are integrable, but not all integrable bounded functions are continuous, this means that the subset of C_b of functions for which this theorem applies is between

C_b and C_0 (since not all bounded functions are integrable. If $\mathcal{R}$ represent all integrable funcitons:

$$C_b \supset \mathcal{R} \supset C^0$$

Now, using the fact that a limit can pass through integrable functions, we can show that infnite sums of of integrated functions is the sme as the integral of the infinite sum of functions:

Theorem 1.2.3: Term By Term Integration Theorem

A uniformly convergetn series of integrable functions $\sum f - k$ can be integrated term-by-term in the sense that

$$\int_a^b \sum_{k=0}^{\infty} f_k(x)dx = \sum_{k=0}^{\infty} \int_a^b f_k(x)dx$$

***Proof*:**
Use the definition, then apply theorem 6.

Now, we move on to differentiable functions

Theorem 1.2.4: Uniform Limit Of Differentiable Functions Is A Differentiable Function

The uniform limit of a sequecne of idfferentiable functiosn is a differentiable function provided that the sequence of derivaties also converges uniformally

***Proof*:**
p.219-220

Assuming f' is continuous makes the proof easy enough using the fundamental theorem of calculus by splitting up $f_n(x) = f_n(a) + \int_a^x f_n'(t)dt$.

For f' not continuous, more to check. Check the book, it's not too bad to follow

This also looks cool: here

Note that this can fail if the derivative doesn't converge uniformally

Example 1.2.5: derivative not converge uniformally
Take

$$f_n(x) = \sqrt{x^2 + \frac{1}{n}}$$

Then this series converges to $f(x) = |x|$, which is not differentiable! The derivatiev converges point-wise, but not uniformally.

Note that in complex analysis, this will not be the case! the uniform limit of a complex differentiable function is complex differentiable, and the sequence of derivatives will automatically be unifomrally convergent to a limit!

LIke before, infinite sums work well

Corollary 1.2.6: Series Of Infinite Sum And Uniform Convergence

A uniformly convergent sereis of differentaible functiosn can be differentiated term-by-term, provided that the derivative series converges uniformly,

$$\left(\sum_{k=0}^{\infty} f_k(x)\right)' = \sum_{k=0}^{\infty} f_k'(x)$$

***Proof*:**
Using the definition of convergence for infinite series, apply theroem 1.2.4 on the partial sums.

1.3 Compactness and equicontinuity

We now want to apply the Heine-Borel theorem to C^0, because checking something is closed and bounded is relatively easy, and for some reason, we want to able to easily find convergent-subsequence of convergent sequences (will come back here once I figure out why TBD)

In C^0, closed and bounded sets are usually not compact. Take

$$B = \left\{ f \in C^0([0,1], \mathbb{R}) | \, \|f\| \leq 1 \right\}$$

that is, teh set of all functions who's supremum over $[0,1]$ is 1. This set is not closed: it contains convergent sequences that do not lie inside B, however the value it converges to is *not* continuous

$$f(x) = \begin{cases} 0 & x < 1 \\ 1 & x = 1 \end{cases}$$

This comes down to the fact that $f(x)$ is infinite dimensional: Heine-Borel will applies to finite-dimensional spaces!

To show that C^0 is an infinite dimensional vector-space over $\mathbb{R}$, I'll show that it has an infinite dimensional linearly independent set (I'll not be showing it's a basis here, but since a set exists, the basis must at least have this cardinatliy) Take the set of "smooth bump" funtions, where f_0 start it's bump at 0 and ends at 1. Then $\{f_n | n \in \mathbb{Z}\}$ is a linearly independent set of continuous (even smooth) functions. Thus, the space must at least have a countably infinite dimension.

Now back to finding the appropriate conditions for the Heine-Borel property. To do this, we need a new condition on our functions:

Definition 1.3.1: [Uniform] Equicontinuous

A sequence of functions (f_n) in C^0 is **equicontinuous** if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |s - t| < \delta, \forall n \in \mathbb{N} \Rightarrow |f_n(s) - f_n(t)| < \epsilon$$

The functions f_n are *equally continuous.*

Note that the δ depends on teh ϵ, but *not* on the n I put [uniform] in the definition because there are definition of equicontinuity which are not "uniform", that is, you pick a point for which the conditions applies:

Definition 1.3.2: Point-Wise Equicontinuous

A sequence (f_n) in C^0 is *pointwise equicontintuous* if

$$\forall \epsilon > 0 \forall x \in [a,b], \exists \delta > 0 \text{ such that } |x-t| < \delta, n \in \mathbb{N}, |f_n(x) - f_n(t)| < \epsilon$$

From here on out, we will say equicontinuity instead of uniform equicontinuity.

Example 1.3.3: Example

1. The set of function $f : \mathbb{R} \to \mathbb{R}$, $f_n(x) = \frac{x}{n}$ are equicontinuous. However, they are not uniformally continuous.
2. Take $f : [0,1] \to \mathbb{R}$, $f_n(x) = n$. That is, a constant function. Then $\{f_n\}$ is an equicontinuous sequence. However $\{f_n\}$ is not a convergent sequence (and so not uniformally convergent).
3. let $\{f\}$ be a equicontinuous set with a single function. Then f is uniformally continuous by definition. In this sense, equicontinuity is intimately related to uniform continuity.

<++>

This definition can also be used to define an equicontinuous set:

Definition 1.3.4: Equicontinuous Set

The set $\mathcal{E} \subset C^0$ is equicontinuous if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |s-t| < \delta \text{ and } f \in \mathcal{E} \Rightarrow |f_(s) - f(t)| < \epsilon$$

Essentially, if a function is continuous, than for all $\epsilon > 0$, there exists an open neighborhood U such that $d(f(x), f(y)) < \epsilon$ for all $y \in U$. Equicontinuity means that a single neighborhood U can be chosen that will work for all the functions f in the collection $\mathcal{E}$.

You might recall Bolzano–Weierstrass theorem:

Theorem 1.3.5: Bolzano–Weierstrass Theorem

Let $M = \mathbb{R}^n$. Then every bounded sequence has a convergent subsequence.

***Proof*:**
Here is the sketch of the proof.

Since it's bounded, put it in a ball, and take the closure of the ball. Then it's closed. So it's compact. Take the cover: $B_1 = \{B_1(x) | x \in \mathbb{R}^n\}$. Then it has a finite sub-cover B_1'. Then one of the ball must have infinitely many points. Let's say $b_1' \in B_1'$. Then cover this ball with $B_{1/2}(x)$ balls, and repeat the same process. Keep doing this for every n. Then you can pick a sub-sequence from each $B_{1/n}(x)$ which will converge.

We'll be proving the analogous of this with equicontinuous sequences in C^0:

Theorem 1.3.6: Arzela-Ascoli Theorem

Let M be a compact metric space. Every uniformally bounded equicontinuous sequence of functions in $C^0(M, \mathbb{R})$ has a uniformly convergent subsequence

***Proof*:**
p.226. Fun to follow!

Notice how this relates to compactness: if every subsequence of f_n converges, then the closure of $\{f_n | n \in \mathbb{N}\}$ is compact!

The last part of the proof also shows that you can weaken the condition a bit, and say that if you converge on a dense subset, then you're uniformally convergent.

Corollary 1.3.7: Arzela-Ascoli Propagation Theorem

Pointwise convergence of an equicontinuous sequence of functions on a dense subset of the domain *propagates* to uniform convergence on the whole domain

***Proof*:**
last part of the proof for Ascoli

Another thing that I don't konw where to put, but I think is neat to consider:

1. Let's say that you have a sequence of function where $f_n \to f$ pointwise. you also have sequences of functions $f_{n,m} \to f_n$, that is, every f_n has a sequnce of function's converging to it. Then is there a subseuqence of $f_{n,m}$ (for varaible n, m) that converges to f?

<++>

If you're derivative is bounded, we cna actually prove this with a much weaker claim: only 1 point must be convergent for the whole thing to be uniformally continuous!!

Corollary 1.3.8: Differentiable Convergent Subsequence

Assume that $f_n : [a, b] \to \mathbb{R}$ is a sequence of differentiable functions whose derivatives are uniformly bounded. If for one point x_0, the sequence $(f_n(x_0))$ is bounded as $n \to \infty$, then the sequence (f_n) has a subsequence that converges uniformly on the whole interval $[a, b]$.

***Proof*:**
Let M be the bound of $|f_n'(x)|$ for all $n \in \mathbb{N}$ and $x \in [a, b]$. Equicontinuity of (f_n) then follows from the MVT:

$$|s - t| < \delta \Rightarrow |f_n(s) - f_n(t) = |f_n'(\alpha)||s - t| \leq M\delta$$

and then set $\delta = \frac{\epsilon}{(M+1)}$. THis shows that (f_n) is equicontinuous!

Now to show that the sequence is uniformally bounded, let C be a bound for $|f_n(x_0)|$ for all $n \in \mathbb{N}$. THen

$$\begin{aligned} |f_n(x) \leq |f_n(x) - f_n(x_0)| + f_n(x_0)| &\leq M|x - x_0| + C \\ &\leq M|b - a| + C \end{aligned}$$

showing that the sequence (f_n) is bouded in C^0! Since it's uniformally bounded and equicontinuous, by Arzela-Ascoli's theorem (1.3.6), $f(f_n)$ has a uniformally convergent subsequence!

As a quick asside, this theorem has two uses in other domain we will not discuss;

1. A sequecne of solutions to a continuous ordinary differential equation in $\mathbb{R}^M$ has a subsequence that converges to a limit, and that limit is also a solutionof the ODE
2. A sequence of complex analytic functions that converges pointwise, converges uniformly (on compact subsets of the domain of definition) and the limit is complex analytic.

However, we'll be using Arzela-Ascoli for Heine-Borel:

Theorem 1.3.9: Heine-Borel Theroem In A Function Space

A subset $\mathcal{E} \subseteq C^0$ is compact if and only if it is closed, bounded, and equicontinuous

***Proof*:**

($\Rightarrow$) Since C^0 is a complete metric space, it is totally bounded (it is automatically closed since C^0 is clopen). This means for some $\epsilon > 0$, ew can cover $\mathcal{E}$ with finitely many balls of size $\epsilon/3$

$$\mathcal{C} = \left\{B_{\epsilon/3}(f_k) | \text{appropriate } f_k\right\}$$

Note that each f_k is uniformally continuous since the domain is compact. Thus, there exists a $\delta > 0$ st

$$|s - t| < \delta \Rightarrow |f_k(s) f_k(t)| < \frac{\epsilon}{3}$$

Now, if $f \in \mathcal{E}$, the for some k we hvae $f \in B_{\epsilon/3}(f_k)$, and $|s - t| < \delta$ will give us

$$|f(s) - f(t)| \leq |f(s) - f_k(s)| + (f_k(s) - f_k(t)| + |f_k(t) - f(t)| < 3\frac{\epsilon}{3} = \epsilon$$

showing $\mathcal{E}$ is equicontinuous.

($\Leftarrow$) Assume $\mathcal{E}$ is closed, bounded, and equicontinuous. Then if (f_n) is a sequence in $\mathcal{E}$, by Arzela-Ascoli's theorem, some subsequence converges uniformly to a limit. This limit lies in $\mathcal{E}$ since $\mathcal{E}$ is clsoed. Thus, by ref:HERE, $\mathcal{E}$ is compact

1.4 Uniform Approximation in C^0

We now move onto a really interesting part of analysis: Given a continuous but nondifferentiable function f (so, any function in C^0), we we can "arbitrarily" approximate f with a smooth function

g! Even better, this smooth function will be a polynomial!!

In order for this to be possible, it would mean that for any $f \in C^0$, there is a polynomial close to it. That is what we're about to prove:

Theorem 1.4.1: Weierstrass Approximation Theorem

The set of polynomials is dense in $C^0([a,b],\mathbb{R})$

Note that since the function is continuous on a compact set, it is uniformally continuous (a fact we'll use in the 2nd proof).

Decomposing what density means for a moment: for each $f \in C^0$, and each $\epsilon > 0$, there is a polynomial function $p(x)$ such that for all $x \in [a,b]$

$$|f(x) - p(x)| < \epsilon$$

meaning that we can arbitrarily approximate $f(x)$!!

***Proof*:**
#1 The key idea to building $p(x)$ from f is to sample values of f and recombining them in a clever way.

Without loss of generality, let's say $[a,b] = [0,1]$. For each $n \in \mathbb{N}$, consider

$$p_n(x) = \sum_{k=0}^{n} \binom{n}{k} c_k x^k (1-x)^{n-k}$$

where $c = f\left(\frac{k}{n}\right)$ and $\binom{n}{k}$ is the binomial coefficient. This polynomial is called the **Bernstein polynomial**. We'll show that (p_n) converges uniformally to f as $n \to \infty$, i.e., $(p_n) \rightrightarrows f$.

***Proof*:**
#2 This proof uses convolution. to link the nice properties of polynomials with the less nice propertries of general continuous functions.

Without loss of generality, let's let $[a,b] = [0,1]$ (they are homeomorphic). Let $f \in C^0([0,1],\mathbb{R})$. What we'll do is manipulate this functions to a nicer homeomorphic equivalent, and approximate that function with a polynomial.

Let $g(x) = f(x) - (mx + b)$, where

$$m = \frac{f(1) - f(0)}{1} = f(1) - f(0) \qquad \text{and} \qquad b = f(0)$$

Since we're adding a polynomial to $f(x)$, $g \in C^0$. Notice that

$$g(0) = f(0) - (f(1) - f(0))\,0 - f(0) = 0 \qquad g(1) = f(1) - f(1) + f(0) - f(0) = 0$$

so $g(0) = 0 = g(1)$. This shows that given any f, it is equivalent to consider f in the "form" of g (the two being homeomorphic), so let $f(0) = 0 = f(1)$. In our construction, we'll sometimes need the domain to extend beyond $[0,1]$ for technical reasons, so let's say $f(x) = 0$ for $x \in \mathbb{R} - [0,1]$ (this is a smooth extension with zero measure, so nothing bad can come from this).

Next, we will construct the nice function we'll convolute with. Let

$$\beta_n(t) = b_n(1-t^2)^n \qquad -1 \leq t \leq 1$$

where the constant b_n is chosen so that $\int_{-1}^{1} b_n(t)dt = 1$. For $0 \leq x \leq 1$, set

$$P_n(x) = \int_{-1}^{1} f(x+t)\beta_n(t)dt$$

This is a weighted average of the value of f using the weight function β_n. This is where we use convlution to get the nice properties of $\beta_n(t)$ to f. We claim that P_n is a polynomial, and $P_n(x) \rightrightarrows f(x)$ as $n \to \infty$.

To check that P_n is a polynomial, we use a change of variables: $u = x + t$. Then for $0 \leq u \leq 1$ we have

$$P_n(x) = \int_{x-1}^{x+1} f(u)\beta_n(u-x))du = \int_{0}^{1} f(u)\beta_n(u-x)du$$

where the last integral is because $f = 0$ outside $[0,1]$. The function $\beta_n(u-x) = b_n(1-(u-x^2))^n$ is a polynomial in x whose coefficients are polynomials in u. We can pull out all the x terms, since the integral is with respect to x, and since the integral is definite with concrete endpoints, it is a constant, and so we have a polynomial!! In other words, we managed to "eliminate" the complexity of f.

To check that $P_n \rightrightarrows f$ as $n \to \infty$, we need to estimate $\beta_n(t)$. We claim that if $\delta > 0$ then

$$\beta_n(t) \rightrightarrows 0 \text{ as } n \to \infty \text{ and } \delta \leq |t| \leq 1$$

Notice that $\delta > 0$, so $0 \leq |t| \leq 1$, and so we are ignoring the fact that $x = 0$ diverges.

You can prove this directly, but I'll skip it for now (p.233)

Now, we'll deduce $P_n \rightrightarrows f$. Let $\epsilon > 0$. Since f is uniformally continuous, let $\delta > 0$ such that $|t| < \delta$ imples $f(x+t) - f(x)| < \frac{\epsilon}{2}$. Since β_n has integral 1 on $[-1,1]$, we can slip in β_n in the following equality:

$$\begin{aligned}
|P_n(x) - f(x)| &= \left| \int_{-1}^{1} \left(f(x+t) - f(x)\right) \beta_n(t)dt \right| \\
&\leq \int_{-1}^{1} |f(x+t) - f(x)|\beta_n(t)dt \\
&= \int_{|t|<\delta} |f(x+t) - f(x)|\beta_n(t)dt + \int_{|t|\geq\delta} |f(x+t) - f(x)|\beta_n(t)dt \\
&< \epsilon/2 + 2M \int_{|t|\geq\delta} B_n(t)dt
\end{aligned}$$

where the 1st integral is less than $\frac{\epsilon}{2}$ becuase HERE, and the 2nd because HERE. Since δ is positive, and $\beta_n \rightrightarrows 0$, then the 2nd integral becomes $\frac{\epsilon}{2}$ as well.

Thus, $P_n \rightrightarrows f$ – as we claimed

With this, we can prove the general Stone-Weierstrass theroem on compact metric spaces: that is, change $[0,1]$ into a general comapct metric space.

To do this, we'll first define an algebra on the functions. Let $A \subseteq C^0(M, \mathbb{R})$ such that if $f, g \in A$, $c \in \mathbb{R}$, then

$$f + g \in C^0(M, \mathbb{R}),\ cf \in C^0(M, \mathbb{R}),\ f \cdot g \in C^0(M, \mathbb{R})$$

where each operation is defined pointwise (ex. $(f + g)(x) = f(x) + g(x)$). For examples, polynoimals form an algebra.

The function algbera *vanishes at point* p if $f(p) = 0$ for all $f \in A$. FOr example, A with polynoimals with zero constatn term vanishes at $x = 0$.

THe function algebra *seperates poitns* if for each pari of distinct points $p_1, p_2 \in M$, ther eis a function $f \in A$ such that $f(p_1) \neq f(p_2)$. FOr example, the function algebra of all tirgonometric polynoimals seperates the points $[0, 2\pi)$ and vanishes nowhere

Theorem 1.4.2: Stone-Weierstrass Theorem

If M is a compact metric sapce and A is a functino-algebra in $C^0(M, \mathbb{R})$ that vanishes nowhere and seperates points, then A is dense in $C^0(M, \mathbb{R})$

Note that the Weierstrass theorem is a specia lcase of this one, but it will be used to prove this one

***Proof*:**
We first prove 2 lemmas to help us

If A vanisehs nowhere and seperates points then there exists $f \in A$ with specified values at any pari of disitnct ponts

Proof. □

1. power series: definition, integration, differentiation
2. define analytic
3. Weierstrass aproximation theroem(set of polynomials is dense in $C^0([a, b], \mathbb{R})$)
4. Stone-Weierstrass Theorem
5. Banach Contraction Principle?
6. Picard's Theorem?

<++>

1.5 Ordinary Differential Equations

Definition 1.5.1: Fixed Point

Let $f : M \to M$. a fixed point is a point $p \in M$ such that $f(p) = p$

Definition 1.5.2: Contraction

Let M be a metric space. A *contraction* of M is a mapping $f : M \to M$ such that for some $k < 1$ and all $x, y \in M$ $d(f(x), f(y)) \leq kd(x, y)$

Theorem 1.5.3: Banach Contraction Principle

Supose that $f : M \to M$ is a contraction and the metric space M is complete. Then$f$ has a unique fixed point p and for any $x \in M$ the iterate $f^n(x) = f \circ f \circ \cdots \circ f(x)$ converges to p as $n \to \infty$

***Proof*:**
p.240

Theorem 1.5.4: Brouwer Fixed-Point Theroem

Suppose that $f : B^m \to B^m$ is continuous where B^m is a closed unit ball in $\mathbb{R}^m$. Then f has a fixed point $p \in B^m$

***Proof*:**
see topology notes

Let $U \subseteq \mathbb{R}^m$. A *vector ODE* on U is given as m simultaneous scalar equations

$$\begin{aligned} x_1' &= f_1(x_1, x_2, ..., x_m) \\ x_2' &= f_2(x_1, x_2, ..., x_m) \\ &... \\ x_m' &= f_m(x_1, x_2, ..., x_m) \end{aligned}$$

where each f_i is a function from U to $\mathbb{R}$. One seeks m real-valued functions $x_1(t)$, ..., $x_m(t)$ such that

$$\begin{aligned} \frac{dx_1(t)}{dt} &= f_1(x_1(t), x_2(t), ..., x_m(t)) \\ \frac{dx_2(t)}{dt} &= f_1 2x_1(t), x_2(t), ..., x_m(t)) \\ &... \\ \frac{dx_m(t)}{dt} &= f_m(x_1(t), x_2(t), ..., x_m(t)) \end{aligned}$$

hold identically and simultaneously. Notice that $x_1, ..., x_n$ are all inside the $f_i(t)$. The functions $x_1(t), ..., x_m(t)$ add said to solve the ode with the initial condition

$$(x_1(0), x_2(0), ..., x_m(0))$$

the ode can be thought geometrically if we cake

$$f(x) = (f_1(x), ..., f_m(x))$$

where $x = (x_1, ..., x_m)$, making f a vector field. thus, what we seek is a path γ such that

$$\gamma'(t) = f(\gamma(t))$$

some more build-up

The way of thinking about ODE's is a vector-fields. When you have an ODE, you'll get

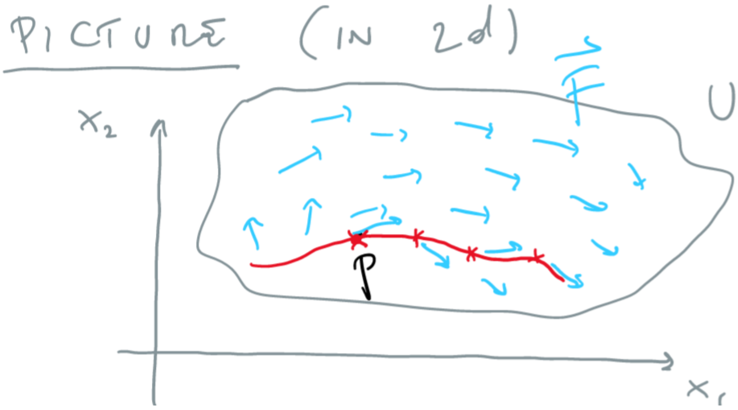


Figure 1.3: ODE as a Vector Field

Theorem 1.5.5: picard's theorem

given $p \in u$ there exists an F-trajectory $\gamma(t)$ in u through p. this means that $\gamma : (a, b) \to u$ solves

$$\gamma'(t) = F(\gamma(t))$$

locally, γ is unique

The locallness is very important. If it wasn't local, you can have functions with grow infinitely fast, or curl around to itself. It is the localness that will let us use contractions.

Proof. p.245 Pugh

First, since F is continuous, there exists a compact neighborhood $N = \overline{N}_r(p) \subseteq U$ and a constant M such that $|F(x)| \le M$. Choose a t small enough so that

$$tM \le r \qquad tL < 1$$

so that that tM is within the epsilon ball and tL will satisfy the contraction condition.

Considder teh set $\mathcal{C}$ of all continuous functions $\gamma, \sigma : [-t, t] \to N$ with respect to the metric

$$d(\gamma, \sigma) = \sup \{|\gamma(t) - \sigma(t)||t \in [-t, t]\}$$

the set $\mathcal{C}$ is a complete metric space. Given $\gamma \in \mathcal{C}$, define a new curve $\Phi(\gamma)$ as

$$\Phi(\gamma)(t) = p + \int_0^t F(\gamma(s))ds$$

Solving for this is a continuous curve γ that will also be differentiable (TBD). We just need a fixed point of Φ

To do this, we just check that Φ is a cntraction of $\mathcal{C}$. First, does Φ send $\mathcal{C}$ to itself? Well, given $\gamma \in \mathcal{C}$, we see that $\Phi(\gamma)(t)$ is a continuous (differentiable) vector-valued function of t and that

$$|\Phi(\gamma)(t) - p| = \left| \int_0^t F(\gamma(s))ds \right| \leq tM \leq r$$

implying Φ does send $\mathcal{C}$ to itself. Next, Φ contracts $\mathcal{C}$ since

$$\begin{aligned} d(\Phi(\gamma), \Phi(\sigma)) &= \sup_t \left| \int_0^t F(\gamma(s)) - F(\sigma(s))ds \right| \\ &= t \sup_s |F(\gamma(s)) - F(\sigma(s))| \\ &= t \sup_s L|\gamma(S) - \sigma(S)| \leq tLd(\gamma, \sigma) \end{aligned}$$

and $tL < 1$! Therefore Φ has a fixed point γ, and $\Phi(\gamma) = \gamma$, and therefore, $\gamma(t)$ sovles oru integral, which implies γ is differentiable and solve (20) (TBD) $\square$

1.6 Nowhere Differentiable Continuous Functions

Theorem 1.6.1: Nowhere Differentiable Continuous Functions

There exists a continuous function $f : \mathbb{R} \to \mathbb{R}$ that has a derivative at no point whatsoever

***Proof*:**
We'll construct a function by taking an infinite series of sawtooth functions wich will sum to this:

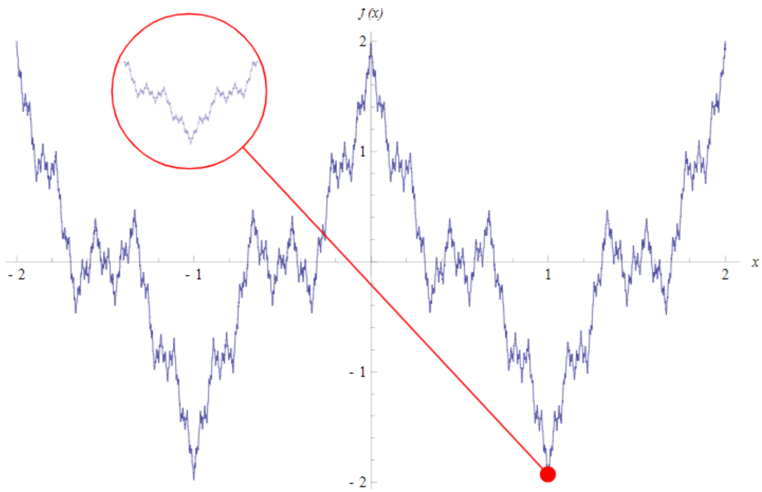


Figure 1.4: Nowehere Differential, yet Continuous Functions

Let the letters k, m, n dentoes integers. Start with the sawtooth function $\sigma_0 : \mathbb{R} \to \mathbb{R}$ defined as

$$\sigma_0(x) = \begin{cases} x - 2n & \text{if } 2n \leq x \leq 2n+1 \\ 2n+2-x & \text{if } 2n+1 \leq x \leq 2n+2 \end{cases}$$

Note that σ_0 is periodic with period 2

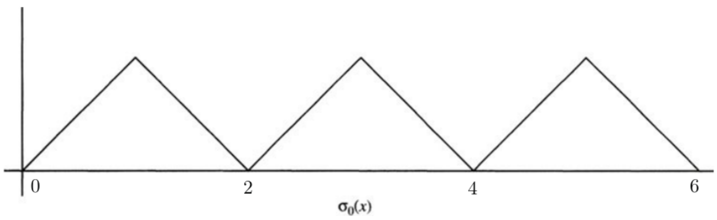


Figure 1.5: σ_0

Next, defined the next sawtooth functions by compression σ_0:

$$\sigma_k(x) = \left(\frac{3}{4}\right)^k \sigma_0(4^k x)$$

the period will be $\frac{2}{4^k}$, since $\sigma(x) = \sigma(x + m\left(\frac{2}{4^k}\right))$ for any $m \in \mathbb{Z}$.

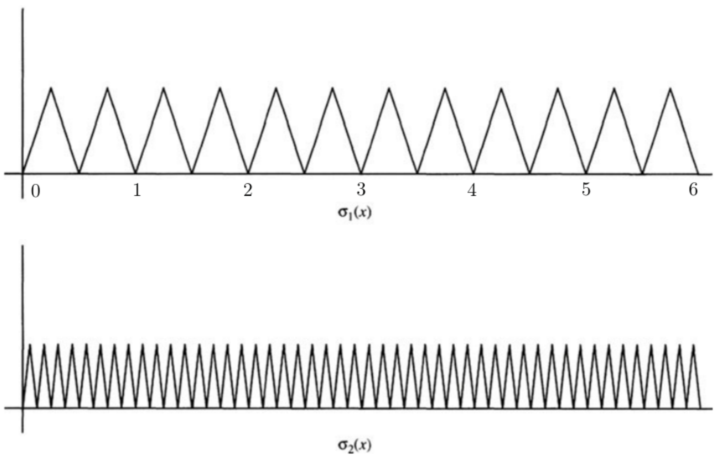


Figure 1.6: two examples of the itteration

By the M-test, $\sum \sigma_k(x0)$ converges uniformly to a limit f since $|\sigma_k| \leq \left(\frac{3}{4}\right)^k$ and

$$f(x) = \sum_{k=0}^{\infty} \sigma_k(x)$$

is continuous. We'll show that f is nowhere differentiable, that is

$$\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

does not exist. Since this must work no matter how $h \to 0$, pick $\delta_m = \frac{1}{2\cdot 4^m}$. Notice that $\delta_m \to 0$ as $m \to \infty$. We'll show that this limit does not exist:

$$\lim_{m\to 0} \frac{f(x \pm \delta_m - f(x)}{\delta_m}$$

where the $\pm$ is because we want this limit to reprent approach from either the left or right.

Using proposition 1.2.6, we get the derivative is

$$\sum_{k=0}^{\infty} \frac{\sigma_k(x \pm \delta_m) - \sigma(x)}{\delta_m}$$

We'll split this sum into 3 cases: $k > m$, $k = m$, $k < m$

($k > m$) Since $\delta_m = \frac{2}{4^k}$

$$\delta_m = \frac{1}{2}\frac{1}{4^m} = m\sigma_k \qquad m \in \mathbb{Z}$$

and thus

$$\sigma_k(x \pm \delta_m) - \sigma_k(x) = \sigma_k(x) - \sigma_k(x) = 0$$

meaning these terms do not contribute to the sum. We can essentially re-write our sum to

$$\sum_{k=0}^{n-1} \frac{\sigma_k(x \pm \delta_m) - \sigma(x)}{\delta_m}$$

($k = m$) σ_m will be monotone on either $[x, x+\delta_m]$ or $x - \delta - m, x]$:

Figure 1.7: Demonstration of monotonicity

and therefore

$$|frac\sigma_m(x+\delta_m) - \sigma_m(x)\delta_m| = 3^m$$

$(k < m)$ using what we've just learn from the previous exercice, we can crudely estimate each σ_k:

$$\left|\frac{\sigma_k(x \pm \delta_m) - \sigma_k}{\delta_m}\right| \leq 3^k$$

giving us

$$\sum k < m \left|\frac{\sigma_k(x \pm \delta_m) - \sigma_k}{\delta_m}\right| \leq \sum_{k<m} 3^k = \frac{3^m - 1}{3 - 1}$$

Thus, this gives us

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{m \to \infty} \sum_{k=0}^{\infty} \frac{\sigma_k(x \pm \delta_m) - \sigma(x)}{\delta_m} = \lim_{m \to \infty} 0 \pm 3^m + (3) \geq 3^m - \frac{3^m - 1}{2} = \frac{3^m}{2} + \frac{1}{2}$$

which diverges as $m \to \infty$! Therefore, this function is *not* differentiable at any point

1.7 Baire Theory

This construction might look like a particular case we need to construct to find, but it turns out that almost all continuous functions are nowhere differentiable! If you define your notion of probability right, then the chance of picking a continuous differentiable function form the set of continuous function is 0%! The way we'll make this rigorous is by taking a detour into Baire spaces.

A set D is dense in M if $\overline{D} = M$. Equivalently, it is dense in M if D for any open $W \subseteq M$, $W \cap M \neq \emptyset$. The intersection of dense sense need not be dense (ex. $\mathbb{Q}$ and $\mathbb{Q}^c$ in $\mathbb{R}$). However, if U, V are *open*-dense sets in M, then $U \cap V$ will also be open-dense in M. If $W \subseteq M$ is an arbitrary open subset. Then $U \cap W$ is also open, and so by denseness of V, $V \cap (U \cap W)$ is also non-empty! re-arranging using

1.8 Cardinality quick

I'm not sure where to add this yet, but I thought it is a good exercise to appreciate the oddity of $\aleph_1$

Pove that C^0 and $\mathbb{R}$ have the same cardinality

It is clear that C^0 has at least as many function as real numbers since every constant function is continuous, so it comes down to show there are no more. This is exercise 6 in Pugh's book, and its marked with a double star.

Found this too

https://math.stackexchange.com/questions/477/cardinality-of-set-of-real-continuous-functions

1.9 Condition for Smooth to be Analytic

Let $f : \mathbb{R} \to \mathbb{R}$ be a smooth function. Then we can always take its' tailor approximations at a point $T_a f$, and ask whether there exists an open interval U around a such that $T_a f|_U = f|_U$, namely:

$$f(h) = \sum_{k=1}^{\infty} \frac{a_k (h-a)^k}{k!}$$

Not all smooth functions are analytic, as the following typical example demonstrates:

$$f(x) = \begin{cases} e^{-1/x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

which can be verified by taking it's Taylor series which is

$$T_0(f) = \sum_{k=1}^{\infty} 0x^k = 0$$

However, we can precisely understand when a smooth is analytic by looking at the growth rate of the derivative, as we shall demonstrate in this chapter.

We can take a partial Taylor approximation and add on to it the remainder:

$$f(t) = \underbrace{\sum_{j=0}^{k-1} \frac{f^{(j)}(x)}{j!}(t-x)^j}_{\text{Polynomial}} + \underbrace{R_{k-1}(t)}_{\text{Remainder}}$$

Let us estimate $R_k(t)$.

Theorem 1.9.1: Lagrange Remainder Theorem

Let f be a smooth function, or at least $C^{k-1}(\mathbb{R})$ and let U be some interval. Let $P_{k-1}(t)$ be the partial Taylor series around a. Then for fixed t:

$$f(t) = P_{k-1}(t) + \frac{f^{(k)}(\xi_t)}{k!}(t-a)^k$$

***Proof*:**
Given the partial Taylor series $P_{k-1}(t)$ around a, fix a t_0 (without loss of generality $t_0 > a$). Then there always exists a constant C such that:

$$f(t_0) = P_{k-1}(t_0) + \frac{Q}{k!}(t_0 - a)^k$$

Define the function:

$$F(z) = f(t_0) - \sum_{j=0}^{k-1} \frac{f^{(j)}(z)}{j!}(t_0 - z)^j - \frac{Q}{k!}(t_0 - z)^k$$

Then $F(a) = F(t_0) = 0$ and F is differentiable, hence by Roles theorem there exists a ξ_t such that:

$$F'(\xi_t) = 0$$

Taking the actual derivative we get:

$$\frac{d}{dz}\left(\frac{f^{(j)}(z)}{j!}(t-z)^j\right) = \frac{f^{(j+1)}(z)}{j!}(t-z)^j - \frac{f^{(j)}(z)}{j!}j(t-z)^{j-1}$$

this creates a telescoping sum gives:

$$F'(z) = -\frac{f^{(k)}(z)}{(k-1)!}(t-z)^{k-1} + Q\frac{k(t-z)^{k-1}}{k!} = \left(-f^{(k)}(z) + Q\right)\frac{(t-z)^{k-1}}{(k-1)!}$$

where the equation is 0 at ξ_t. Now, as $a \neq t_0$, $(t-\xi_t)^{k-1} \neq 0$, hence:

$$(-f^{(k)}(\xi_t) + Q) = 0 \qquad \Rightarrow \qquad f^{(k)}(\xi_t) = Q$$

as we sought to show.

Using this, we can bound the remainder term. Given a interval $(r-a, r+a) = U$, for any $t \in U$, $|a-t| \leq r$. Then, for any choice of t, we have $f^{(k)}(\xi_t)$, and we can define:

$$M_k = \sup_{t\in U}|f^{(k)}(t)|$$

Hence:

$$|R_{k-1}(t)| = \left|\frac{f^{(k)}(\xi)}{k!}(t-a)^k\right| \leq \frac{M_k}{k!}r^k$$

This is the worst the function can grow. Then by the root test (which compares the growth rate of of function to $\sum_k \frac{1}{k^n}$[2]) if the quantity:

$$L = \limsup_{k\to\infty}\sqrt[k]{\frac{M_k}{k!}r^k} = r\limsup_{k\to\infty}\sqrt[k]{\frac{M_k}{k!}}$$

is strictly less than 1, it converges, if it is strictly greater than 1 it diverges, and if $L = 1$ then it is indeterminate. Note how R_{k-1} is dependent on the radius r.

Definition 1.9.2: Derivative Growth Rate

Let $I = (x-r, x+r)$ be an interval. Then the *bounded derivative growth rate* of a smooth function f on I is:

$$\alpha_r = \limsup_{k\to\infty}\sqrt[k]{\frac{M_k}{k!}}$$

Using definition 1.9.2, we have:

$$L = r \cdot \alpha_r$$

[2] $L < \rho, 1$, taking nth powers we see by the comparison test that L converges

Note that α_r is dependent on the radius r as we take the max over an interval of radius r. Now, since $|f^{(k)}(a)| \leq M_k$, we have that

$$R_a = \frac{1}{\limsup_{k\to\infty} \sqrt[k]{\frac{|f^{(k)}(a)|}{k!}}} \leq \frac{1}{\alpha_r}$$

Hence:

$$\frac{1}{\alpha_r} \leq R_a$$

Theorem 1.9.3: Taylor Convergence

Let I be an interval. If $\alpha \cdot r < 1$ then the Taylor series converges uniformly to f on the interval I.

***Proof*:**
Choose $\delta > 0$ such that $(\alpha + \delta)r < 1$. The Taylor remainder formula from Chapter 3, applied to the $(r-1)$st-order remainder, gives

$$f(x+h) - \sum_{k=0}^{r-1} \frac{f^{(k)}(x)}{k!} h^k = \frac{f^{(r)}(\theta)}{r!} h^r$$

for some θ between x and $x+h$. Thus, for r large we have

$$\left| f(x+h) - \sum_{k=0}^{r-1} \frac{f^{(k)}(x)}{k!} h^k \right| \leq \frac{M_r}{r!} r^r = \left(\left(\frac{M_r}{r!} \right)^{1/r} r \right)^r \leq ((\alpha+\delta)r)^r.$$

Since $(\alpha + \delta)r < 1$, the Taylor series converges uniformly to $f(x+h)$ on I.

Theorem 1.9.4: Analytic then bounded derivative growth

If f is expressed as a convergent power series $f(x+h) = \sum c_k h^k$ with radius of convergence $R > \sigma$ then f has bounded derivative growth rate on I.

***Proof*:**
The proof of theorem 1.9.4 uses two estimates about the growth rate of factorials. If you know Stirling's formula they are easy, but we prove them directly.

$$\lim_{r\to\infty} \frac{r}{\sqrt[r]{r!}} = e \tag{1.1}$$

$$0 < \lambda < 1 \implies \limsup_{r\to\infty} \sqrt[r]{\sum_{k=r}^{\infty} \binom{k}{r} \lambda^k} < \infty. \tag{1.2}$$

Taking logarithms, applying the integral test, and ignoring terms that tend to zero as $r \to \infty$

gives

$$\begin{aligned}\frac{1}{r}(\log r! - \log r!) &= \log r - \frac{1}{r}(\log r + \log(r-1) + \cdots + \log 1) \\ &\sim \log r - \frac{1}{r}\int_1^r \log x\, dx = \log r - \frac{1}{r}(x\log x - x)\Big|_1^r \\ &= 1 - \frac{1}{r},\end{aligned}$$

which tends to 1 as $r \to \infty$. This proves (1.1).

To prove (1.2) we write $\lambda = e^{-\mu}$ for $\mu > 0$, and reason similarly:

$$\begin{aligned}\sum_{k=r}^{\infty}\binom{k}{r}\lambda^k &= \sum_{k=r}^{\infty}\frac{k(k-1)\dots(k-r+1)}{r!}e^{-k\mu} \\ &\le \frac{1}{r!}\sum_{k=r}^{\infty}k^r e^{-k\mu} \approx \frac{1}{r!}\int_r^{\infty} x^r e^{-\mu x}\, dx \\ &= \frac{-1}{r!}e^{-\mu x}\left(\frac{x^r}{\mu} + \frac{rx^{r-1}}{\mu^2} + \frac{r(r-1)x^{r-2}}{\mu^3} + \cdots + \frac{r!}{\mu^{r+1}}\right)\Big|_r^{\infty} \\ &\le \frac{1}{r!}e^{-\mu r}(r+1)r^r\frac{1}{\min(1,\mu)^{r+1}}.\end{aligned}$$

According to (1.1) the rth root of this quantity tends to $e^{1-\mu}/\min(1,\mu)$ as $r \to \infty$, completing the proof of (1.2).

By assumption the power series $\sum c_k h^k$ has radius of convergence R and $\sigma < R$. Since $1/R$ is the lim sup of $\sqrt[k]{|c_k|}$ as $k \to \infty$, there is a number $\lambda < 1$ such that for all large k we have $|c_k\sigma^k| \le \lambda^k$. Differentiating the series term by term with $|h| \le \sigma$ gives

$$\begin{aligned}|f^{(r)}(x+h)| &\le \sum_{k=r}^{\infty} k(k-1)(k-2)\dots(k-r+1)|c_k h^{k-r}| \\ &\le \frac{r!}{\sigma^r}\sum_{k=r}^{\infty}\binom{k}{r}|c_k\sigma^k| \le \frac{r!}{\sigma^r}\sum_{k=r}^{\infty}\binom{k}{r}\lambda^k.\end{aligned}$$

for r large. Thus,

$$M_r = \sup_{|h|\le\sigma}|f^{(r)}(x+h)| \le \frac{r!}{\sigma^r}\sum_{k=r}^{\infty}\binom{k}{r}\lambda^k.$$

According to (1.2),

$$\alpha = \limsup_{r\to\infty}\sqrt[r]{\frac{M_r}{r!}} \le \frac{1}{\sigma}\limsup_{r\to\infty}\sqrt[r]{\sum_{k=r}^{\infty}\binom{k}{r}\lambda^k} < \infty,$$

and f has bounded derivative growth rate on I.

Theorem 1.9.5: Analyticity Theorem

A smooth function is analytic if and only if it has locally bounded derivative growth rate.

***Proof*:**
Assume that $f : (a, b) \to \mathbb{R}$ is smooth and has locally bounded derivative growth rate. Then $x \in (a, b)$ has a neighborhood N on which the derivative growth rate α is finite. Choose $\sigma > 0$ such that $I = [x - \sigma, x + \sigma] \subset N$ and $\alpha\sigma < 1$. We infer from theorem 1.9.3 that the Taylor series for f at x converges uniformly to f on I. Hence f is analytic.

Conversely, assume that f is analytic and let $x \in (a, b)$ be given. There is a power series $\sum c_k h^k$ that converges to $f(x + h)$ for all h in some interval $(-R, R)$ with $R > 0$. Choose σ with $0 < \sigma < R$. We infer from theorem 1.9.4 that f has bounded derivative growth rate on I.

Thus, while smooth functions have no bounds for the derivative:

$$|f^{(k)}(x)| \leq \text{anything, no bound}$$

analytic functions satisfy:

$$|f^{(k)}(x)| \leq C \cdot \frac{k!}{R^k}$$

2

Banach and Fréchet Spaces

(these are all spaces key spaces to consider with normed

(alternative title: Fréchet, Banach, and Hilbert Spaces)

2.1 Banach Spaces

Let k be either $\mathbb{R}$ or $\mathbb{C}$, let V or X denote a vector-space over k, with 0 representing the zero vector. Let kx denote the subspace of X spanned by x. If $M, N \subseteq X$, then $M + N = \{m + n \mid m \in M, n \in N\}$.

Definition 2.1.1: Semi-Norm And Norm

Let V be a vector-space. A *semi-norm* on V is a function $x \mapsto \|x$ from V to $[0, \infty)$ satisfying the propeties:

1. $\|x + y\| \leq \|x\| + \|y\|$ (the triangle inequality)
2. $\|\lambda x\| = |\lambda| \|x\|$ (homogeneity of scalars)

If furthermore:

(3) $\|x\| = 0$ if and only if $x = 0$

then the function is called a *norm*

Definition 2.1.2: Normed Vector-Space

Let V be a vector space. Then if there exists a function $\|\cdot\|$ that is defined on V, then $(V, \|\cdot\|)$ is called a *normed vector-space.*

If V is a normed vector-space, we can induce a metric $\rho(x,y) = \|x-y\|$, namely since:

$$\|x - z \leq \|x-y\| + \|y-z\| \qquad \|x-y\| = \|(-1)(x-y)\| = \|y-x\|$$

Since any metric induces a topology, the induced topology by the metric induced by a norm $\|\cdot\|$ is called the *norm topology* on V. A nice consequence of the induced norm topology is that addition and scalar multiplication is continuous. From the axioms of a semi-norm, $\|0\| = 0$. There are norms for which $x \neq 0$ but $\|x\| = 0$, as we'll soon show. Note that the norm $\|\cdot\| : V \to \mathbb{F}$ is evidently continuous in the topology it induces[1]. If $x_n \to x$ in X, then for all $\epsilon > 0$, there is an $N \in \mathbb{N}$ such that for all $n \geq N$, $\|x_n - x\| < \epsilon$. By the reverse triangle inequality:

$$|\|x_n\| - \|x\|| \leq \|x_n - x\| < \epsilon$$

showing that $|\|x_n\| - \|x\||$ get's arbitrarily close, and so $\|x_n\| \to \|x\|$, i.e. $\|\cdot\|$ is continuous. This means that the pre-image of $\{0\}$ of $\|\cdot\|$ is closed. Furthermore, it is a vector-space. Denoting

$$K = \{x \in V \mid \|x\| = 0\}$$

Then if $x, y \in K$, $0 \leq \|x+y\| \leq \|x\| + \|y\| = 0$, and $\|cx\| = |c|\|x\| = |c|0 = 0$. Hence, we may consider V/K. What norm to put on V/K that is natural with respect to the projection map $\pi : V \to V/K$ will be addressed soon.

Two norms $\|\cdot\|_1$ and $\|\cdot\|_1$ are said to be *equivalent* if there exits some $C_1, C_2 > 0$ st

$$C_1\|\cdot\|_1 \leq \|\cdot\|_2 \leq C_2\|\cdot\|_1$$

Equivalent norms define equivalent metrics and the same topology, and so the same Cauchy sequences. Recall that in finite dimensions, all norms are equivalent, meaning there is only one notion of distance. As we'll see, this is not the case in infinite dimensions.

Since we have a metric, we have a notion of completeness. As is the case when working with analysis, we will require our spaces to be complete:

Definition 2.1.3: Banach Space

Let V be a normed vector space. Then V is a *Banach space* if it is complete with respect to it's induced metric.

Example 2.1.4: Banach Spaces

1. Every finite dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$ is a Banach space. In this space, all linear map are continuous.
2. Let K be a compact space, and take the set of all continuous function, $C_0(K)$. Then $C(K)$ is a Banach space with the supremum norm $\|f\|_\infty = \sup_{x \in K} |f(x)|$ (also called uniform

[1]Note that in the metric space topology, continuous if and only if sequentially continuous

norm). We will later show (Banach–Mazur theorem) that all real separable Banach spaces are isometric (the isomorphisms in the category of Banach spaces) to a subspace of $C([0,1])$, giving us a theoretical universal way of thinking about Banach spaces. We shall later see that $C([0,1])$ has the Approximation property and has a Schauder basis, but there exists Banach spaces without these properties. This shows that for Banach spaces, it is possible to have nice total spaces with "bad" subspaces. In particular, if X is an isometric embedding into $C([0,1])$ there are not always nl nice projection $P : C([0,1]) \to X$.

If the space is not separable, if the smallest dense set D of a Banach space X has density of cardinality α, then X is isometric to $C_0([0,1]^\alpha, \mathbb{R})$[a]

3. If X is a topological space, then $B(X,k)$ and $BC(X,k)$ (all bounded functionals and bounded continuous functionals) form a Banach space with the uniform norm $\|f\|_u = \sup_{x\in X} |f(x)|$.

4. For p where $1 \le p \le \infty$, the space $L^p(X)$ for some measurable space X is a Banach space with the L^p-norm (recall that $L^p(\mu)$ is the set of equivalence classes functions that agree a.e.; if $L^p(\mu)$ represented only functions, then $\|\cdot\|_p$ would be a semi-norm)

5. Two other common examples of Banach spaces we will encounter later will be special subspaces spaces of differentiable functions: Banach Algebras, and Hilbert Spaces.

There are also examples of spaces that are not Banach Spaces:

1. Let X be a locally compact Hausdorff space and consider $C_c(X)$ with the ∞-norm. Then it is not a Banach space, not being complete. It is easy to see that $C_c(X)$, the set of continuous bounded function, is a Banach space and contains $C_c(X)$, and so it suffices to find the closure of $C_c(X)$ in $C_b(X)$. The closure turns out to be $C_0(X)$.

2. Let $U \subseteq \mathbb{R}^n$ be open and take $C_0(U,\mathbb{C})$. Then This space is *not* a Banach space

3. The space $H(U,\mathbb{C})$ of all holomorphic functions from U to $\mathbb{C}$ is *not* a Banach Space

4. The space $C_c^\infty(\mathbb{R}^n)$ is *not* a Banach space

5. (If you know some PDE's) The space of distributions $\mathcal{D}'$ is *not* a Banach Space

These are all examples of a weaker structure known as a *topological vector space* which we shall cover in section 2.4.

[a] An interesting result using Banach-Mazur's theorem by Luis Rodríguez-Piazza shows that the space of smooth functions with the uniform norm is isometric to the space of nowhere differentiable functions

There is a nice way to check whether a vector space V is a Banach space which requires the following definition (this essentially follows from the fact that continuity if and only if sequential continuity in a metric space and that $\|\cdot\|$ is continuous)

Definition 2.1.5: Absolutely Convergent

Let V be a normed vector space. and $\{x_k\}_k^\infty \subseteq V$ a sequence. Then the sequence is said to *converge absolutely* if

$$\sum_k^\infty \|x_k\| < \infty$$

Lemma 2.1.6: Cauchy and Absolute Convergence

Let V be a normed vector space. Then V is complete if and only if every absolutely convergent series in V converges.

The idea is that $\mathbb{R}$ has an order, making it easier to produce many arguments about sequences.

***Proof*:**
Let's first say that X is complete and that $\sum_{n=1}^\infty \|x\| < \infty$. To show it converges, we take advantage of completeness and show there is a Cauchy sequence. In particular, let $S_n = \sum_{k=1}^n x_k$ be the partial sum. Then since the series is absolutely convergent, there for all $n > m$,

$$\|S_n - S_m\| \leq \sum_{m+1}^n \|x_k\| \to 0 \quad \text{as } m, n \to \infty$$

so the sequence $\{S_n\}$ is Cauchy, and so converges, so $\{x_k\}$ converges.

Conversely, let's say an absolutely convergent series converges, and choose any Cauchy sequence $\{x_k\}$. Since it's a Cauchy sequence, choose a subsequence such that

$$\left\|x_{k_{i+1}} - x_{k_i}\right\| < 2^{-i}$$

Take $y_1 = x_{k_1}$ and $y_i = x_{i+1} - x_i$ for $i > 1$ so that $\sum_{i=1}^n y_i = x_{k_n}$. Then notice that $\{y_i\}$ is in fact absolutely convergent:

$$\sum_{i=1}^\infty \|y_i\| \leq \|y_1\| + \sum_{i=1}^\infty 2^{-i} = \|y_1\| + 1 < \infty$$

So $\sum_{i=1}^\infty y_i = \lim x_{k_n}$ exists. Since the subsequence of every Cauchy sequence approaches the same "point" (just like convergent sequences), we see that $\{x_n\}$ also converges to the same point, showing that Cauchy sequences are in fact convergent sequences, as we sought to show.

Next, we would want to study which linear functions between two vector-spaces are *continuous*. It will *not* be the case that all linear functions will be continuous. The following builds up to the classification of continuous linear functions:

Definition 2.1.7: Bounded Linear Map

Let $T : X \to Y$ be a linear map between two normed vector space. Then if there exists some C such that for all $x \in X$

$$\|T(x)\| \leq C\|x\|$$

Then T is said to be *bounded*, or is a *bounded linear operator*.

Note how this is different then boundedness of a set (ex. the range is bounded if $\|T(x)\| \leq C$). This definition is would be useless to us since no nonzero linear map would satisfy it. All linear map over finite dimensional vector spaces clearly bounded (can you find the C?), so this concept gets interesting in infinite dimesion

Proposition 2.1.8: Bounded $\Leftrightarrow$ Continuous

Let $T : X \to Y$ be a linear map between two normed vector spaces. Then the following are equivalent:

1. T is continuous
2. T is continuous at 0
3. T is bounded

***Proof*:**
If T is the zero map, then we're done, so assume $T \neq 0$

(1) implies (2) by definition. For (2) implies (3), Since T is continuous at 0, for for all $\epsilon > 0$, there exists a $\delta > 0$ such that if $\|x - 0\| < \|x\| < \delta$, then $\|T(x) - T(0)\| = \|T(x) - 0\| = \|T(x)\| < \epsilon$. Choosing $\epsilon = 1$, we get that there exists a δ_1 such that

$$\|x\| < \delta_1 \to \|T(x)\| < 1$$

we know want to show there exists a C such that $\|T(y)\| \leq C\|y\|$ for all y in the domain. We will do this by taking advantage of linearity and shrinking element so that we can take advantage of continuity.
Define $x(y) = \frac{\delta_1}{2\|y\|}y$ so that

$$\|x(y)\| = \left\|\frac{\delta_1}{2\|y\|}y\right\| = \frac{\delta_1}{2}\frac{1}{\|y\|}\|y\| = \frac{\delta_1}{2} < \delta_1$$

Therefore, by continuity and linearity:

$$\|T(x(y))\| = \left\|T\left(\frac{\delta_1}{2\|y\|}y\right)\right\| = \frac{\delta_1}{2\|y\|}\|T(y)\| < 1$$

since this is true for all y, we if we re-arrange we have:

$$\|T(y)\| < \frac{2}{\delta_1}\|y\|$$

showing that T is indeed bounded.

Finally, if $\|T(x)\| \leq C\|x\|$, then for all $\epsilon > 0$, choose $\delta = C^{-1}\epsilon$ so that

$$\|T(y) - T(x)\| = \|T(y-x)\| \leq \epsilon$$

completing the proof.

Note that for $T \in B(X,Y)$ to be an isomorphism, we require that T^{-1} also be bounded[2]. If $\|T(x)\| = \|x\|$ for all x, then T is called an *isometry.*

Since the topology is that of norms, then continuous linear functions between normed vector-spaces are the homomorphisms, meaning they preserve the norm (to the degree of boundedness mentioned in the above proposition) and the vector-space structure (to the degree the map is injective). We will denote the set of bounded linear functions (which we will call linear operators) as $B(X,Y)$, $\mathcal{L}(X,Y)$, or $\mathrm{Hom}_k(X,Y)$, where we understand that since X and Y are normed spaces $\mathrm{Hom}_k(X,Y)$ refers to continuous lienar maps.

Like with most fields of mathematics, we are more often more interested in the functions between spaces rather than the spaces itself, for it is function that ultimatly give us information about the structure we are working with. Thus, if we can use the technics we will be developing on $B(X,Y)$, we may gain many interesting results. For this to be the case, we require to introduce a norm on $B(X,Y)$. Naturally, we want that the norm on $B(X,Y)$ to be compatible with X and Y in some way, especially the codomain Y (as we usually investigate the structure of the image of a map). Sicne we are working with bounded operators, we have that $\|T(x)\| \leq C\|x\|$. Furthermore, since we are working with linear operators, by some scaling we see that we only need to consider when $\|x\| = 1$. This moivates the following norm on $B(X,Y)$:

Definition 2.1.9: Operator Norm

$B(X,Y)$ is a normed vector space with norm[a]

$$\begin{aligned}\|T\| &= \sup_{\|x\|=1} \|T(x)\| \\ &= \inf\{C \mid \|T(x)\| \leq C\|x\| \text{ for all } x\} \\ &= \sup_{\|x\|\neq 0} \frac{\|T(x)\|}{\|x\|}\end{aligned}$$

If $\|T(x)\| = \|x\|$, then T is called an *isometry*

[a]the third equality holds if $V \neq \{0\}$

The operator norm can be thought of as the smallest consant that will satisfy $\|T(x)\| \leq C\|x\|$ when T is bounded, that is $\|T(x)\| \leq \|T\|\|x\|$. In the spacial case where we have $B(X,k)$, we write V^* and call it the dual space with the *norm topology.* It is useful to establish when is $B(X,Y)$ a Banach space

[2]We will show using the open mapping theorem that it suffices to show T is continuous and bijective

Proposition 2.1.10: Completness Of $B(X, Y)$

Let X, Y be normed vector-space and $B(X, Y)$ the normed vector-space of all bounded linear functions from X to Y. If Y is complete, so is $B(X, Y)$.

***Proof*:**
Let (T_n) be a cauchy sequence in $B(X, Y)$. For each $x \in X$, we have a cauchy sequence $(T_n(x)) \subseteq Y$, and thus a limit vector which we will denote $T(x)$. This defines a function $T : X \to Y$ which is clearly linear. The inequality

$$\begin{aligned} \|T(x) - T_n(x)\| &= \lim_m \|T_m(x) - T_n(x)\| \\ &\leq (\limsup_m \|T_m - T_n\|)\|x\| \end{aligned}$$

shows that T is bounded and that $T_n \to T$.

Hence, V^* is always a banach space, since $\mathbb{R}$ and $\mathbb{C}$ are complete. Next, we see what happens with common vector-space constructions like products and quotients.

Subspaces, Product spaces, Quotient Spaces

There are 3 equivalent norms we can put on the product of two normed vector-spaces:

1. $\|(x, y)\| = \max(\|x\|, \|y\|)$
2. $\|(x, y)\| = \|x\| + \|y\|$
3. $\|(x, y)\| = (\|x\|^2 + \|y\|^2)^{1/2}$

The key to these being equivalent is that we are only multiplying by a finite number of spaces, a different strategy would have to be implemented for norms on infinite dimensional vector spaces (recall the different ℓ^p−norms put on the set of all sequences). For subspace, it is clear that if $W \subseteq V$ is a subspace of a normed vector space V, then W has an induced norm with respect to V. What is more interesting is if V is a banach space and we want W to be a Banach space. We then require all sequences to converge. Since V is Hausdorff (even metrizable since there is a norm on V), we see that W is complete if W is a *closed subspace* with respect to the topology of V. For quotients V/W, we would like to define a new norm on V/W so that the natural projeciton map $\pi : V \to V/W$ is continuous. This leads us to the following definition:

Proposition 2.1.11: Banach Quotient Space

Let X be a normed vector space and $Y \subseteq X$ a subspace of X. Then $\pi : X \to X/Y$ is the quotient map. This induces the topology on X/Y by defining:

$$\|\pi(x)\| = \|x + Y\| = \inf_{y \in Y} \|x - y\|$$

which is a *seminorm* on X/Y. If Y is closed, then this is in fact a norm, and if furthermer X is Banach, then X/Y is a banach space.

The intuition in my mind for this definition of quotient norm is to "eliminate" any effect of the quotienting space Y on any element of $x+Y$. Take for example $\mathbb{R}^2/\mathbb{R}$. Then the the norm of $\|(x,y)+\mathbb{R}\|$ should really be $|x|$. Furthermore, π is naturally a continuous map since it is norm-decreasing, meaning $\|\pi(x)\| \leq \|x\|$ (simply choose $C=1$).

***Proof*:**
For subadditivity, we take advantage of the definition of infimum: for every $x_1, x_2 \in X$, there exists an $\epsilon > 0$ and elements $y_1, y_2 \in Y$ such that

$$\begin{aligned}\|\pi(x_1)\| + \|\pi(x_2)\| + \epsilon &\geq \|x_1 - y_2\| + \|x_2 - y_2\| \\ &\geq \|x_1 + x_2 - (y_1 + y_2)\| \\ &\geq \|\pi(x_1 + x_2)\|\end{aligned}$$

Since ϵ was arbitrary, we have subaditivity. Homogeneity is immediate since π is linear, and hence we have a seminorm.

If $\|\pi(x)\| = \|x+Y\| = 0$, there is a sequence $(y_n) \subseteq Y$ such that $\|x - y_n\| \to 0$. If Y is closed, $\lim_n(y_n) = x \in Y$, meaning $\pi(x) = 0$. It follows that X/Y is a normed space if and only if Y is closed in X.

Now, say $(z_n) \subseteq X/Y$ is a cauchy sequence. Then we can find a subsequence (z_n') were

$$\|z_{n+1}' - z_n'\| < 2^{-n}$$

By induction we can choose $x_n \in X$ such that $\pi(x_n) = z_n'$ and $\|x_{n+1} - x_n\| < 2^{-n}$. If X is banach, (x_n) converges to an element $x \in X$, and since π is "norm decreasing" (the norm the same or smaller), then (z_n') converges to $\pi(x)$. Hence, (z_n) converges to $\pi(x)$, and so X/Y is complete, as we sought to show.

Proposition 2.1.12: Universal Property of Quotient Maps

If $T \in B(X,Y)$ where X and Y are normed spaces, and if $W \subseteq X$ is a closed subspace of X containing $\ker(T)$, then then there exists an operator $\tilde{T} \in B(X/W, Y)$ defined pointwise as $\tilde{T}(\pi(x)) := T(x)$ such that

$$\|\tilde{T}\| = \|T\|$$

in particular, the following diagram commutes in **Ban**:

$$\begin{tikzcd} X \arrow[r, "T"] \arrow[d, "\pi"'] & Y \\ X/W \arrow[ur, "\tilde{T}"'] & \end{tikzcd}$$

***Proof*:**
First, by elementary linear algebra we have that $\tilde{T}$ is an operator on X/W, which is a normed space by the earlier proposition. Since

$$\|\tilde{T}(\pi(x))\| = \|T(x)\| = \|T(x-z)\| \leq \|T\|\|x-z\|$$

for every $z \in W$, it follows form the definition of the quotient norm that

$$\|\tilde{T}(\pi(x)) \leq \|T\|\|\pi(x)\|$$

hence, $\|\tilde{T}\| \leq \|T\|$. The reverse inequality $\|\tilde{T}\| \geq \|T\|$ is clear since Q is norm decreasing.

Proposition 2.1.13: Extending Banach Spaces

Let X be a normed space and Y a closed subspace. Then if Y and X/Y are Banach spaces, then X is a Banach space.

***Proof*:**
Let $(x_n) \subseteq X$ be a cauchy sequence. Since $\pi : X \to X/Y$ is norm-decreasing, $(\pi(x_n)) = (y_n) \subseteq X/Y$ is a cauchy sequence. Since X/Y is complete, it converges to a point $\overline{x} \in X/Y$. Next, by the definition of the quotient norm, for every x_n, there exists an n st

$$\|(x - x_n) - y_n\| \leq \frac{1}{n} + \|\pi(x - x_n)\|$$

Then we get:

$$\begin{aligned}\|y_n - y_m\| &= \|(y_n + x - x_n) - (y_m + x - x_m) + x_n - x_m\| \\ &\geq \frac{1}{n}\|\pi(x - x_n)\| + \frac{1}{m}\|\pi(x - x_m)\| + \|x_n + x_m\|\end{aligned}$$

which is clearly cauchy, and hence since Y is banach it is complete and so this sequence converges to y. Finally, it is easy to see that

$$\|x_n - (x + y)\| \leq \|x_n - (x - y_n)\| + \|y - y_n\|$$

showing that $x - n \to x + y$, as we sought to show.

Finite space naturally have none of the problems introduced in infinite dimensions and preserve all of their nice properties even when subspace of infinite dimensinoal vectorspace:

Proposition 2.1.14: Finite Dimensional Normed Space

Every finite-dimensional subspace Y of a normed space X is a Banach space, and consequently is closed in X. Moreover, if $\dim(Y) = n$, every linear isomrophism of $\mathbb{F}^n$ onto Y is a homeomorphism

***Proof*:**
p. 46 Analysis now.

Working with infinite dimensional space dense subsets become of key interest. As a simple example, the smooth bump functions form a dense subset of L^p for all $1 \leq p < \infty$. The following simplifies for us how we may define functions from spaces given a known dense set:

Proposition 2.1.15: Defining Operator using Dense Sets

Let X and Y be banach spaces and X_0 is a dense subspace of X. Then every operator $T_0 \in B(X_0, Y)$ has a unique extension to an operator $T \in B(X, Y)$ and $\|T\| = \|T_0\|$

***Proof*:**
For each $x \in X$, there is a sequence $(x_0) \subseteq X_0$ that converges to x: $x_n \to x$. Since T_0 is a bounded operator ($\|T_0(x_n)\| \leq C\|x_n\|$), it is clear that $(T_0(x_0))$ is a convergent sequence in Y; let's say it converges to $T(x)$. It should be clear that if we choose another sequence converging to x and do this procedure, then we will get the same value of $T(x)$. Thus, the map $x \to T(x)$ is a operator that is linear, extends T_0, and by construciton

$$\|T\| = \|T_0\|$$

an excellent example would be polynomial over a compact set are dense over continuous functions over compact sets, which themselves are dense over all L^p spaces. Hence in the compact case, for real or complex functions, it suffices to define our function over a the polynomials. For the case of locally compact spaces, Note that smooth bump functions are dense in L^p, and since they have compact support a polynomial sequnce uniformally converges to them. Hence, we may define a lot of operators on different real/complex functions spaces over locally compact spaces using polynomals.

For future reference, it will be important to remember that all normed spaces can be completed and that the norm between a normed space X and its completion $\overline{X}$ is isometric:

Proposition 2.1.16: Unique Extension To Banach Space

For each normed space X, there is a Banach space $\overline{X}$, uniquely determiend up to isometric isomoprhism, such that $\overline{X}$ contains X as a dense subspace

***Proof*:**
p. 47 Analysis now

An important concept in any infinite dimensional space is to figure out which sets are compact. For example, you may recall in real analysis that we proved the Stone-Weistrass theorem, or even simpler that in $\mathbb{R}^n$ a set is compact if it is closed and bounded. The following shows that subsets that are "almost finite-dimensional" are compact.

Proposition 2.1.17: Compact Sets In Banach Spaces

Let $K \subseteq V$ be a subset of a Banach Space V. Then the following are equivalent:

1. K is compact
2. K is sequentially compact
3. K is closed and bounded, and for every $\epsilon > 0$, K lies in the ϵ-neighborhood

$$\{x \in V \mid \|x - y\| < \epsilon \text{ for some } y \in W\}$$

of a finite dimensional subspace W of V.

Suppose further that there is a nested sequence $V_1 \subseteq V_2 \subseteq \cdots$ of finite dimensional subspaces of V such that $\bigcup_{n=1}^{\infty} V_n$ is dense. Then the following statement is equivalent to the first three:

4. K is closed and bounded, and for every $\epsilon > 0$, there exits an n such that K lies in the ϵ-neighborhood of V_n

***Proof*:**
exercise

Example 2.1.18: Compact Sets In Banach Spaces

Let $1 \leq p < \infty$ and take a set $K \subseteq \ell^p(\mathbb{N})$. In order for it to be compact in the norm topology, it needs to be closed, bounded, and also uniformly p^{th}-power integrable at spatial infinity, that is for every $\epsilon > 0$, there exists an $n > 0$ such that

$$\left(\sum_{m>n} |f(m)|^p\right)^{1/p} \leq \epsilon$$

for all $f \in K$. This shows that the moving bump set is not uniformly p^{th} power integrale and thus not a compact subset, even though it's closed and bounded.

This condition doesn't translate to the continuous equivalent $L^p(\mathbb{R})$. We also need some uniform integrability at a "finte scale", which would be described using harmonic analysis tools like Fourier transforms, and hence we simply mention it here without developing it further.

Given the Banach space $\text{End}_k(V)$, it is natural have a second binary operation of composition and ask if it is also continuous (hence bounded). More generally, if $T \in B(X,Y)$, and $S \in L(Y,Z)$, then

$$\|ST(x)\| \leq \|S\|\|T(x)\| \leq \|S\|\|T\|\|x\|$$

and so $ST \in B(X,Z)$. In particular, that makes $B(X,X) = \text{End}_k(X)$ into an algebra, and when X is complete, $B(X,X)$ is a *Banach Algebra*. More generally:

Definition 2.1.19: Banach Algebra

Let X be a normed k-algebra where addition, multiplication, and scalar multiplication are continuous with respect to the induced topology. Then if X is complete, X is a *Banach Algebra*.

For the multiplication to be continuous, it suffices to check that $\|xy\| \leq \|x\|\|y\|$ (can you see why?). If X has a multiplicative unit, it is called *unital*. Not all Banach algebra's have a multiplicative unit, $C_0(X)$ where X is a locally compact (hausdroff) space and $C_0(X)$ is the set of functions from X to $\mathbb{R}$ (or $\mathbb{C}$) that vanish at infinity is *not* unital. On the otherhand, $C_b(X)$, the set of bounded functions with pointwise operations with the supremum norm *is* unital. We will work more with Banach Algebras later.

The Local Geometry of Banach Spaces

While infinite-dimensional Banach spaces can be geometrically wild (as we will see with L^1 and L^∞), their finite-dimensional subspaces exhibit a surprising regularity. We previously noted that all norms on a finite-dimensional space are equivalent. The following theorem, a cornerstone of the "Local Theory of Banach Spaces," makes a much stronger claim: Euclidean geometry is "locally universal".

Theorem 2.1.20: Dvoretzky's Theorem

Let $(X, \|\cdot\|)$ be an infinite-dimensional Banach space. For every integer $n \geq 1$ and every $\epsilon > 0$, there exists a subspace $E_n \subset X$ with $\dim(E_n) = n$ such that E_n is $(1+\epsilon)$-isomorphic to the Euclidean space ℓ_2^n.
In terms of the Banach-Mazur distance, this means:

$$d(E_n, \ell_2^n) = \inf \|T\|\|T^{-1}\| \leq 1 + \epsilon$$

where the infimum is taken over all isomorphism $T : E_n \to \ell_2^n$.

Geometrically, this is often described as the "slicing problem." If you visualize the unit ball of a Banach space X as a high-dimensional convex body K, Dvoretzky's theorem states that there is always a slice through the center of K (a subspace) that is nearly spherical.

This theorem highlights a tension between local and global structure:

1. *Globally*, X might be very far from a Hilbert space (like ℓ_∞ or the Enflo space).

2. *Locally*, X contains "copies" of Hilbert spaces of all finite dimensions.

Thus, while the "global" geometry of Banach spaces is a vast zoo of structures, the "atomic" local geometry is always Euclidean.

Exercise 2.1.1

1. Let $A, B \subseteq V$ are two closed subspaces. Show that $A + B$ need not be a closed subspace! However, if one of A or B is finite dimensional, then $A + B$ is closed.

2. Show that the open unit ball in X maps onto (by π) the open unit ball of X/Y, but a similar statement about closed unit balls holds only in special cases.

3. For the categorically inclined, it should be verified that the topology on X/Y is the quotient topology corresponding to the equivalent relation $\sim$ where $x_1 \sim x_2$ if and only if $x_1 - x_2 \in Y$.

2.2 Baire Category Theorem

In this section, we introduce some of the basic tools that allow us to work with functions between Banach spaces. Of keen interest to us is the *open mapping theorem*, the *closed graph theorem*, and the *uniform boundedness theorem*. The open mapping theorem makes it sufficient that a surjective continuous linear operator is open, which in particular means that if it is also bijective it is an isomorphism. The closed graph theorem allows us to show that a linear operator $T : X \to Y$ is continuous if the graph $\Gamma(T)$ is closed, i.e. if all sequences in the graph converge. This will lead to great simplifications in showing a function is continuous. Finally, the uniform boundedness theorem will give a powerful criterion for upgrading pointwise boundedness of a family of operators to uniform boundedness.

The underpinning results we need are about dense sets. Let's say we have a countable family of dense sets, which can be thought of as our "generalized basis". Then it is natural that their intersection may be empty, for example taking linearly independent transcendental elements $r_1, r_2, ... \in \mathbb{R}$, then $\bigcap_r \mathbb{Q} + r = \emptyset$. What we need is to ensure that the intersection is non-empty. For that, we require that the limit of an infinite intersection exists (ex. our space is LCH or a complete metric space), and that our sets have a nice property that ensures non-emptiness. The following deals with the case of the space being a complete metric space:

Proposition 2.2.1: Baire Category Theorem I

Let (A_n) be a sequence of open dense subset of a complete metric space (X, d). Then $\bigcap_n A_n$ is dense in X.

***Proof*:**
We want to show that $\bigcap_n A_n$ is dense, so for every ball $B_0 \subseteq X$, $B_0 \cap \bigcap_n A_n \neq \emptyset$.

Let B_0 be any closed ball in X with radius $r > 0$. Then since A_1 is dense in X and is open, the set $A_1 \cap \operatorname{int}(B_0)$ is a nonempty open set, and hence contains a closed ball B_1 with radius at most $r/2$. Since A_2 is dense and open, we can repeat this with $A_2 \cap \operatorname{int}(B_1)$ and find a closed ball B_2 with radius at most $r/2^2$. By induction, we get a sequence of closed balls (B_k) in X such that $B_k \subseteq A_k \cap \operatorname{int}(B_{k-1})$, each with radius at most $r/2^k$. The sequence of centers is Cauchy (since the radii tend to zero and the balls are nested), and since X is complete, there exists a unique point $\{x\}$ such that

$$\{x\} = \bigcap_k B_k \subseteq B_0 \cap \left(\bigcap_k A_k \right)$$

This shows that $\bigcap_k A_k$ intersects every nontrivial ball in X, which implies density, completing the proof.

Note that this is a fully topological proof, and hence is invariant under homeomorphism. In particular, completeness is *not* a topological property (it is a uniform property), however $(0,1) \cong_{\mathbf{Top}} \mathbb{R}$. Furthermore, since each U_n is open, its complement must be nowhere dense. For if some non-empty open set O satisfies $O \subseteq \overline{U_n^c} = U_n^c$ (since U_n^c is closed), then $U_n \subseteq O^c$, and so

$$X = \overline{U_n} \subseteq O^c \subsetneq X$$

a contradiction. This leads to the following very useful corollary:

Corollary 2.2.2: Countable union of Empty Interior Sets

The countable union of closed sets with empty interior has empty interior. If X is a complete metric space, $X \neq \bigcup_i \overline{X_i}$ for nowhere dense sets X_i.

As a consequence, if $X = \bigcup_n X_n$, then there exists some n_0 such that X_{n_0} has nonempty interior.

***Proof*:**
Let (F_n) be a countable collection of closed sets with empty interior. Then each F_n^c is open and dense (open since F_n is closed; dense since F_n has empty interior, meaning F_n^c is dense). By the Baire Category Theorem, $\bigcap_n F_n^c$ is dense in X, and in particular nonempty. Hence $\bigcup_n F_n \neq X$, and since $\bigcup_n F_n$ is contained in a countable union of closed sets with empty interior, it itself has empty interior.

If each X_i is nowhere dense, then each $\overline{X_i}$ is closed with empty interior, so $X \neq \bigcup_i \overline{X_i} \supseteq \bigcup_i X_i$. Contrapositively, if $X = \bigcup_n X_n$, then some X_{n_0} must have nonempty interior, for otherwise each $\overline{X_n}$ would be closed with empty interior, contradicting the above.

This consequence is very interesting, since it shows that through a countable process, we require a set with nonempty interior to form our entire space. This motivates the following definition:

Definition 2.2.3: Meager and Generic Set

Let X be a topological space. Then $A \subseteq X$ is said to be *meager* (or of first-category) if it is a countable union of nowhere dense sets. The complement of a meager set is a *residual* (or co-meager, or generic, or of second-category).

Example 2.2.4: Interesting Meager Sets

1. The Cantor set $\mathcal{C}$ is meager in $\mathbb{R}$. Indeed, $\mathcal{C}$ is closed (as the intersection of closed sets in its construction) and has empty interior (it contains no intervals, since at each stage of the construction every remaining interval is split and its middle third removed). Hence $\mathcal{C}$ is nowhere dense, and in particular meager (as a single-term union of a nowhere dense set).

2. Take $C([0,1])$ with the uniform convergence topology. Let E be the set of all continuous functions that are differentiable somewhere. Then E is in fact a *meager* set in $C([0,1])$. To see this, take:

$$E_n = \{f \mid \exists x_0 \in [0,1],\ |f(x) - f(x_0)| < n|x - x_0|\ \forall x \in [0,1]\}$$

Then all differentiable functions are contained in $\bigcup_n E_n$, and it should be shown that this is a closed set and that there exists functions g such that $\|f-g\|_u < \epsilon$ for any $f \in E_n$ but $g \notin E_n$. Hence, we satisfied the conditions of the Baire Category theorem. As a consequence, the set of continuous nowhere differentiable functions is residual! Thus more generally, smooth and analytic functions are meager.

3. $\mathbb{Q}$ is not a G_δ subset of $\mathbb{R}$. Suppose for contradiction that $\mathbb{Q} = \bigcap_n U_n$ for open sets U_n. Since $\mathbb{Q}$ is dense, each U_n is open and dense. The irrationals $\mathbb{R} \setminus \mathbb{Q}$ can also be written as a countable intersection of open dense sets: $\mathbb{R} \setminus \mathbb{Q} = \bigcap_q (\mathbb{R} \setminus \{q\})$ where q ranges over $\mathbb{Q}$. Then

$$\emptyset = \mathbb{Q} \cap (\mathbb{R} \setminus \mathbb{Q}) = \left(\bigcap_n U_n\right) \cap \left(\bigcap_q (\mathbb{R} \setminus \{q\})\right)$$

would be a countable intersection of open dense subsets of the complete metric space $\mathbb{R}$. By the Baire Category Theorem, this intersection must be dense, contradicting the fact that it is empty. Hence $\mathbb{Q}$ is *not* a G_δ. Note however that $\mathbb{R} \setminus \mathbb{Q}$ *is* a G_δ, i.e. the irrationals form a residual set.

The 2nd example above is wroth proving in full: the set of continuous functions on $[0,1]$ that are differentiable at *even one point* is meager in $C([0,1])$. Equivalently, the "generic" continuous function is nowhere differentiable. Compare this with theorem 1.6.1, which gives an explicit construction of a single such function; the Baire category argument produces no explicit example, but shows that *almost every* continuous function (in the sense of category) has this pathological behaviour.

Theorem 2.2.5: Genericity of Nowhere Differentiable Functions

Let $X = C([0,1])$ equipped with the uniform norm $\|\cdot\|_u$. Then the set of functions in X that are differentiable at some point of $[0,1]$ is meager. Consequently, the set of continuous nowhere differentiable functions is residual in X.

***Proof*:**
For each $n \in \mathbb{N}$, define:

$$E_n = \{f \in X \mid \exists x_0 \in [0,1],\ |f(x) - f(x_0)| \le n|x - x_0| \ \forall x \in [0,1]\}$$

We first check that every function differentiable at some point belongs to $\bigcup_n E_n$. If $f'(x_0)$ exists with $|f'(x_0)| = L$, then there exists $\delta > 0$ such that $|f(x) - f(x_0)| \le (L+1)|x - x_0|$ whenever $|x - x_0| < \delta$. For $|x - x_0| \ge \delta$, the bound $|f(x) - f(x_0)| \le 2\|f\|_u \le (2\|f\|_u/\delta)|x - x_0|$ holds trivially. Taking $n > \max(L+1,\ 2\|f\|_u/\delta)$ gives $f \in E_n$. It now suffices to show each E_n is closed with empty interior.

E_n is closed. Let $(f_k) \subseteq E_n$ converge uniformly to $f \in X$. For each k, choose $x_k \in [0,1]$ witnessing $f_k \in E_n$, i.e. $|f_k(x) - f_k(x_k)| \le n|x - x_k|$ for all x. By compactness of $[0,1]$, pass to a subsequence (still called f_k) with $x_k \to x_0$. Fix any $x \in [0,1]$. Then:

$$\begin{aligned} |f(x) - f(x_0)| &\le |f(x) - f_k(x)| + |f_k(x) - f_k(x_k)| + |f_k(x_k) - f(x_0)| \\ &\le \|f - f_k\|_u + n|x - x_k| + |f_k(x_k) - f(x_0)| \end{aligned}$$

As $k \to \infty$, $\|f - f_k\|_u \to 0$ and $|f_k(x_k) - f(x_0)| \le \|f_k - f\|_u + |f(x_k) - f(x_0)| \to 0$ by uniform convergence and continuity of f. Hence $|f(x) - f(x_0)| \le n|x - x_0|$ for all x, showing $f \in E_n$.

E_n has empty interior. Let $f \in X$ and $\epsilon > 0$. We construct g with $\|f - g\|_u < \epsilon$ and $g \notin E_n$. Since f is uniformly continuous on $[0, 1]$, there exists $\delta > 0$ such that $|f(x) - f(y)| < \epsilon/8$ whenever $|x - y| < \delta$. Choose an integer $M > 4n$ large enough that $w := \epsilon/(2M) < \delta$. Define the *zigzag function* $\varphi : [0, 1] \to \mathbb{R}$ to be the continuous piecewise linear function with slope alternating between $+M$ and $-M$ on consecutive intervals of width w, starting from $\varphi(0) = 0$. Since φ oscillates between 0 and $Mw = \epsilon/2$, we have $\|\varphi\|_u \leq \epsilon/2$.

Set $g = f + \varphi$, so $\|f - g\|_u = \|\varphi\|_u \leq \epsilon/2 < \epsilon$. Suppose for contradiction that $g \in E_n$, witnessed by some $x_0 \in [0, 1]$. Then x_0 lies in some monotone piece of φ of width w. Pick x to be the endpoint of this piece farther from x_0, so that $|x - x_0| \geq w/2$ and φ has constant slope $\pm M$ on the segment from x_0 to x. Since $|x - x_0| \leq w < \delta$:

$$\begin{aligned}
|g(x) - g(x_0)| &\geq |\varphi(x) - \varphi(x_0)| - |f(x) - f(x_0)| \\
&= M|x - x_0| - |f(x) - f(x_0)| \\
&> M \cdot \frac{w}{2} - \frac{\epsilon}{8} = \frac{\epsilon}{4} - \frac{\epsilon}{8} = \frac{\epsilon}{8}
\end{aligned}$$

On the other hand, $n|x - x_0| \leq nw = n\epsilon/(2M) < \epsilon/8$ since $M > 4n$. Hence $|g(x) - g(x_0)| > n|x - x_0|$, contradicting $g \in E_n$.

Since each E_n is closed with empty interior, $\bigcup_n E_n$ is meager. The set of nowhere differentiable functions contains the complement of $\bigcup_n E_n$ and is therefore residual, completing the proof.

Hence the vast majority (in the sense of Baire category) of continuous functions have this "wild" behaviour: they oscillate so violently at every scale that no tangent line can be drawn at any point. The zigzag perturbation in the proof is a shadow of this oscillation, forced into existence by the Baire Category Theorem without any explicit fractal construction.

The following lemma is the key technical ingredient for the open mapping theorem. It shows that if the image of the unit ball is *dense* in some ball, then it actually *contains* a slightly smaller ball:

Proposition 2.2.6: Open Mapping Lemma

Let $T : V \to W$ be a bounded linear operator between two Banach spaces, $T \in B(V, W)$. Then if the image of the unit ball $B_1(0)$ is dense in some ball $B_r(0)$ with $r > 0$, i.e. $B_r(0) \subseteq \overline{T(B_1(0))}$, then

$$B_{(1-\epsilon)r}(0) \subseteq T(B_1(0))$$

for every $\epsilon > 0$.

***Proof*:**
Let $U = T(B_1(0))$. Since $T(B_1(0))$ is dense in $B_r(0)$, for any $y \in B_r(0)$ and $\epsilon > 0$, there is a $y_1 \in U$ such that $\|y - y_1\| < \epsilon r$. Consider now ϵU. This set is dense in $B_{\epsilon r}(0)$. Hence, there exists a $y_2 \in \epsilon U$ such that $\|(y - y_1) - y_2\| < \epsilon^2 r$. Continuing by induction, we get a sequence (y_n) where $y_n \in \epsilon^{n-1} U$ and

$$\left\| y - \sum_{k=1}^{n} y_k \right\| < \epsilon^n r$$

Choose now $x_n \in B_1(0)$ such that $T(x_n) = y_n$ and $\|x_n\| \leq \epsilon^{n-1}$. Then $\sum x_n = x \in X$ converges, and by definition $T(x) = y$. Since $\|x\| \le \sum \epsilon^{n-1} = (1-\epsilon)^{-1}$ [a] we have

$$(1-\epsilon)^{-1}U \supseteq B_r(0)$$

completing the proof.

[a] recall that $1 - b^n = (1-b)(1 + b + b^2 + \cdots + b^{n-1})$

Theorem 2.2.7: Open Mapping Theorem

Let X and Y be Banach spaces and $T \in B(X,Y)$ be a bounded linear operator with $T(X) = Y$. Then T is an open map (i.e. T maps open sets to open sets)

***Proof*:**
Since T is surjective:

$$Y = T(X) = \bigcup_n T(B_n(0)) \subseteq \bigcup_n \overline{T(B_n(0))} = \bigcup_n n\overline{T(B_1(0))}$$

Then by corollary 2.2.2, not all closed sets $n\overline{T(B_1(0))}$ have an empty interior. Since scaling by n is a homeomorphism, it follows that $\overline{T(B_1(0))}$ itself has nonempty interior. Hence, there exist $y_0 \in Y$ and $\delta > 0$ such that

$$B_\delta(y_0) \subseteq \overline{T(B_1(0))}$$

We now show $B_{\delta/2}(0) \subseteq \overline{T(B_1(0))}$. For any z with $\|z\| < \delta$, both y_0 and $y_0 + z$ lie in $B_\delta(y_0) \subseteq \overline{T(B_1(0))}$. Hence:

$$z = (y_0 + z) - y_0 \in \overline{T(B_1(0))} - \overline{T(B_1(0))} \subseteq \overline{T(B_1(0)) - T(B_1(0))} \subseteq \overline{T(B_2(0))} = 2\overline{T(B_1(0))}$$

where we used linearity of T in the last two steps. It follows that $B_\delta(0) \subseteq 2\overline{T(B_1(0))}$, i.e. $B_{\delta/2}(0) \subseteq \overline{T(B_1(0))}$, so $T(B_1(0))$ is dense in $B_{\delta/2}(0)$.

By the Open Mapping Lemma (proposition 2.2.6), $B_s(0) \subseteq T(B_1(0))$ for every $s < \delta/2$. Since every open set $U \subseteq X$ is a union of open balls, we conclude from linearity of T that $T(U)$ contains a neighborhood around each of its points, thus $T(U)$ is open, completing the proof.

Corollary 2.2.8: Bijective And Continuous

Let $T : X \to Y$ be a bijective bounded operator. Then it has a bounded inverse, that is T is in fact a Banach-space isomorphism

The following also lets us recover a result similar to the one showing us that there is only one norm on a finite dimensional vector space up to equivalence, namely no (non-equivalent) norm on a Banach space can dominate another:

Corollary 2.2.9: Equivalent Banach Spaces

Let X be a vector-space, and a Banach space under two norms $\|\cdot\|_a$ and $\|\cdot\|_b$. Then if $\|\cdot\|_a \le \alpha\|\cdot\|_b$ for some $\alpha > 0$, then there is a $\beta > 0$ such that $\|\cdot\|_b \le \beta\|\cdot\|_a$

***Proof*:**
Take $\mathrm{id} : (X, \|\cdot\|_b) \to (X, \|\cdot\|_a)$. This is certainly bijective, continuous by the boundedness assumption, and open by the open mapping theorem. Hence it is a homeomorphism, meaning it has a bounded inverse, giving the desired result.

The next powerful result we get is a quick way of proving a function is bounded:

Theorem 2.2.10: Closed Graph Theorem

Let $T : X \to Y$ be an operator between Banach spaces. Then if the graph

$$\Gamma(T) = \{(x, y) \in X \times Y \mid T(x) = y\}$$

is closed in $X \times Y$, then T is bounded

***Proof*:**
Using the ∞-norm $\|(x, y)\| = \max(\|x\|_X, \|y\|_Y)$ on $X \times Y$, assume that $\Gamma(T)$ is closed in $X \times Y$. Since $X \times Y$ is a Banach space (as both X and Y are), and $\Gamma(T)$ is a closed linear subspace, $\Gamma(T)$ is itself a Banach space. Now consider the projection map $\pi_1 : \Gamma(T) \to X$ given by $\pi_1(x, T(x)) = x$. This is bijective (since T is a function) and norm-decreasing: $\|\pi_1(x, T(x))\| = \|x\| \leq \max(\|x\|, \|T(x)\|)$. Since π_1 is a bounded bijection between Banach spaces, the Banach isomorphism theorem (corollary 2.2.8) gives that π_1^{-1} is bounded. The other projection $\pi_2 : \Gamma(T) \to Y$ given by $\pi_2(x, T(x)) = T(x)$ is also norm-decreasing by the same reasoning. Hence:

$$T = \pi_2 \circ \pi_1^{-1}$$

is a composition of bounded operators, and therefore bounded, completing the proof.

The key take-away is that if we are verifying the continuity of a linear map $T : X \to Y$ (so $x_n \to x$ implies $T(x_n) \to T(x)$), we just need to verify the closedness of the graph. The closedness of the graph implies that if $x_n \to x$ and $T(x_n) \to y$, then $T(x) = y$. Notice how this simplifies proving a function is continuous, since we did not need to show that $\lim_n T(x_n) = T(x)$, simply that it converges to *something*, namely if $T(x_n) \to y$, then $T(x) = y$. Note how completeness is crucial in both these theorems (exercise 29-31 folland for counter-examples).

The converse of the closed graph theorem, that if $T : X \to Y$ is bounded between Banach spaces, then $\Gamma(T)$ is closed, is elementary: let's say T is bounded and that $T(x_n) \to y$. Since $x_n \to x$, $\lim_n T(x_n) = T(x)$. Then:

$$\begin{aligned}\|T(x) - y\|_Y &\leq \|T(x) - y + T(x_n) - T(x_n)\|_Y \\ &\leq \|T(x - x_n)\|_Y + \|T(x_n) - y\|_Y \\ &\leq C\|x - x_n\|_X + \|T(x_n) - y\| \xrightarrow{n\to\infty} 0\end{aligned}$$

hence, $\|T(x) - y\| = 0$, implying $T(x) = y$ and showing $\Gamma(T)$ is closed.

Finally, the following result is a way to pass from pointwise boundedness to uniform boundedness, only requiring the completeness of our domain! It can be thought of as one of the "pillars" of functional analysis:

Theorem 2.2.11: Principle Of Uniform Boundedness

Let X, Y be normed vector spaces and $A \subseteq L(X, Y)$. Then if X is a Banach space and $\sup_{T\in A} \|T(x)\| < \infty$ for all $x \in X$, then $\sup_{T\in A} \|T\| < \infty$

This theorem is also called by some the Banach–Steinhaus Theorem.

***Proof*:**
We want to upgrade from a pointwise bound to a uniform bound, showing that $\sup_{T\in A} \|T\| < \infty$. Let

$$E_n = \left\{ x \in X \;\middle|\; \sup_{T\in A} \|T(x)\| \leq n \right\} = \bigcap_{T\in A} \{x \in X \mid \|T(x)\| \leq n\}$$

Each E_n is closed (as an intersection of closed sets, since each T is continuous) and $X = \bigcup_n E_n$ (since $\sup_{T\in A} \|T(x)\| < \infty$ for each x). Then by the Baire Category Theorem, there must exist some $n \in \mathbb{N}$ such that E_n has nonempty interior. Hence, for some $x_0 \in E_n$ and $r > 0$, $\overline{B_r}(x_0) \subseteq E_n$.

Now, consider $u \in X$ such that $\|u\| \leq 1$ and take any $T \in A$. Since $x_0 + ru \in \overline{B_r}(x_0) \subseteq E_n$, we have $\|T(x_0 + ru)\| \leq n$. Also $x_0 \in E_n$, so $\|T(x_0)\| \leq n$. Then:

$$\begin{aligned}
\|T(u)\|_Y = r^{-1}\|T(ru)\|_Y &= r^{-1}\|T(x_0 + ru) - T(x_0)\|_Y \\
&\leq r^{-1}(\|T(x_0 + ru)\|_Y + \|T(x_0)\|_Y) \\
&\leq r^{-1}(n + n)
\end{aligned}$$

Hence, taking the supremum over u in the unit ball of X and over $T \in A$, we get:

$$\sup_{T\in A} \|T\|_{B(X,Y)} \leq 2r^{-1}n < \infty$$

completing the proof.

If X is simply a normed vector-space, we may keep this theorem by simply taking the set supremum of the set of nonmeager points. The key realization is that $\sup_{T\in A} \|T(x)\|$ is a different supremum for each x, but is finite for all x! That is, we are not working with $\sup_{\substack{T\in A \\ x\in X}} \|T(x)\|$ since that would make the result trivial:

$$\sup_{T\in A} \|T\| = \sup_{\substack{T\in A \\ \|x\|=1}} \|T(x)\| \leq \sup_{\substack{T\in A \\ x\in X}} \|T(x)\|$$

This is what makes the Uniform boundedness principle so strong, we got a *uniform* bound by having a *local* bound for all x!

Example 2.2.12: Divergence of Fourier Series
One of the most celebrated applications of the uniform boundedness principle is the existence of continuous functions whose Fourier series diverges at a given point. Consider $X = C(\mathbb{T})$, the Banach space of continuous 2π-periodic functions with the supremum norm. For each n, define the functional $\Lambda_n : C(\mathbb{T}) \to \mathbb{C}$ by

$$\Lambda_n(f) = S_n f(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) D_n(t)\, dt$$

where $S_n f(0)$ is the n-th partial sum of the Fourier series of f evaluated at 0, and $D_n(t) = \sum_{k=-n}^{n} e^{ikt}$ is the Dirichlet kernel. Each Λ_n is a bounded linear functional with $\|\Lambda_n\| = \frac{1}{2\pi}\int_{-\pi}^{\pi} |D_n(t)|\,dt = \|D_n\|_{L^1}$. It is a classical computation that $\|D_n\|_{L^1} \sim \frac{4}{\pi^2}\log n \to \infty$. Hence $\sup_n \|\Lambda_n\| = \infty$. By the *contrapositive* of the uniform boundedness principle, there must exist some $f \in C(\mathbb{T})$ for which $\sup_n |\Lambda_n(f)| = \sup_n |S_n f(0)| = \infty$, i.e. the Fourier series of f diverges at 0. By translation, the same holds at any prescribed point.

Note how the principle is used here: we showed $\sup_n \|\Lambda_n\| = \infty$, and concluded that the pointwise boundedness hypothesis must fail for some f, even though we never constructed such an f explicitly. This is a quintessential non-constructive existence proof via Baire category methods.

Corollary 2.2.13: Pointwise Convergence of Operators

Let X be a Banach space, Y a normed vector space, and $(T_n)_{n\in\mathbb{N}} \subseteq B(X,Y)$ a sequence of bounded operators such that $(T_n(x))_{n\in\mathbb{N}}$ converges in Y for every $x \in X$. Define $T(x) = \lim_n T_n(x)$. Then $T \in B(X,Y)$, i.e. the pointwise limit is itself a bounded linear operator.

***Proof*:**
First, T is linear: for $x, y \in X$ and α, β scalars, $T(\alpha x+\beta y) = \lim_n T_n(\alpha x+\beta y) = \alpha \lim_n T_n(x) + \beta \lim_n T_n(y) = \alpha T(x) + \beta T(y)$.

Since $(T_n(x))$ converges for each x, it is bounded: $\sup_n \|T_n(x)\| < \infty$ for all $x \in X$. Hence, by the principle of uniform boundedness, $\sup_n \|T_n\| = C < \infty$. That is, $\|T_n(x)\| \leq C\|x\|$ for all n and all $x \in X$, where C is independent of n. Taking $n \to \infty$ we get:

$$\|T(x)\| \leq C\|x\|$$

which shows $T \in B(X,Y)$, completing the proof.

More generally, if $(T_\lambda)_{\lambda\in\Lambda} \subseteq B(X,Y)$ is a net such that each $(T_\lambda(x))$ is bounded and converges for every $x \in X$, the same conclusion holds. For sequences, convergence automatically implies boundedness. It also suffices in the principle of uniform boundedness to know that each sequence $(T_n(x))$ is bounded and that the sequence converges for a dense set of elements in X.

We conclude this section with an application of the Baire Category Theorem that does not involve linear operators. It shows that the set of discontinuities of a pointwise limit of continuous functions is, in a precise sense, small:

Theorem 2.2.14: Continuity of Pointwise Limits

Let (f_n) be a sequence of continuous (complex-valued) functions on a complete metric space X such that $f(x) = \lim_{n\to\infty} f_n(x)$ exists for every $x \in X$. Then the set of points at which f is continuous is a dense G_δ (and hence residual) in X.

***Proof*:**
Recall that f is continuous at x if and only if the *oscillation* of f at x is zero, where

$$\omega_f(x) = \inf_{\delta>0} \sup_{y,z\in B_\delta(x)} |f(y) - f(z)|$$

Thus the set of continuity points of f is $\{x : \omega_f(x) = 0\} = \bigcap_{m=1}^{\infty}\{x : \omega_f(x) < 1/m\}$. Each set $O_m = \{x : \omega_f(x) < 1/m\}$ is open (this follows from the definition of oscillation), so the set of continuity points is a G_δ. It remains to show that each O_m is dense.

Fix $m \geq 1$ and let B be any nonempty open ball in X. We must find a point in $B \cap O_m$. For each $n, k \in \mathbb{N}$, define

$$F_{n,k} = \left\{ x \in \overline{B} \;\middle|\; |f_n(x) - f_k(x)| \leq \frac{1}{3m} \right\}$$

Each $F_{n,k}$ is closed (since $f_n - f_k$ is continuous). Since $f_n(x) \to f(x)$ for every x, for each $x \in \overline{B}$ there exists N such that $|f_n(x) - f_k(x)| \leq 1/(3m)$ for all $n, k \geq N$. Hence:

$$\overline{B} = \bigcup_{N=1}^{\infty} \bigcap_{n,k \geq N} F_{n,k}$$

The sets $G_N = \bigcap_{n,k\geq N} F_{n,k}$ are closed in $\overline{B}$. Since $\overline{B}$ is a complete metric space (closed subset of X), the Baire Category Theorem implies that some G_{N_0} has nonempty interior in $\overline{B}$. Hence there exists an open ball $B' \subseteq B$ such that $|f_n(x) - f_k(x)| \leq 1/(3m)$ for all $n, k \geq N_0$ and all $x \in B'$.

Taking $k \to \infty$, we get $|f_n(x) - f(x)| \leq 1/(3m)$ for all $n \geq N_0$ and $x \in B'$. In particular, f_{N_0} is continuous, so we may shrink B' so that f_{N_0} varies by at most $1/(3m)$ on B'. Then for any $y, z \in B'$:

$$|f(y) - f(z)| \leq |f(y) - f_{N_0}(y)| + |f_{N_0}(y) - f_{N_0}(z)| + |f_{N_0}(z) - f(z)| \leq \frac{1}{3m} + \frac{1}{3m} + \frac{1}{3m} = \frac{1}{m}$$

Hence any point of B' lies in O_m, showing that O_m is dense. Since each O_m is open and dense, $\bigcap_m O_m$ is a dense G_δ by the Baire Category Theorem.

As a consequence, if $f = \lim f_n$ is a pointwise limit of continuous functions (such an f is called a *Baire class 1* function), then f cannot be discontinuous everywhere: its set of continuity points is always residual. This provides another perspective on why "most" functions are badly behaved: a generic continuous function is nowhere differentiable (example 2.2.4), but any pointwise limit of continuous functions must still be continuous on a large set.

2.3 Dual Spaces

Let V be a Banach space over k (k being either $\mathbb{R}$ or $\mathbb{C}$). Since we are in the category of Banach spaces, we'll consider $V^* := \text{Hom}(V, k)$ to the collection of all continuous linear functionals (i.e. bounded linear functionals). If V is finite dimensional, then it is clear what the structure of V^*, namely since $V \cong V^*$. In infinite dimensions, it is in fact hard to identify when a funcitonal is continuous and bounded. Famously, d/dx is an unbounded operator, and $\int$ is bounded given an appropriate restriction of the domain. It would be good if we can find some easier type of function that we know can be extended to a functional. We start by doing just that, in particular we start by proving the Hahn-Banach extension theorem, which shows that there exists "enough" linear functions in X^* to make their study intersting (this theorem is the equivalent of the Uryohn's Theorem from topology).

The key "simpler" function is the following:

Definition 2.3.1: Linear Subfunctional

Let $m : V \to \mathbb{R}$ be a subatitive function $m(x+y) \le m(x)+m(y)$ that is positive homogeneous $m(tx) = tm(x)$ for all $t \ge 0$ in $\mathbb{R}$. Then m is called a *linear subfunctional*.

Note that it is not a semi-norm since homogeneity only needs to hold for positive scalars. On the otherhand, every semi-norm (and hence norms) are example of minkowski functionals. Note that the result works just as well over real or complex fields:

Proposition 2.3.2: Converting Complex And Real Linear Functionals

Let X be a vector space over $\mathbb{C}$. If f is a complex linear functional on X, and $u = \text{Re} f$, the nu is a real linear functional, and $f(x) = u(x) - iu(ix)$ for all $x \in X$. Conversely, if u is a real lienar functional on X and $f : X \to \mathbb{C}$ is defiend by $f(x) = u(x) - iu(ix)$, then f is a complex linear functional. In this case, if X is normed, $\|u\| = \|f\|$

***Proof*:**
If f is complex linear then $u = \text{Re} f$ is clealry real linear and $\text{Im} f(x) = -\text{Re}[if(x)] = -u(ix)$, so $f(x) = u(x) - iu(ix)$. On ther other hiand, if u is real linea, then $f(x) = u(x) - iu(ix)$ is clearly R liner and

$$f(ix) = u(ix) - iu(-x) = u(ix) + iu(x) = if(x)$$

thus f is $\mathbb{C}$-linaer. Finally, if X is norme,d since $|u(x)| = |\text{Re} f(x)| \le |f(x)|$, $\|u\| \le \|f$
. On the other ahdn, if $f(x) \neq 0$, let $\alpha = \overline{\text{sgn}(f(x))}$. Then

$$|f(x)| = \alpha f(x) = f(\alpha x) = u(\alpha x)$$

since $f(\alpha x)$ is real. Hence $|f(x)| \le \|U\|\|\alpha x\| = \|u\|\|x\|$, thus $\|f\| \le \|u$
, hence

$$\|f\| = \|u\|$$

completing the proof.

Theorem 2.3.3: Hahn-Banach Extension theorem

Let X be a real vector space, p a sublinear functional on X, $M \subseteq X$ a subspace of X, and f a linear funtional on M such that $f(x) \le p(x)$ for all $x \in M$. Then there exitss a lienar functional F on X such that $F(x) \le p(x)$ for all $x \in X$ and $F|_M = f$

***Proof*:**
We shall start with a "base case" and then use Zorn's lemma (in this context, think of it as transfinite induction). Let $x \in X \setminus M$. We shall extend f to a functional g on $M + \mathbb{R}x$ that satisfies $g(y) \le p(y)$ on $M + \mathbb{R}x$. First, first any $y_1, y_1 \in M$, we have

$$f(y_1) + f(y_2) = f(y_1 + y_2) \le p(y_1 + y_2) \le p(y_1 - x) + p(x + y_2)$$

where x is our above chosen $x \in X$. Re-arranging, we get:

$$f(y_1) - p(y_1 - x) \leq p(x + y_2) - f(y_2)$$

Since this is true for all $y_1, y_2 \in M$, we have:

$$\sup_{y \in M}\{f(y) - p(y - x)\} \leq \inf_{y \in M}\{p(x + y) - f(y)\}$$

Thus, we can find an α satisfying:

$$\sup_{y \in M}\{f(y) - p(y - x)\} \leq \alpha \leq \inf_{y \in M}\{p(x + y) - f(y)\}$$

Define

$$g : M + \mathbb{R}x \to \mathbb{R} \qquad g(y + \lambda x) = f(y) + \lambda\alpha$$

Clearly g is linear, $g|_M = f$, and on M $g(y) \leq p(y)$ (equivalently when $\lambda = 0$). When $\lambda > 0$, then for any $y \in M$:

$$g(y + \lambda x) = \lambda[f(y/\lambda) + \alpha] \leq \lambda[f(y/\lambda) + p(x + (y/\lambda)) - f(y/\lambda)] = p(y + \lambda)$$

and when $\lambda = -\mu < 0$:

$$g(y + \lambda x) = \mu[f(y/\mu) - \alpha] \leq \mu[f(y/\mu) - f(y/\mu) + p((y/\mu) - x)] = p(y + \lambda x)$$

Thus, $g(z) \leq p(z)$ for all $z \in M + \mathbb{R}x$.

But now, notice that g satisfies the same condition as in the hypothesis: g is a linaer functional on a subspace $M + \mathbb{R}x \subseteq X$ such that $g(z) \leq p(z)$ for all $z \in M + \mathbb{R}x$. We can thus finish off by noticing that we can create a partial order given by inclusion of the domains (since the domains resrct appropraitely), and we clearly see that this partial is closed under unions of chains. Hence, it has a maximal, which through an easy proof by contradiction can be shown to be X, completing the proof.

Before moving on to some of the fascinating consequence, we first insure the we understand how to move this result to complex functionals. For order to be well-defined, we must insure that we are using seminorms instead of general lienar subfunctionals, that is if p is a seminorm and $f : X \to \mathbb{R}$ is lienra, then the inequality $f \leq p$ is equivalent to $|f| \leq p$ since$|f(x)| = \pm(x) = f(\pm x)$ and $p(-x) = p(x)$.

Proposition 2.3.4: Complex Hahn-Banach Theorem

Let X be a vetor space over $\mathbb{C}$, p a seminorm on X, $M \subseteq X$ a subspace of X and f a compelx lienar functional on M such that $|f(x)| \leq p(x)$ for all $x \in M$. Then there exists a compelx linear functional F on X such that $|F(x)| \leq p(x)$ for all $x \in X$ and $F|_M = f$

***Proof*:**

Let $u = \text{Re}f$. By The Hahn-Banach Theorem there exists a linear extension of u to X such that $|U(x)| \leq p(x)$ for all $x \in X$. Let $F(x) = \underline{U(x) - iU(ix)}$. Then F is a complex linear extension of f, and by proposition 2.3.2, if $\alpha = \underline{\text{sgn}(f(x))}$ we have $|f(x)| = \alpha F(x) = F(\alpha x) = U(\alpha x) \leq p(\alpha x) = p(x)$, completing the proof.

Proposition 2.3.5: Hahn-Banach Consequences

Let X be a normed vector-spae. Then:

1. If $M \subseteq X$ is a closed subpsace and $x \in X \setminus M$, there exists a $f \in X^*$ such that $f(x) \neq 0$ and $f|_M = 0$. In particulr, if $\delta = \inf_{y \in M} \|x - y\|$, there exists an f such that
$$\|f\| = 1 \qquad f(x) = \delta$$
2. If $0 \neq x \in X$, there exists $f \in X^*$ such that $\|f\| = 1$ and $f(x) = \|x\|$
3. The bounded lienar functionals on X seperate points
4. if $x \in X$, the double dual $\hat{x} : X^* \to \mathbb{C}$ given by $\hat{x}(f) = f(x)$ is alienar isometry from X to X^{**}

***Proof*:**

1. $M \subseteq X$ be a closed subspace and $\delta = \inf_{y \in M} \|x - y\|$. Define
$$f : M + \mathbb{C}x \qquad f(y + \lambda x) = \lambda \delta$$
Then $f(x) = \delta$, $f|_M = 0$, and for $\lambda \neq 0$ by the infinum proeprty:
$$|f(y + \lambda x)| = |\lambda|\delta \leq |\lambda|\|\lambda^{-1}y + x\| = \|y + \lambda x\|$$
Hence, taking $p(x) = \|x\|$ to be the norm (which is a lienar subfunctional defined on X), by the Hahn-Banach theorem F on X where $F(x) = \delta$ and $F|_M = 0$. Finally, if $\|y + \lambda x\| = 1$, then certainly the supremum of all such values when plugged into F must be 1, hence $\|F\| = 1$
Alternatively, apply the result from part (a) to X/Y, then note that functionals in $(X/Y)^*$ may be regarded as elements in X^* that annihilate Y
2. In the above, let $M = \{0\}$ be the trivial vector-space and define $f(x) = \|x\|$.
3. If $x \neq y$, by (b) there exists a $f \in X^*$ such that $f(x - y) \neq 0$, hence $f(x) \neq f(y)$
4. Certainly $\hat{x}$ is a lienar functioanl onX^* and $x \mapsto \hat{x}$ is linaer. Furthermore
$$|\hat{x}(f)| = |f(x)| \leq \|f\|\|x\|$$
hence $\|\hat{x}\| \leq \|x\|$. On the other hand, by (b), we get $\|\hat{x}\| \geq \|x\|$, hence we have an isometry

The first (and second) point can be thought of as a generalization of Uryshon's lemma, and also inspires the following definition:

Definition 2.3.6: Annihilator

For a subspace $Y \subseteq X$ of a normed space X, the *annihilator* of Y is defined as:

$$Y^\perp = \{\varphi \in X^* \mid \varphi(y) = 0, \ \forall y \in Y\}$$

Similarly, we define the annihilator of a subspace $Y^* \subseteq X^*$ to be

$$(Y^*)^\perp = \{x \in X \mid \varphi(x) = 0, \ \forall \varphi \in Y^*\}$$

It is immedaite that $Y \subseteq (Y^\perp)^\perp$, and it follows that in fact $Y = (Y^\perp)^\perp$. for every norm closed subspace $Y^* \subseteq X^*$.

The third point is useful if we have setup a senario where for all $f \in X^*$:

$$f(T(x)) = f(y) \qquad f \in X^*$$

Since the f separate points, it must be that

$$T(x) = y$$

in other words, separation of points can be thought of as as sort of "injectivity". As an exercise, show that if $T : X \to Y$ is a linear operator and for eac $f \in Y^*$, $f \circ T \in T^*$, then T is bounded (hint: used closed graph theorem and the above remark).

The 4th point inspires a quick look at double duals. If we let $\hat{X} = \{\hat{x} \mid x \in X\}$ and let X^{**} be the double dual, then since X^{**} is always complete, the closure of $\hat{X}$ in X^{**}, $\overline{\hat{X}}$ is banach space, and $x \mapsto \hat{x}$ embeds X into $\overline{\hat{X}}$ as a dense subspace. The sapce $\overline{\hat{X}}$ is called the *completion* of X, and if X Is itself Banah then $\overline{\hat{X}} = \hat{X}$. If X is finite dimensional, then $\hat{X} = X^{**}$, since they have the same dimension. This is not guaranteed in infinite dimension, hecne if $\hat{X} = X^{**}$, we shall call X a *reflexive space*. For example, $C_c^\infty([0,1])$ will be shown to not be reflexive. We shall see later that L^p spaces are reflexive.

2.4 Weak Topologies and TVS's

(This post seems like an excellent post to follow! link here)

There are many topologies on which the operations are continuous (i.e. + and scalar multiplication is continuous) that do not arrise from norms. The topology on a vector-space induced by a norm is in fact quite strong. In general, such a space (in infinite dimensions) doesn't have many compact set, and so we loose many of the nice results of working on comapct or pre-compact sets. This is naturally unfavourable in analysis, since many arguments rely on properties of sequences converging. Looking at functional analysis as the study of sequences, norms induce a particular definition of convergence: $\|x - x_n\|_V \to 0$. However, there are many other notions of convergence that are weaker that are quite useful, for example:

1. pointwise convergence
2. convergence in measure

3. smooth convergence

4. convergence on compact sets

None of these types of convergence are induced by a norm. We will instead be induced new types of topologies on our vector-spaces. The two types of topologies we will usually impose on V will be known as the *weak topology* and the *weak* topology*. We will define analogous topologies on $\operatorname{Hom}_k(V, W)$ which somewhat confusingly will be named *strong operator topology* and *weak operator topology*.

Definition 2.4.1: Topological Vector Space (TVS)

Let V be a vector-space over $\mathbb{R}$ or $\mathbb{C}$. Then V is a *topological vector space* (TVS) if the operation of addition and scalar multiplication are continuous. A topological vector-space is said to be *locally convex* if there is a base for the topology consisting of convex sets.

It should be verified that the translation map $x \mapsto x_0 + x$ and the dilation map $x \mapsto \lambda x$ are both homeomorphissm for any scalar λ and vector x_0.

Example 2.4.2: Topological Vector Spaces

1. Let V be a normed vector-space with the topology induced by the norm. Then $\|c \cdot x\| \leq |c|\|x\|$ (in fact $=$) and $\| + (x, y)\| \overset{!}{\leq} 2(\|x\| + \|y\|)$ where $\overset{!}{\leq}$ comes from the triangle inequality and the fact that $\|(x, y)\| = \max(\|x\|, \|y\|)$. Hence, V is a topologicla vector space. If $\| - \|$ was a quasi-norm so that we replace the triangle inequality with $\|x + y\| \leq K(\|x\| + \|y\|)$ then a quasi-normed space V is still a topological vector space.

2. Let V be a semi-normed space so that $\|x\| = 0$ does not necessarily imply $x = 0$. Then V is a topological vector space with the topology induced by the semi-norm. V is Hausdroff if and only if the semi-norm is a norm.

3. Any subbspace $W \subseteq V$ of a topological vector space V is a topological vector space with the subspace topology. Similarly for products and quotients.

4. Let V be a vector-space and $(\mathcal{T}_\alpha)_{\alpha \in A}$ be a collection of topologies on V that makes V a topological vector space. Let $\mathcal{T} = \langle \mathcal{T}_\alpha \rangle_{\alpha \in A}$ be the topology generated by $\bigcup_{\alpha \in A} \mathcal{T}_\alpha$ (this is the weakest topology that contains each $\mathcal{T}_\alpha$). Then $(V, \mathcal{T})$ is also a topological vector space. Importantly, a sequence (resp. net) $x_n \to x$ in $\mathcal{T}$ if and only if $x_n \to x$ in $\mathcal{T}_\alpha$ for all $\alpha \in A$ (recall that $x_n \to x$ in a general topology space if for any neighborhood U of x, there exists an N such that for all $n \geq N$, $x_n \in U$). Commonly, we will have $(\mathcal{T}_\alpha)_{i=1}^\infty$ be generated by countable collection of semi-norms. A complete space generated by a countable collection of semi-norms is called a *Fréchet space*.

5. (Pointwise Convergence) Let X be a set and $\mathbb{C}^X$ the collection of all complex valued functions $f : X \to \mathbb{C}$. Then $\mathbb{C}^X$ is a complex-vector-space. For each $x \in X$, define the seminorm:

$$\|f\|_x := |f(x)|$$

Then the topology generated by all of these seminorms is the *topology of pointwise convergence* on $\mathbb{C}^X$ (it is also the product topology on this space). A sequence $(f_n) \subseteq \mathbb{C}^X$ converges to

f in this topology if and only if it converges pointwise. Note that if X has more than one point, then none of the semi-norms individually generate a Hausdroff topology, but taken all together the induced topology is Haudsorff.

6. (Uniform Convergence) Let X be a topological space and $C(X)$ the set of complex-valued continuous function $f : X \to \mathbb{C}$. If X is not compact, then in general we would not expect functions in $C(X)$ to be bounded, so the sup-norm will not necessarily make $C(X)$ into a normed vector-space. Nonetheless, we can still define "balls" $B_r(f) \subseteq C(X)$ by:

$$B_r(f) := \left\{ g \in C(X) \;\middle|\; \sup_{x \in X} |f(x) - g(x)| \leq r \right\}$$

these do form a topological structure on $C(X)$, but not a topological vector space structure since scalar multiplication is no longer continuous. Nonetheless, we have that $(f_n) \subseteq C(X)$ converges in this topology to a limit $f \in C(X)$ if and only if f_n converge uniformaly to f. Thus $\sup_{x \in X} |f_n(x) - f(x)|$ is finite for sufficinetly large n and converges to 0 as $n \to \infty$.

7. (Uniform Convergence on Compact sets) Let X and $C(X)$ be as in the above example. For every compact subset $K \subseteq X$, define a seminorm:

$$\|f\|_{C(K)} = \sup_{x \in K} |f(x)|$$

The topology generated by all these seminorms as k ranges over all compact subsets of X is called the *topology of uniform convergence on compact sets*. It is stronger than the topology of pointwise convergence but weaker than the topology of uniform convergence, and is frequently used in complex analysis. A sequence $(f_n) \subseteq C(X)$ converges to $f \in C(X)$ in this toplogy if and only if f_n converges uniformly to f on each compact set. Is the resulting space a topological vector space?

8. (Smooth Convergence) Let $C^\infty([0,1])$ be the space of smooth functions $f : [0,1] \to \mathbb{C}$. Then we can define the C^k-norm on this space for any non-negative integer k by the formula:

$$\|f\|_{C^k} := \sum_{i=1}^{k} \sup_{x \in [0,1]} |f^{(i)}(x)|$$

where $f^{(i)}$ is the ith derivative of f. The topology generated by all C^k norms $k = 0, 1, 2, ...,$ is the *smooth topology*. A sequence f_n converges in this topology to a limit f if $f_n^{(i)}$ converges uniformally to $f^{(i)}$ for each $i \geq 0$. Is this a topological vector space?

9. (Convergence in measure) Let $(X, \mathcal{M}, \mu)$ be a measure space and $L(X)$ the set of measurable functions $f : X \to \mathbb{C}$. Then the sets

$$B_{r,\epsilon}(f) := \{g \in L(X) \mid \mu(\{x \mid |f(x) - g(x)| \geq r\} < \epsilon)\}$$

for $f \in L(X)$, $\epsilon, r > 0$ forms the basis for a topology that makes $L(X)$ into a topological vector space. A sequence $(f_n) \subseteq L(X)$ converges to f in this topology if and only if it converges in measure.

10. Let $[0,1]$ be given the usual Lebesgue measure. Then show that $L^\infty([0,1])$ cannot be given a topological vector space structure in which a sequence $(f_n) \subseteq L^\infty([0,1])$ converges to f in this topology if and only if it converges a.e.

11. Let V be an $\mathbb{R}$-vector space. a subset $U \subseteq V$ is called *algebraically open* if
$$\{t \in \mathbb{R} \mid x + tv \in U\}$$
is open for all $x, v \in V$. Then any open set in a topological vector space is algebraically open. Note that there are algebraically open sets that are not open (take any dense subset of S^1, say D, and $\mathbb{R} \setminus \{D\}$.) This larger set of algebraically open sets forms a topology, but this topology does not give the $\mathbb{R}^2$ structure of a topological vector space.

12. Let X be a LCH space. On $\mathbb{C}^X$, the topology of uniform convergence on compact sets is defined by the seminorms $p_K(f) = \sup_{x \in K} |f(x)|$ as K rangers over compact subsets of X. If X is σ-compact and $\{U_n\}$ are as in proposition ref;HERE (folland p. 168), this topology is defined by the seminorms $p_n(f) = \sup_{x \in \overline{U_n}} |f(x)|$. In this case, $\mathbb{C}^X$ is easly seen to be complete, so it is a Fréchet space. By proposition ref:HERE, so is $C(X)$.

13. The space $L^1_{\mathrm{loc}}(\mathbb{R}^n)$ is a Fréchet space with the topology defined by the seminorms $p_k(f) = \int_{|x| \leq k} |f(x)| dx$ (completeness follows from completness of L^1).

14. Here is another useful place where TVS are defined. Let's say we want to study a space in which d/dx is continuous (and hence bounded) on $C^\infty([0,1])$. However, this is essentially impossible to define the topology through a norm. Indeed, if $f_\lambda(x) = e^{\lambda x}$, then $(d/dx) f_\lambda = \lambda f_\lambda$, and so $\|d/dx\| \geq \lambda$ for all λ, no matter what norm is used on $C^\infty([0,1])$. There are a few ways to rectify this:
    (a) One can consider differentiation as an unbounded operator form V to W where W is a suitable banace space and X is a dense subpsac eof W (exercise 30 in Folland)
    (b) One can consider differentiation as a bounded linear map from one Banach space V to a different one W such that $V = C^k([0,10])$ and $W = C^{k-1}([0,1])$ (execise 9)
    (c) One can consider differentiation as a continuous operator on a locally convex space V whose topology is not given by a norm. It is easy to construct families of seminorms on the space of smooth functions st differentiation becomes continuous almost by definition. For example, the seminorms:
    $$p_k(f) = \sup_{0 \leq x \leq 1} |f^{(k)}(x)|$$
    where $k \in \{0, 1, 2, ...\}$ makes $C^\infty([0,1])$ into a Fréchet space (the completeness is proved in exercise 9), and d/dx is continuous by proposition 5.15, since $p_k(f') = p_{k+1}(f)$. Other examples are in exer exercsie 45 in chatper 9 of Folland.

15. The space $\mathbb{R}^\infty = \varinjlim \mathbb{R}^n$ is a Fréchet space; can you see the norms?

Proposition 2.4.3: Topological Vector Space And Hausdorff

Let V be a topological vector space. Then V is Hausdroff if and only if $\{0\}$ is closed

***Proof*:**
exercise

We may ask what properties of Banach spaces extend to topological vector space's. Some nice properties of Banach spaces include the uniform boundedness principle and the open mapping theorem. These can be extended to the slighty larger class of *Fréchet spaces* that we mentioned in the examples above. We will single them out and talk about them in the next section.

If V is a topological vector space, we can take V^* to be the set of all continuous functionals from V to the base field $k \in \{\mathbb{R}, \mathbb{C}\}$. Since V has no norm, V^* has no norm topology. We can however still define interesting topologies on it. This topology is dual to a weaker topology we can put on V:

Definition 2.4.4: Weak And Weak* Topologies

Let V be a topological vector space and V^* its dual. Then

1. the *weak* topology* on V^* is the topology generated by the seminorms $\|\lambda\|_x := |\lambda(x)|$ for all $x \in V$
2. the *weak topology* on V is the topology generated by the seminorms $\|x\|_\lambda = |\lambda(x)|$ for all $\lambda \in V^*$

If V is also a normed vector-space, then the weak topology on V is weaker than the norm topology. If V is a normed space, then so is V^* and $(V^*)^*$. the weak* topology on V^* makes V^* into a topological vector space, and is a topology weaker than the weak topology on V^* (which is defined using the double dual $(V^*)^*$). When V is reflexive ($V \cong V^{**}$), then the weak and weak* topologies on V^* are in fact equivalent.

Definition 2.4.5: Weak Convergence

Let V a vector-space endowed with the weak topology. Then a sequence $(x_n) \subseteq V$ that converges in V with the weak topology is called *weakly convergent* and is said to *weakly converge* to a point x. A sequence weakly converges $x_n \to x$ if and only if

$$\lambda(x_n) \to \lambda(x) \qquad \forall \lambda \in V^*$$

and is usually denoted as $x_n \rightharpoonup x$. Similarly, $\lambda_n \rightharpoonup \lambda$ if $\lambda_n(x) \to \lambda(x)$ for all $x \in V$ (λ_n, as a function of V, converges pointwise to λ)

Proposition 2.4.6: Normed Vector Spaces Weak Topology

Let V be a normed vector-space. Then the weak topology on V and the weak* topology on V^* are both Hausdorff. In particular, we conclude that in the weak and weak* limits, when they exist they are unique

***Proof*:**
Use the Hahn-Banach Theorem

Example 2.4.7: Norm, Weak, Weak* Convergence
Let $V = c_0(\mathbb{N})$, $V^* = \ell^1(\mathbb{N})$, and $V^{**} = \ell^\infty(\mathbb{N})$. Let $e_1, e_2, ...$ be the standard basis of any of these vetor-spaces.

1. The sequence (e_i) converges weakly in V to zero, but not not converge in norm in V
2. (easy to think up some examples)

A natural generalization of the weak* topology on X^* can be imposed $B(X,Y)$ for some TVS Y. Just like for X^*, we can impose a weak topology on it, hewer there is another natural topology we may induce on $B(X,Y)$:

Definition 2.4.8: Strong And Weak Operator Topology

Let $B(X,Y)$ be a all continuous operators between Fréchet spaces. Then

1. the *weak operator topology* is the topology induced by the seminorm $T \mapsto |\lambda(T(x))|$ for all $x \in X$ and $\lambda \in Y^*$
2. If furthermore Y is a Banach space, then the *strong operator topology* (or *operator norm topolgoy*) is the topology induced by the seminorms $T \mapsto \|T(x)\|_Y$ for all $x \in X$

Hence, a sequence $(T_n) \subseteq B(X,Y)$ converges in the strong operator topology to a limit T if and only if $T_n(x) \to T(x)$ in the norm topology on Y, and T_n converges to T in the weak operator topology if $T_n(x) \to T(x)$ in e weak topology on x. Hence, we see that the weaker operator topology is weaker than the strong operator topology, which in turn is (somewaht confusingly) weaker than the operator norm topology.

A final result we shall point out is hat we can reprhase the uniform boundedness principle for convergence now as follows:

Proposition 2.4.9: Uniform Boundedness Principle Reformulation

Let $(T_N) \subseteq B(X,Y)$ be a sequence of bounded linear operators from a Banach space X to a normed space Y, and let $T \in B(X,Y)$ be another bounded lienar operator, and let D be a dense subspace of X. Then the following are equivalent:

1. T_n converges in the strong operator topology fo $B(X,Y)$ to T
2. T_n is bounded in the operator norm (i.e. $||T_n||_{op}$ is bounded) and the restriction of T_n to D converges in the strong operator topology of $B(D,Y)$ to the restriction of T to D

2.5 Fréchet Spaces

When constructing a space, we often cannot define a space simply through a norm; not all elements may be "finite". However, we can often construct a space using countably many normed spaces, or usually semi-normed space $\|x\| = 0$ does not imply $x = 0$, for example if f is an integrable function). Due to the countable nature of this construction, most types of convergences are accomplished by a family of semi-norms. For example, convergence on compact sets, pointwise convergnce, uniform convergence, and smooth convergence are all convergence in Fréchet spaces. In analysis, there will be many spaces that will Fréchet spaces, notably when you encounter distributions and require to define a topology on the dual space. Hence, a moment to get a better undersatnding of them is worth it.

Theorem 2.5.1: Semi-Norms Generate TVS

Let $\{p_\alpha\}_{\alpha\in A}$ be a family of norms on a vector space V. If $x \in X$, $\alpha \in A$, and $\epsilon > 0$, let

$$U_{x,\alpha,\epsilon} = \{y \in V \mid p_\alpha(y-x) < \epsilon\}$$

and let $\mathcal{T}$ be the topology generated by the sets $U_{x,\alpha,\epsilon}$. Then:

1. For each $x \in X<$ the inite intersections of the sets $U_{x,\alpha,\epsilon}$ $(\alpha \in A, \epsilon > 0)$ form a neigborhood base at x
2. if $\langle x_i\rangle_{i\in I}$ is a net in V ,then $x_i \to x$ if and only if $p_\alpha(x_i - x) \to 0$ for all $\alpha \in A$.
3. $(V, \mathcal{T})$ is a locally convex topological vector-space

***Proof*:**

1. If $x \in \bigcap_1^k U_{x_j,\alpha_j,\epsilon_j}$, let $\delta_j = \epsilon_j - p_\alpha(x - x_j)$. By the triangle inequality

$$x \in \cap_1^k U_{x,\alpha_j,\delta_j} \subseteq \cap_1^k U_{x-j,\alpha_j,\epsilon_j}$$

   Thus, the assertion now follows from the fact that the topology generated consists of $\emptyset$, X, and all unions of finite intersections.

2. By part (1), it suffices to see that $p_\alpha(x_i - x) \to 0$ if and only if $\langle x_i\rangle$ is eventually in $U_{x,\alpha,\epsilon}$ for all $\epsilon > 0$.

3. Continuity is easy from part (b), for if $x_i \to x$ nd $y_i \to y$, then

$$p_\alpha((x_i + y_i) - (x + y)) \leq p_\alpha(x_i - x) + p_\alpha(y_i - y) \to 0$$

   so $x_i + y_i \to x + y$. If furthermore $\lambda_i \to \lambda$, then eventually $\lambda_i \leq C + |\lambda| + 1$, so

$$p + \alpha(\lambda_i x_i - \lambda x) \leq p_\alpha(\lambda_i(x_i - x)) + p_\alpha((\lambda_i - \lambda)x) \leq CP_\alpha(x_i - x) + |\lambda_i - \lambda| p_\alpha(x)$$

   Hence, $\lambda_i x_i \to \lambda x$. Furthermore, the sets $U_{x,\alpha,\epsilon}$ are convex, for if $y, z \in U_{x,\alpha,\epsilon}$, then

$$p_\alpha(x - [t + (1-t)z]) \leq p_\alpha(tx - ty) + p_\alpha((1-t)x + (1-t)z) < t\epsilon + (1-t)\epsilon = \epsilon$$

   Then local convexity follows from (1)

Definition 2.5.2: Fréchet Space

A complete Hausdorff topological vector-space whose topology is defined by a countable family of seminroms is called a *Fréchet space.*

Proposition 2.5.3: Continuity in Fréchet Space

Let $T : V \to W$ be a linear map between two vector spaces. Let V be a topology induced by a family of semi-norms $(\| - \|_{V_\alpha})_{\alpha \in A}$ and W the topology induced by a family of semi-norm $(\| - \|_{W_\beta})_{\beta \in B}$. Then T is continuous if and only if for each $\beta \in B$, there exists a finite subsets $A_\beta \subseteq A$ and a constant C_β such that

$$\|T(x)\|_{W_\beta} \leq C_\beta \sum_{\alpha \in A_\beta} \|x\|_{V_\alpha} \tag{2.1}$$

for all $x \in V$.

***Proof*:**
Let equation 2.1 hold and consider a convergent net $\langle x_i \rangle \subseteq X$. Then by theorem 2.5.1 $p_\alpha(x_i - x) \to 0$ for all α, thus by the inequality $\|T(x_i) - T(x)\|_{S_\beta} \to 0$ for all β, which implies $T(x_i) \to T(x)$. Since T preserves convergent nets, T is continuous.

Conversely, if T is continuous for every $\beta \in B$ choose a neighbor U_β of 0 in V such that $\|T(x)\|_{W_\beta} < 1$ for all $x \in U_\beta$. By theorem 2.5.1, we may assume that $U_\beta = \cap_{1k} U_{x \alpha_{\epsilon_j}}$. Let $\epsilon = \min(\epsilon_1, \epsilon_2, ..., \epsilon_k)$ so that $\|T(x)\|_{W_\beta} < 1$ whenever $p_{\alpha_j}(x) < \epsilon$ for all j. Now, for $x \in X$, either:

1. $p_{\alpha_j})(x) > 0$ for some j. Then let $y = \epsilon x / (\sum_1^k p_{\alpha_j}(x))$ so tht $p_{\alpha_j}(y) < \epsilon$. Then:

$$\|T(x)\|_{W_\beta} = \sum_1^k \epsilon^{-1} p_{\alpha_j}(x) \|T(y)\|_{W_\beta} \leq \epsilon^{-1} \sum_1^k p_{\alpha_j}(x)$$

2. If $p_{\alpha_j}(x) = 0$ for all j, then $p_{\alpha_j}(rx) = 0$ for all j and all $r > 0$. Thus $r\|T(x)\|_{W_\beta} = \|T(rx)\|_{W_\beta} < 1$ for all $r > 0$, and os $\|T(x)\|_{W_\beta}$. Then certainly

$$\|T(x)\|_{W_\beta} \leq \epsilon^{-1} \sum_1^k p_{\alpha_j}(x)$$

covering both cases, we are done.

The following proposition is an interesting characteriztion of Fréchtet spaces:

Proposition 2.5.4: Hausdorff and invariant Metric

Let X be Frćhet space given by $\{p_\alpha\}_{\alpha \in A}$. Then

1. X is Haudorff if and only if for each $x \neq 0$, there exists $\alpha \in A$ such that $p_\alpha(x) \neq 0$
2. If X is Hausdorff and A is countable, then X is metrizable with a trnslation invariant metric (i.e. $\rho(x, y) = \rho(x + z, y + z)$ for all $x, y, z \in X$)

***Proof*:**
exercise 43 Folland chatper 5

Note that topologically, all these spaces are the same (as the bellow theorem will sketch). So what really matters is what extra information these semi-norms (or this invariant metric) provides:

Theorem 2.5.5: Anderson-Kadec Theorem

Every infinite-dimensional separable Fréchet space is homeomorphic to the product space $\mathbb{R}^{\mathbb{N}}$ (the countable product of real lines equipped with the product topology).

Note that L^∞ is *inseparable* and hence is not included in this theorem.

***Proof*:**
(Sketch). The proof is historically a combination of results by Kadec (1966) and Anderson (1966), establishing a chain of homeomorphisms. We outline the two main steps:

1. The Banach Case: Kadec proved that every infinite-dimensional separable Banach space is homeomorphic to the Hilbert space ℓ_2. The construction typically involves re-norming the space with a strictly convex equivalent norm (often called a Kadec norm) and defining a homeomorphism via radial projection and coordinate mapping.
2. The Product Case: Anderson proved that ℓ_2 is homeomorphic to $\mathbb{R}^{\mathbb{N}}$. This is often shown using "separable absorption" theorems or the "notback-and-forth" method to manage the coordinate systems.

Since every Fréchet space is a projective limit of Banach spaces (and using the assumption of separability), one can extend the homeomorphism to the general case. Consequently, despite their distinct linear and geometric properties, spaces such as the Schwartz space $\mathscr{S}(\mathbb{R})$, the space of holomorphic functions $H(\mathbb{C})$, and $C^\infty[0,1]$ are all topologically indistinguishable from $\mathbb{R}^{\mathbb{N}}$.

Proposition 2.5.6: dense convergence, then strong convergence

Let $(T_n)_1^\infty \subseteq L(V,W)$ where $\sup_n \|T_n\| < \infty$ and $T \in L(V,W)$. If $\|T_n(x) - T(x)\| \to 0$ for all x in a dense subset D of X, then $T_n \to T$ strongly.

***Proof*:**
We want to show that $\|T_n(x) - T(x)\| \to 0$ for all x. Let $C = \sup\{\|T\|, \|T_1\|, \|T_2\|, ...\}$. Then given $x \in X$ and $\epsilon > 0$, choose $x' \in D$ such that $\|x - x'\| < \epsilon/3C$. Since $T_n(x') \to T(x)$, choose n large enough so that $\|T_n(x') - T(x')\| < \epsilon/3$. Then

$$\begin{aligned}\|T_n(x) - T(x)\| &\leq \|T_n(x) - T_n(x')\| + \|T_n(x') - T(x')\| + \|T(x') - T(x)\| \\ &\leq 2C\|x - x'\| + \frac{1}{3}\epsilon \\ &< \epsilon\end{aligned}$$

but then $T_n(x) \to T(x)$ strongly, as we sought to show.

One important result is that in the weak operator topology, unit balls are always compact:

Theorem 2.5.7: Alaoglu's Theorem

Let X be a normed space, and B^* the closed unit ball of X^*, $B* = \{f \in X^* \mid \|f\| \leq 1\}$. Then B^* is compact in the weak*-topology.

***Proof*:**
We shall show that B^* is a closed subspace of a compact space. For each $x \in X$, define

$$D_x = \{z \in \mathbb{C} \mid \|z\| \leq \|x\|\}$$

and take $D = \prod_{x \in X} D_x$. Then by Tychnoff's Theorem, D is compact. Now, we may exactly identify the elements of D the complex-valued function $\varphi : X \to \mathbb{C}$ such that $|\varphi(x)| \leq \|x\|$ for all $x \in X$. Then B^* would conssit of the elements of D that are linear. Hence, we must just show that B^* is closed. The induced topology B^* from the product topology on D and the weak* topology on X^* are both the topology of pointwise convergence (for D by the product construction), and hence checking for closedness in X is equivalence to checking closedness of D. Take a net $\langle f_\alpha \rangle$ in B^* that converges to $f \in D$ for any $x, y \in X$ and $a, b \in \mathbb{C}$. Then:

$$f(ax + by) = \lim f_\alpha(ax + by) = \lim[af_\alpha(x) + bf_\alpha(y)] = af(x) + bf(y)$$

showing that f is linear, and certainly $\|f\| \leq 1$, hence $f \in B^*$, making it compact and completing the proof.

2.5.8 Note Since X^* is a vector space, one might hope to conclude that it is locally compact (via the "it suffices to show the result at 0" reduction). This is *not* the case, as Exercise 49(b) of Folland's *Real Analysis* shows.

(The following was copied from nLab, and is very informative!)

Every complete locally convex topological vector space T is the cofiltered projective limit of Banach spaces in the category of locally convex spaces. (Note that Fréchet spaces are additionally required to be metrisable, so this is more general.)

(see this here) Now given that a Fréchet space admits a decreasing sequence of convex balanced and absorbing neighborhoods, it follows immediately that: Every Fréchet space is a sequential projective limit of Banach spaces.

(this is also really cool on homomorphisms of Fréchet space here)

3

L^p space

Look at this blog!! https://terrytao.wordpress.com/2009/01/09/245b-notes-3-lp-spaces/
In this chapter, we will be studying Banach spaces which are defined through the integral: L^p spaces. These spaces are a generalization of L^1 spaces, where if $f \in L^p$, then it must decay "faster" at infinity, and f doesn't blow up too fast at any finite point. In other words, L^p spaces capture a type of "speed of growth". As we have seen in the previous chapters, limits of integrals are central to many constructions. For some constructions, being in L^1 was enough, but for others (like Fourier series) we will require that function be more than finite in integral (i.e. more than $f \in L^1$). L^p spaces will give a nice set of examples of Banach spaces that are relatively simple to define.

My current intuition is that they are treated as the intermediate to being able to bound a function: it is sometimes nontrivial to immediately say a function is bounded, but we can build-up to it by saying it's L^3, which will imply L^5, which will imply L^7, and so on allowing us to conclude it's L^∞, which means a functions is bounded up to a zero-measure set. This is a common phenomena in analysis where we may not be able to bet a function to fit exactly in one space or another (in our case L^1 or L^∞ which are bounded a.e. functions), but we can put the function in intermediary buckets.

3.1 Basics

So far, we have limited our attention to measurable functions f such that $\int |f| < \infty$ and have labelled them L^1. Often, what we are looking are function with more precise "convergent properties", in particular:

$$\int |f|^p < \infty$$

for some $p \geq 1$, including ∞ (which we will define shortly). We start by labelling the colletion of all these functions:

Definition 3.1.1: L^p Space

Let f be measurable. for $p \in (1, \infty)$[a], define the *p-norm* to be:

$$\|f\|_p = \left[\int |f|^p \, d\mu\right]^{1/p}$$

From this norm, define:

$$L^p(X, \mathcal{M}, \mu) := \left\{f : X \to \mathbb{C} \;\middle|\; f \text{ is measurable and } \|f\|_p < \infty\right\}$$

[a] we will define L^∞ later

As with L^1, we often abbreviate to $L^p(\mu)$, $L^p(X)$, or just L^p. As with L^1, we'll let L^p be the equivalence class of the set of functions such that $|f|^p$ is integrable where $f \sim g$ if and only if $\int |f - g|^p = 0$. Just like L^1, two functions will be equal in L^p (i.e. belong in the same equivalence class) if they are equal almost everywhere. Furthermore, we claimed that $\|\cdot\|_p$ is a norm. That $\|f\|_p = 0$ if and only if $f = 0$ a.e. comes almost immediately, since

$$\left(\int |f|^p\right)^{1/p} = 0 \quad \text{iff} \quad \int |f|^p = 0 \quad \text{iff} \quad \int |f| = 0 \quad \text{iff} \quad f = 0 \text{ a.e.}$$

For $\|cf\|_p = |c| \, \|f\|_p$, we take advantage of our normalisation:

$$\|cf\|_p = \left(\int |cf|^p\right)^{1/p} = \left(\int |c|^p |f|^p\right)^{1/p} = (|c|^p)^{1/p} \left(\int |f|^p\right)^{1/p} = |c| \, \|f\|_p$$

However, the triangle inequality is not so trivial. We will in fact take good part of this section to show that the triangle inequality is satisfied. For now, we take a moment to explore an example of L^p spaces and L^p functions to gain an intuition:

Example 3.1.2: L^p spaces

1. Let $f : [0, \infty) \to \mathbb{R}$ $f(x) = \frac{1}{x^{1/p}}$. Then $f \in L^q$ for $q > p$. This comes down to the fact that:

$$x > x^{1/p} \;\Leftrightarrow\; \frac{1}{x} < \frac{1}{x^{1/p}}$$

and by the fact that $\int \left|\frac{1}{x^n}\right| < \infty$ if $n > 1$. We have therefore found a class of functions that fits into all possible $p \in [1, \infty)$. In fact, we can find functions that fit into any connected subinterval of $[1, \infty)$! In particular, let $(X, \mathcal{M}, \mu)$ be a measure space and $f : X \to \mathbb{C}$ a complex measurable function on X. Let

$$E := \left\{p \in [1, \infty)| \;\middle|\; \int_X |f|^p < \infty\right\}$$

(a) Then E is connected, that is, if $p, q \in E$ such that $p < r < q$ for some r, then $r \in E$

Proof. Clearly, if $E = \emptyset$ or $|E| = 1$, then E is connected because the only non-trivial clopen set is $\emptyset$ and E, so let $|E| > 1$.

Pick $p, q \in E$ such that $p < q$, and consider a point r such that $p < r < q$. We'll show that $\int_X |f|^r < \infty$ using the following trick:

$$|f| = \chi_{|f|\geq 1}|f| + \chi_{|f|<1}|f|$$

that is, we're splitting $|f|$ to where f is greater than or equal to 1 and strictly less than 1. The equality holds since $A = \{x \mid |f(x)| \geq 1\}$ and $B = \{x \mid |f(x)| < 1\}$ are disjoint. Importantly, notice that:

$$|f|^p = \chi_{|f|\geq 1}|f|^p + \chi_{|f|<1}|f|^p \qquad |f|^q = \chi_{|f|\geq 1}|f|^q + \chi_{|f|<1}|f|^q$$

Thus, by the linearity of the integral, we get that:

$$\int |f|^p = \int \chi_{|f|\geq 1}|f|^p + \int \chi_{|f|<1}|f|^p \qquad \int |f|^q = \int \chi_{|f|\geq 1}|f|^q + \int \chi_{|f|<1}|f|^q$$

It thus suffice show that $\int \chi_{|f|\geq 1}|f|^r < \infty$ and $\int \chi_{|f|<1}|f|^r < \infty$. This is essentially immediate:

$$\chi_{|f|\geq 1}|f|^r \leq \chi_{|f|\geq 1}|f|^q < \infty \qquad \chi_{|f|<1}|f|^r \leq \chi_{|f|<1}|f|^p < \infty$$

Therefore

$$\int |f|^r < \infty$$

showing that $r \in E$, as we sought to show. $\square$

(b) Show that we can pick a measure and an f such that $E = (a, b), [a, b), [a, b]$, or $\{a\}$ for $1 \leq a \leq b \leq \infty$. In other words, all connected subsets of $[1, \infty)$ are possible for E (You may freely use the fact that any Riemann integrable function is also integrable with respect to the Lebesgue measure and its Riemann integral is equal to the Lebesgue integral)
Hint: Think about integrability properties of functions like x^{-a} and $\frac{-1}{\log^2(x)x} = \frac{d}{dx}\frac{1}{\log(x)}$

Proof. Throughout this hole proof we'll be using Riemann integration as a substitute for Lebesgue integration (we showed earlier that this is allowed). We'll be using these function's to construct our intervals:

i. $f(x) = x^{-1/a}$ on $(1, \infty)$: Notice that

$$\int_1^\infty x^{-1/a} dx = \lim_{x\to\infty} \frac{x^{-1/a+1}}{-1/a+1} - \frac{1}{-1/a+1}$$

so $\|f\|_p < \infty$ if and only if $p \in (a, \infty]$.

ii. $f(x) = x^{-1/a}$ on $(0, 1)$: Notice that

$$\int_0^1 x^{-1/a} dx = \frac{1}{-1/a+1} - \lim_{x\to 0} \frac{x^{-1/a+1}}{-1/a+1}$$

so $\|f\|_p < \infty$ if and only if $p \in [1, a)$.

iii. $f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}$ on $(1, \infty)$: This integral doesn't have an elementary solution, however we can still ascertain some basic convergence properties. In particular, if $p = a$, then:

$$\int_1^\infty |f(x)|^a dx = \lim_{x\to\infty} \left[-\frac{1}{\ln(x)}\right] - \lim_{x\to 1} \left[-\frac{1}{\ln(x)}\right]$$

so $\|f\|_p < \infty$ if and only if $p \in [a, \infty]$

iv. $f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}$ on $(0, 1)$: This result is similar to the other 3 cases we've proven. The result for this integral shows that $\|f\|_p < \infty$ if and only if $p \in [1, a]$.

Thus, we have proven all possible "sub-rays" of $[1, \infty]$ can exist. To show all possible sub-intervals (closed, open, or half), we simply add the appropriate function's to produce the result. For example, if we want to have the interval $[a, b)$, simply take:

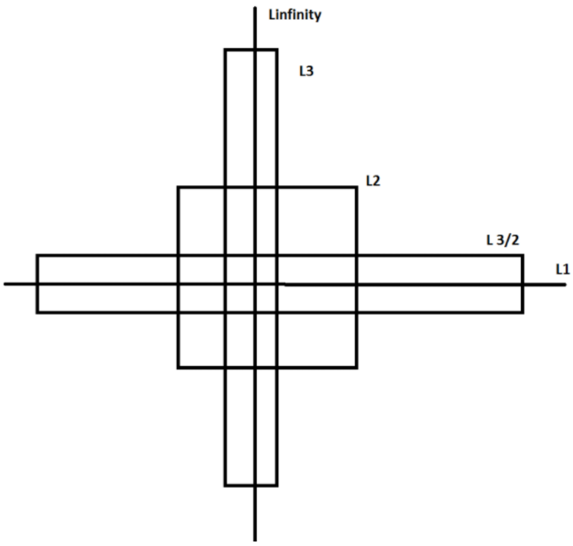
$$f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}\chi_{(1,\infty)} + x^{-1/b}\chi_{(0,1)}$$

which clearly has interval of convergence of $[a, b)$. Similarly, to get a singleton $\{a\}$, take:

$$f(x) = \frac{1}{(\ln^2(x)x)^{1/a}}\chi_{(1,\infty)} + \frac{1}{(\ln^2(x)x)^{1/a}}\chi_{(0,1)}$$

Thus, we have all possible connected subintervals of $[1, \infty]$. □

So if $f \in L^p$, it does not imply that $f \in L^q$ for $p > q$ or $p < q$! Later on, we will show some of the possible relations between L^p spaces. A more accurate intuition is the following [a]: If $f \in L^p$ for any p, then f must decays fast enough at infinity, and f must not blow up too fast at any finite point. As p increases, the first requirement gets less strict, and the second gets more strict. If X is bounded, the first requirement is vacuous, and if X is discrete, the second is vacuous. Sometimes this visual is used to represent these relations:



2. A particular case that will be common for analyzing sequences is when μ is the counting measure on A. In that case, we usually denote $L^p(\mu)$ by $\ell^p(A)$. If $A = \mathbb{N}$, we often simply

abbreviate notation to ℓ^p. If we write $f : \mathbb{N} \to \mathbb{R},\ f(k) = a_k$, then the sequence f (i.e. $\{a_k\}$ is in ℓ^p if:

$$\sum_{k=1}^{\infty} |a_k|^p < \infty$$

Thus, ℓ^1 is the set of convergence series.

[a] credit to this post on stack-exchange: https://mathoverflow.net/questions/314050/intuition-about-lp-spaces

Notice that we are "normalizing" each integral by a factor of $1/p$. This normalization is done to make sure that the constant can be brought out of the norm. It might seem tempted to say that:

$$\left(\int |f|^p\right)^{1/p} \text{ ``='' } \int |f|$$

However, this is most certainly not true (simply take $f(x) = x$). As mentioned before, this makes it non-trivial to show that $\|\cdot\|_p$ is a norm on L^p. However, it is easy to see that L^p is a vector-space. If $f, g \in L^p$, then:

$$|f + g|^p \leq [2\max(|f|, |g|)]^p \leq 2^p(|f|^p + |g|^p)$$

which is finite by assumption, and so $f + g \in L^p$. It is also immediate that $cf \in L^p$. Thus, we may freely add and multiply by L^p functions without worrying about leaving L^p. Because of this, it makes it meaningful to ask whether

$$\|f_1 + f_2\|_p \leq \|f_1\|_p + \|f_2\|_p$$

What we will now do is show that $\|\cdot\|_p$ is does indeed satisfy the triangle inequality for $p \geq 1$. If $p < 1$, then the triangle inequality fails in general:

Example 3.1.3: $p < 1$ then no Triangle Inequality

Take $a, b > 0$ and $0 < p < 1$. Then for any $t \in \mathbb{R}$, we have that:

$$t^{p-1} > (a + t)^{p-1}$$

Integrating both sides from 0 to b, we get:

$$\begin{aligned}
\int_0^b t^{p-1} &> \int_0^b (a+t)^{p-1} \\
\frac{b^p}{p} &> \frac{(a+b)^p}{p} \\
b^p &> (a+b)^p \\
a^p + b^p &> (a+b)^p \\
(a^p + b^p)^{1/p} &> a + b
\end{aligned}$$

Now, if E and F are disjoint union with positive finite measures in X such that $\mu(E)^{1/p} = a$ and $\mu(E)^{1/p} = b$, then $\chi_E, \chi_F \in L^p$. Taking advantage of the fact that $\chi_E\chi_F = 0$ (i.e. the zero function) since $E \cap F = \emptyset$, then:

$$\|\chi_E + \chi_F\|_p = (a^p + b^p)^{1/p} > a + b = \|\chi_E\|_p + \|\chi_F\|_p$$

Thus, the $\|\cdot\|_p$ for $0 < p < 1$ cannot be a norm.

Thus, we stick to showing the case where $p \geq 1$ First, we will state a simple inequality fact similar Jensen's inequality. This is in fact the key idea behind the entire proof:

Lemma 3.1.4: Young's Inequality

If $a, b \geq 0$ and $0 < \lambda < 1$, then:

$$a^{\lambda} b^{1-\lambda} \leq \lambda a + (1-\lambda) b$$

with equality if and only if $a = b$. In particular, notice that if $\frac{1}{p} + \frac{1}{q} = 1$, then if $\lambda = p^{-1}$, then $(1 - \lambda) = q^{-1}$ and so:

$$a^{1/p} b^{1/q} \leq \frac{1}{p} a + \frac{1}{q} b$$

***Proof*:**
This proof is trying to capture what this image is doing in formal language:

https://www.desmos.com/calculator/jdznjzqcju

If $a = b$, equality clearly holds. If $a = 0$ or $b = 0$, inequality clearly holds. If $a, b \neq 0$, then dividing by b and letting $t = a/b$, we get

$$a^{\lambda} b^{-\lambda} \leq \lambda(a/b) + (1-\lambda) \quad \Leftrightarrow \quad t^{\lambda} \leq \lambda t + (1-\lambda) \quad \Leftrightarrow \quad t^{\lambda} - \lambda t \leq (1-\lambda) \tag{3.1}$$

From here, we use some calculus: we see that $t^{\lambda} - \lambda t$ is strictly increasing for $0 < t < 1$ (since $a, b \geq 0$, $t \geq 0$) and strictly decreasing for $t > 1$, so the maximal value occurs at $t = 1$, and in fact we get that the maximal value is $1 - \lambda$:

$$1^{\lambda} - \lambda 1 = 1 - \lambda \leq 1 - \lambda$$

and so the inequalities in equation (3.1) hold! Equality happens when $t = a/b = 1$ which implies $a = b$, completing the proof.

Usually, Young's inequality is stated as

$$a^{\alpha} b^{\beta} \leq \alpha a + \beta b \qquad 0 \leq \alpha, \beta \leq 1, \quad \alpha + \beta = 1$$

which is exactly what we have, since if we fix some α, then it must be that $\beta = 1 - \alpha$. Another formulation of Young's inequality is:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

with $p^{-1} + q^{-1} = 1$. This proof is a "Jensen equality" like proof that more clearly shows the role of concavity and so I'll dedicate a moment to prove this one too:

***Proof*:**
If $a = 0$ or $b = 0$, the result is immediate, so assume $a, b > 0$. Then notice that since log is monotone:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q} \quad \Leftrightarrow \quad \log(ab) \leq \log\left(\frac{a^p}{p} + \frac{b^q}{q}\right)$$

which is great, since we know log is concave and so Jensen's inequality applies. Since $p^{-1}+q^{-1}=1$, we get $t=1/p$ and $(1-t)=1/q$, and so by Jensen's inequality:

$$\log(ta^p+(1-t)b^q)\geq t\log(a^p)+(1-t)\log(b^q)=\frac{p}{p}\log(a)+\frac{q}{q}\log(b)=\log(ab)$$

completing the proof.

The second interpretation of $p^{-1}+q^{-1}=1$ in Young's inequality is a nice way to write the straight line (or sub-straight line) equation without the $1-\lambda$ term, but just two numbers. It furthermore let's us relate two numbers that can be much larger. For example, if I choose $p=100$, then it must be that $q=100/99$ so that

$$p^{-1}+q^{-1}=1/100+99/100=1$$

which, if we consider p and q to be the exponents associated to L^p and L^q, we see that this gives us a way to relate different L^p space!

To see this more succinctly, we use this convexity property given by Young's inequality in the "build-up" lemma towards the triangle inequality. This lemma is important later on in functional analysis since it let's us bound the operation of multiplication of tow functions fg, (namely, $\|fg\|_1$ will be less than a value on the right hand side that seperates f and g). For our current purposess, it is important for the triangle inequality:

Theorem 3.1.5: Hölder's Inequality

Let $1\leq p,q\leq\infty$ satisfy:

$$\frac{1}{p}+\frac{1}{q}=1$$

or equivalently $q=p/(p-1)$, where we will interpret $\frac{1}{\infty}$ as 0. Then if f and g are measurable functions on X then:

$$\|fg\|_1\leq\|f\|_p\,\|g\|_q$$

In particular, if $f\in L^p$ and $g\in L^q$ then $fg\in L^1$. Equality holds if and only if $\alpha|f|^p=\beta|g|^q$ a.e. for some constants α,β with $\alpha\beta\neq 0$

Notice that when $p=q=2$ (if $p=2$, it must be that $q=2$ to satisfy the equation), then this inequality is called the *Cauchy-Schwartz inequality*[1]. Notice too that this is the only time when $p=q$; this will become important when we analyze the $p=2$ case later.

I also found this intuition on a stack-exchange: the slower f decays, the faster g needs to decay, and the faster f blows up, the more we need g to not blow up. The previous sentence is equally true if we switch f and g, which convinces us that L^p spaces should be reflexive, and by considering $f=g$, we see that the restrictions on f and g should balance at exactly $p=2$, so L^2 should be self-dual.

Notice too that since we are taking the absolute value of the function on the inside, the integral can be thought of as a sort of "concave function", where the input would be, say in the special case of $\mathbb{R}$ as the domain, some interval. This makes it easier to see why this inequality holds (notice that convex functions must preserve monotonicity, and so does the integral).

[1]Technically we need to define an inner product on L^2 to satisfy this inequality. However, as well see in chapter 5, L^2 is indeed a Hilbert space

***Proof*:**
If $\|f\|_p = 0$ or $\|g\|_p = 0$, then $f = 0$ or $g = 0$ a.e., and so equality holds. Similarly if $\|f\|_p = \infty$ or $\|g\|_p = \infty$, inequality immediately holds. Furthermore, notice that if we find a particular f and g for which the result holds, then the result holds for af and bg since

$$\begin{aligned}
&\Leftrightarrow \ \|afbg\|_1 \leq |ab| \, \|af\|_p \, \|bg\|_q \\
&\Leftrightarrow \ |ab| \, \|fg\|_1 \leq |ab| \, \|f\|_p \, \|g\|_q \\
&\Leftrightarrow \ \|fg\|_1 \leq \|f\|_p \, \|g\|_q
\end{aligned}$$

So, it suffices to show the result when $\|f\|_p = \|g\|_q = 1$, and equality holds of $|f|^p = |g|^q = 1$. To this end, we use the previous lemma and set $a = |f(x)|^p$ and $b = |g(x)|^q$ and $\lambda = p^{-1}$ (so that $1 - \lambda = 1 - p^{-1} = q^{-1}$). Then by the previous lemma we have:

$$|f(x)|^{p \cdot p^{-1}} |g(x)|^{q \cdot q^{-1}} \leq p^{-1}|f(x)|^p + q^{-1}|g(x)|^q \quad \Leftrightarrow \quad |f(x)g(x)| \leq p^{-1}|f(x)|^p + q^{-1}|g(x)|^q \tag{3.2}$$

Now, integrating both sides, we get:

$$\|fg\|_1 \leq p^{-1} \int |f|^p + q^{-1} \int |f|^q = p^{-1} + q^{-1} = 1 = \|f\|_p \, \|g\|_q$$

where equality holds when equality holds in equation (3.2) a.e., which holds only when $|f|^p = |g|^q$ a.e., as we sought to show.

For a visual conformation, you can check out:

https://www.desmos.com/calculator/gwqvfv4tie

When p and q satisfy the equality in the equation, we call them *conjugate exponents*. As mentioned before, the conjugate exponents is simply another way or writing the equation of a straight line to apply Young's inequality. From this, we can take advantage of this convexity property yet again to prove the triangle inequality of $\|\cdot\|_p$, which has a different name since for the special case of the p-norm due to the fact it is used on other places and needs to be referenced:

Theorem 3.1.6: Minkowski's Inequality

If $1 \leq p < \infty$ and $f, g \in L^p$. Then:

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

Before proceeding with the proof, a quick word on notation to avoid confusion:

$$\|f\|_p^p = \left(\int |f|^p \right)^{p/p} = \int |f|^p$$

We often work with $\|f\|_p^p = \int |f|^p$ to get rid of the $1/p$ exponent to be able to work with the linearity of the integral, and re-introduce it at the end to get the desried result. We did not have to do this for Hölder's inequality since we reduced to a very nice special case.

***Proof*:**
If $p = 1$, then the result immediately follows by the regular triangle inequality and linearity of the integral. If $f + g = 0$ a.e., then the result also follows, so assume that $f + g \neq 0$ a.e., which using the triangle inequality:

$$|f + g|^p \leq (|f| + |g|)|f + g|^{p-1} = |f||f + g|^{p-1} + |g||f + g|^{p-1}$$

So, we will apply Hölder's inequality using the fact that $q = \frac{p}{p-1}$ and integrating:

$$\begin{aligned}
\|f+g\|_p^p &= \int |f+g|^p d\mu \\
&= \int |f+g| \cdot |f+g|^{p-1} d\mu \\
&\leq \int (|f| + |g|)|f+g|^{p-1} d\mu \\
&= \int |f||f+g|^{p-1} d\mu + \int |g||f+g|^{p-1} d\mu \\
&= \left\| |f|(|f+g|^{p-1}) \right\|_1 + \left\| |g|(|f+g|^{p-1}) \right\|_1 \\
&\leq \||f|\|_p \|f+g\|_q^{p-1} + \|g\|_p \|f+g\|_q^{p-1} && \text{Hölder's Inequality} \\
&= \left(\left(\int |f|^p d\mu \right)^{\frac{1}{p}} + \left(\int |g|^p d\mu \right)^{\frac{1}{p}} \right) \left(\int |f+g|^{(p-1)\left(\frac{p}{p-1}\right)} d\mu \right)^{\frac{p-1}{p}} && \text{recall } q = \frac{p}{p-1} \\
&= \left(\left(\int |f|^p d\mu \right)^{\frac{1}{p}} + \left(\int |g|^p d\mu \right)^{\frac{1}{p}} \right) \left(\int |f+g|^p d\mu \right)^{\frac{p-1}{p}} \\
&= (\|f\|_p + \|g\|_p) \frac{\|f+g\|_p^p}{\|f+g\|_p}
\end{aligned}$$

Thus, multiplying both sides by $\frac{\|f+g\|_p}{\|f+g\|_p^p}$, we get

$$\|f+g\|_p \leq \|f\|_p + \|g\|_p$$

As we sought to show

Thus, we get that $\|\cdot\|_p$ is indeed a norm! Thus L^p is a normed vector space. In fact, just like L^1, it is a complete normed vector space, i.e., a Banach space. We first recall a lemma to simply our task. To understand the lemma, notice that every norm defines a metric:

$$\rho(x, y) = \|x - y\|$$

and so defines a topology, in particular a normal space, that has the notion of sequences well-defined. Next, recall that if $\{x_n\}$ is a sequence in V, it's series is said to converge if there exists an $x \in V$ such that $\sum_{n=1}^{\infty} x_n = x$, and is said to be absolutely convergence if $\sum_{n=1}^{\infty} \|x_n\| < \infty$ (this is lemma 2.1.6). If you did not cover this result yet, it is important to go see it now to understand the following result:

Theorem 3.1.7: L^p is a Banach Space

For $1 \le p < \infty$, L^p is a complete normed vector-space, i.e., a Banach space. Moreover, if $f_n \to f$ in L^p, then there exists a subsequence that converges pointwise to f a.e.

***Proof*:**
Let $1 \le p < \infty$. By the previous lemma, it suffices to show that every absolutely convergent series converges:

$$\sum_{k=1}^{\infty} \|f\|_p = B < \infty \quad \Rightarrow \quad \exists f \text{ s.t. } \sum_{k=1}^{\infty} f_k = f \in L^p$$

Let $G_n = \sum_{k=1}^{n} |f_k|$ be the partial sums of the series and $G = \sum_{k=1}^{\infty} |f_k|$ be the series. Then by Minkowski's inequality, we get:

$$\|G_n\|_p \le \sum_{k=1}^{n} \|f_k\|_p \le B \quad \forall n \in \mathbb{N}$$

and so $\|G\|_p = \left(\int |G|^p\right)^{1/p} < \infty$ and so $\left(\int |G|^p\right) < \infty$. Since $G_n \le G_{n+1}$, and $G_n : X \to [0, \infty)$, by the Monotonce Convergence Theorem:

$$\int \lim_{n\to\infty} G_n^p = \lim_{n\to\infty} \int G_n^p \le B^p < \infty$$

and so $G \in L^p$. By proposition **??**, $G(x) < \infty$ a.e., so the series $\sum_{k=1}^{\infty} f_k$ converges a.e. If we let $F = \sum_{k=1}^{\infty} f_k$, we have that $|F| \le G$ a.e., and so $F \in L^p$. From this, we can see that:

$$\left\| F - \sum_{k=1}^{\infty} f_k \right\|_p^p = \int \left| F - \sum_{k=1}^{\infty} f_k \right|^p \to 0$$

Thus, the series $\sum_{k=1}^{\infty} f$ converges (to F) in the p-norm, and so by lemma 2.1.6, L^p is complete, as we sought to show.

This is the proof the prof was doing in class, but I got stuck so found a different proof (the above proof) (Commented it out for now)

This also finally showed that L^1 is a complete space under $\|\cdot\|_1$! Note the importance of taking a subsequence: since $f_n \to f$ in the p-norm, i.e. the "sum-averaging" of the function converges, then there can be lots of wiggle room for hwo the functions f_n approach f:

Example 3.1.8: Taking A Subsequence
Construct a sequence of functions such that all the following hold:

1. $f_n : [0, 1] \to \mathbb{R}$ is continuous
2. $0 \le f_n(x) \le 1$ for every x
3. $\lim_{n\to\infty} \int_0^1 f_n dm = 0$
4. For every $x \in [0, 1]$ we have that $\lim_{n\to\infty} f_n(x)$ does not exist

(that is, it is definitely important to take a subsequence in order to deduce pointwise convergence from converge in L^1)

Proof. Essentially, we'll be taking advantage that $\sum_n \frac{1}{n}$ does not converge. Let $f_1(x) = 1$, that is, that constant function with constant 1.

For g_2, we split the interval into 2 equal parts. Define

$$g_2(x) = \begin{cases} 1 & x \in [0, 1/2] \\ 0 & x \in (1/2, 1] \end{cases}$$

This function is not continuous. We will make it continuous by simply climbing to it (and for other values, we'll be descending back down too). At the point $1/2$, slope down at a speed of 2^2 until we reach 0:

$$f_2(x) = \begin{cases} 1 & x \in [0, 1/2] \\ -2^2 \left(x - \left(1 - \frac{1}{4}\right)\right) & x \in [1/2, 3/4] \\ 0 & x \in (3/4, 1] \end{cases}$$

From here, we will define our function in a sort of "recursive" sense. Starting form the inteval $[0, 1/2)$, we keep the right end point, and add an interval of length $1/3$ to it: $[1/2, 5/6)$. Now, define f_3 to be 1 on $[1/2, 5/6]$, and zero everywhere else. Next, to define f_4, repeat the process by adding $1/4$ to $5/6$, giving us the interveal $[5/6, 13/12]$. Note that $13/12$ goes beyond 1. We thus wrap it around back to 0: $[0, 1/12] \cup [5/6, 1]$. Now, define a function f_3 to be 1 on this set, and zero everywhere else.

In each of these functions, we extend them to be continuous by attaching them a slope which has 2^n as a factor. In this way, the slope shrinks in size quicker than the size of the rectangles, and so we can morally imagine the problem to be one of indicator functions.

We can continue doing this, defining every f_k by appending $1/k$ and wrapping around when we go past 1 and then making the function continuous. Clearly, $\int f_k \to 0$ as $k \to \infty$ since each inteval will be of total length $1/k + 1/2^k$. However, f_k does not converge anywhere: for any $x \in [0, 1]$, we can choose a subsequence from $f_k(x)$ (say $f_{k_l}(x)$) such that $f_{k_l}(x) = 1$. or $f_{k_l}(x) = 0$

To construct such a sequence, it is easier to imagine our construction without the wrapping around, that is, we keep adding intervals of length $1/k$ as though our co-domain was $[0, \infty)$. Then, our first value of l will be the value for which $x + 1$ is in some interval of length $1/k$ for appropriate k. This interval will exist, since $\sum_k 1/k$ diverges. Similarly, our next value of l for which $x + 2$ will be in some interval of length $1/k$ for appropraite k. We can continue doing this indefinitely, showing that there exists a subsequence of f_k that doesn't converge to 0, meaning f_k cannot converge to 0. On the other hand, it is easy to see that we can choose a subsequence that converges to 0 (since f_k is zero after the trapezoid passes by our point).

Since x was arbitrary, this will work any $x \in [0, 1]$. Thus, f does not converge anywhere, as we sought to show. $\square$

Retruning to analyzing L^p spaces, we can now extend theorem **??** to show that simple function are dense in every L^p!

Theorem 3.1.9: Simple Functions Dense in L^p

For $1 \leq p < \infty$, the set of simple functions $f = \sum_{i=1}^{n} a_i \chi_{E_i}$ where $\mu(E_i) < \infty$ for all i is dense in L^p

***Proof*:**
First of all, the set of simple functions is in L^p for every $1 \leq p < \infty$. Next, choosing any $f \in L^p$, choose some standard representation $\{f_n\}$ where $f_n \to f$ a.e[a]. Then since $|f_n| \leq |f|$, we get that $|f_n| \in L^p$. Furthermore

$$|f - f_n|^p \leq |2f|^p = 2^p |f|^p \in L^1$$

and so by the dominated convergence theorem, we get:

$$\int |f - f_n|^p \to 0 \quad \Rightarrow \quad \left(\int |f - f_n|^p \right)^{1/p} = \|f - f_n\|_p \to 0$$

Moreover, if $f_n = \sum_{k=1}^{\infty} a_k \chi_{E_k}$ was a simple function where all E_i are pairwise disjoint and a_i are nonzero, it must be that $\mu(E_i) < \infty$ since

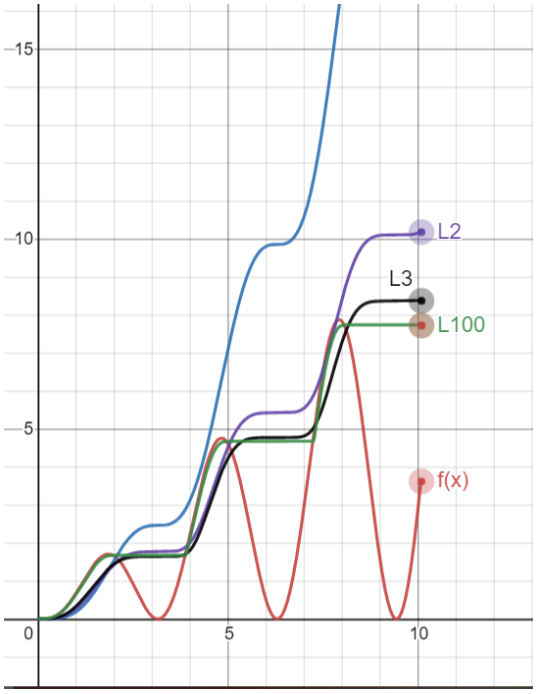
$$\sum_{k=1}^{\infty} |a_k|^p \mu(E_k) = \int |f_n|^p < \infty$$

completing the proof

[a]Recall that the any measurable function has a standard representation, not just integrable functions

Some further remarks about which sets are dense can be enlightening. For $1 \leq p, q < \infty$, L^p is dense in L^q, since each simple set of the form we descriped are subsets of L^p for all $p \geq 1$. If we modify the characteristic function to make them continuous (simply draw a connecting line as we've done in example 3.1.8), then we see that $C_0(\mathbb{R})$ (continuous functions that vanish at infinity) are dense in L^p for every $1 \leq p < \infty$. If we restrict to closed intervals, then $C([a, b])$ is dense in $L^p([a, b])$ since the standard representation becomes uniformaly convergent on bounded domains (see theorem **??**).

So far, we have defined L^p spaces where $0 < p < \infty$ and showed that for $1 \leq p < \infty$, L^p is a Banach space. Interestingly, if we take a look at a graph of $\|f\|_p$ as $p \to \infty$, we'll find that there is a striking pattern that appears:



Notice that as $p \to \infty$, the resulting $\|f\|_p$ value got closer to a max function. In fact, this is almost the case! We might be tempted to then say that $\|f\|_\infty = \sup f$. However, the problem with this is that we might have a measure 0 set that diverges, while the supremum of the rest of the function is attained by a value $k < \infty$. The idea of $\sup f$ on all but a zero measure set will be called the *essential supremum*, and all functions that have a finite essential supremum will be denoted as L^∞. In this way, L^∞ is simply a natural extension of our definition of L^p:

To see the intuition behind this, let us simply consider the p-norm of $\|\vec{v}\|_p = (|3|^p + |10|^p)^{1/p}$

1. Case $p = 1$: $(3^1 + 10^1)^{1/1} = 13$. (The 3 contributes about 23% to the total)
2. Case $p = 2$ (Euclidean): $\sqrt{3^2 + 10^2} = \sqrt{9 + 100} = \sqrt{109} \approx 10.44$. (The 3 matters less now. The 10 is dominating)
3. Case $p = 10$: $(3^{10} + 10^{10})^{1/10} = (59{,}049 + 10{,}000{,}000{,}000)^{1/10}$. Notice that 3^{10} is negligible dust compared to 10^{10}. $\approx (10^{10})^{1/10} = 10.00000\textit{something}$
4. Case $p = 100$: The 3^{100} is mathematically invisible next to 10^{100}. The sum is basically just 10^{100}. When you take the $1/100$ root, you are just returning the base number: 10.

Definition 3.1.10: L^∞ Space

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f : X \to [0, \infty]$ be a measurable function. Let

$$S = \left\{ r \in \mathbb{R} \;\middle|\; \mu(f^{-1}((r, \infty])) = 0 \right\}$$

Then the *essential supremum* of f, denoted $\|f\|_\infty$ or $\operatorname{ess\,sup} f$ is :

$$\|f\|_\infty := \begin{cases} \infty & \text{if } S = \emptyset \\ \inf S & \text{if } S \neq \emptyset \end{cases}$$

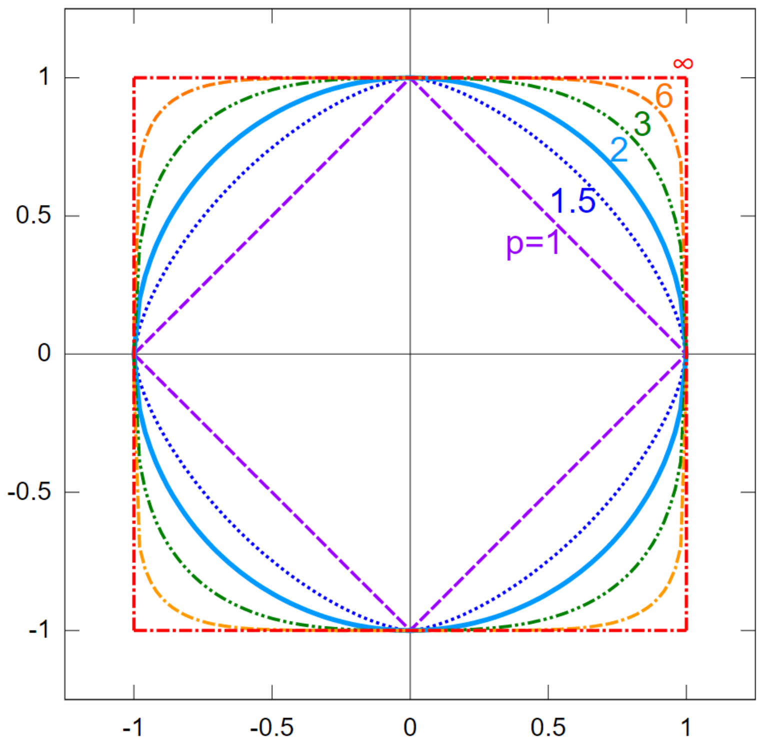
or, equivalently:

$$\|f\|_\infty = \inf_{\substack{E \in \mathcal{M} \\ \mu(X \setminus E) = 0}} \left(\sup_{x \in E} |f(x)| \right)$$

If f is a complex measurable, function, then $\|f\|_\infty = \|\,|f|\,\|_\infty$

Essentially, $\|f\|_\infty$ is the least upper bound of f, ignoring measure zero sets worth of points. Using this definition, we can define L^∞ to be the equivalence classes of functions f such that $\|f\|_\infty < \infty$.

Another quick way of visualizing the relation between all p-norms, $1 \leq p \leq \infty$ is by taking the ℓ^p norm on $\mathbb{R}^2$, taking the function $x \mapsto \|x\|_p = \sqrt[p]{x^p + y^p}$, and considering when $\|x\| = 1$, or in terms of set notation $S_p = \left\{ x \in \mathbb{R}^2 \;\middle|\; \|x\|_p = 1 \right\}$. We'd get:



A couple more quick remarks: $L^\infty(\mu)$ depends on μ only insofar as μ determines which sets have measure zero. If μ and ν are mutually absolutely continuous ($\mu \ll \nu$ and $\nu \ll \mu$[2]) then $L^\infty(\mu) = L^\infty(\nu)$. Secondly, if μ is not semi-finite, it is useful to adpot a slightly useful definition of L^∞ (Folland left this as exercsie 23-25 in chatper 6).

Most properties of L^p for $1 \leq p < \infty$ generalises naturally to the L^∞ case:

Proposition 3.1.11: Properties of L^∞

Let $(X, \mathcal{M}, \mu)$ be a measure and L^∞ the set of measurable function satisfying $\|\cdot\|_\infty < \infty$. Then:

1. If f, g are measurable functions on X, then

$$\|fg\|_1 \leq \|f\|_1 \|g\|_\infty$$

   If $f \in L^1$ and $g \in L^\infty$, then $\|fg\|_1 = \|f\|_1 \|g\|_\infty$ if and only if $|g(x)| = \|g\|_\infty$ a.e. on the set where $f(x) \neq 0$

2. $\|\cdot\|_\infty$ is a norm on L^∞

3. $\|f_n - f\|_\infty \to 0$ if and only if there exists $E \in \mathcal{M}$ such that $\mu(E^c) = 0$ and $f_n \to f$ uniformally on E

4. L^∞ is a Banach space

5. The simple functions are dense in L^∞

As a consequence of the well-definedness of L^∞, $1^{-1} + \infty^{-1} = 1$ is indeed a well-defined notion (so $\infty^{-1} = 1/\infty = 0$). Thus, 1 and ∞ are conjugate exponents!

***Proof*:**

We will prove (1), from which the rest of the results follow.

1. We'll be using the $\|f\|_\infty = \inf\limits_{\substack{E \in \mathcal{M} \\ \mu(X \setminus E) = 0}} (\sup_{x\in E} |f(x)|)$ definition of the essential supremum to take advantage of the supremum within the definition. Our goal is to show that

$$\int |fg| = \|fg\|_1 \leq \|f\|_1 \|g\|_\infty$$

   First, we will limit our attention of $\int |fg|$ to some $E \in \mathcal{M}$ such that $\mu(X \setminus E) = \mu(E^c) = 0$ (i.e. ignoring a zero-measure set). Then:

$$\int |fg| = \int_E |fg| + \int_{E^c} |fg| = \int_E |fg|$$

   Next, we take advantage the fact that we are working with an infimum to see that:

$$\|f\|_\infty \leq \sup_{x \in A} |f(x)| < \|f\|_\infty + \epsilon$$

[2] see definition **??**

Therefore, we can take:

$$\int_E |fg| \leq \int_E \left| f \cdot \sup_{x\in E} |g(x)| \right| = \int_E |f| \cdot \left| \sup_{x\in E} |g(x)| \right| = \left| \sup_{x\in E} |g(x)| \right| \int_E |f|$$

where the last equality comes from the fact that the supremum is a constant. Finally, by the property of the infimum, we have:

$$\left| \sup_{x\in E} |g(x)| \right| \int_E |f| \leq (\|g\|_\infty + \epsilon) \int |f|$$

Notice that this inequality does not depend on ϵ, and so:

$$\|fg\|_1 = \int_E |fg| \leq \|f\|_1 \|g\|_\infty$$

as we sought to show

Notice how similar $\|\cdot\|_\infty$ is to the uniform metric $\|f\|_u = \sup_{x\in X} f(x)$. If we are dealing with Borel measures that assign a positive value to all open sets, then if f is continuous we in fact have $\|f\|_\infty = \|f\|_u$ (since $\{x \mid |f(x)| > a\}$ is open). In this way, we can consider the space of bounded continuous functions as a (closed!) subspace of L^∞.

We conclude with some final remarks about the density of simple functions in L^∞. Notice that for L^p where $p < \infty$, simple functions where $\mu(E_i) < \infty$ where dense. This condition is no longer present for density in L^∞; we in fact require simple functions where $\mu(E) = \infty$ now since the function $1(x) = 1$ is in $L^\infty(\mathbb{R})$. This also means that $C_0(\mathbb{R})$ cannot be dense in L^∞. In the case of $L^\infty([a, b])$ $C([a, b])$, not all funcions can be converged to uniformally dense: consider $f : [0, 1] \to \mathbb{R}$, $f(x) = 0$ for $x \in [0, 1/2]$ and $f(x) = 1$ for $x \in [1/2, 1]$. The no continuous function uniformally converges to f in the ∞-norm.

3.1.1 Relating L^p spaces: Interpolation

In this section, we will show how we can relate L^p spaces with each other. Fundamentally, the way we get any order with L^p spaces is through Hölder's inequality. We take full advantage of it in the following proofs. The end-goal is to show that if $p < q < r$ where we allow $r = \infty$, then:

$$L^p \cap L^r \subseteq L^q \subseteq L^p + L^r$$

This should reflect on the example that we have demonstrated in example 3.1.2. In particular, the larger p is, the less f has decay at infinity, but the more f must not explode at any point. In other words p gets bigger, we can let p decay slower, but it must behave better at points where it can explod3. If our space X is bounded, then we don't have to care about decay (this will give us that $L^p \supseteq L^q$ where $p < q$). If X is discrete, we don't have to care about exploding at any point (this will give us that $\ell^p \subseteq \ell^q$ if $p < q$). We thus explore these relation's in this section. These types of relations bewteen space via addition of functions or intersection of spaces is called *interpolation*.

Proposition 3.1.12: L^p **subset**

Let $0 < p < q < r \leq \infty$. Then

$$L^q \subseteq L^p + L^r$$

In particular, each $f \in L^q$ is the sum of a function in L^p and a function in L^q

***Proof*:**
This proof is really similar to that of example 3.1.2[1]. Let $f \in L^q$. Then consider:

$$|f| = \chi_{|f|\geq 1}|f| + \chi_{|f|<1}|f|$$

that is, we're splitting $|f|$ to where f is greater than or equal to 1 and strictly less than 1. The equality holds since $A = \{x \mid |f(x)| \geq 1\}$ and $B = \{x \mid |f(x)| < 1\}$ are disjoint. Importantly, notice that:

$$|f|^p = \chi_{|f|\geq 1}|f|^p + \chi_{|f|<1}|f|^p \qquad |f|^r = \chi_{|f|\geq 1}|f|^r + \chi_{|f|<1}|f|^r$$

Thus, by the linearity of the integral, we get that:

$$\int |f|^p = \int \chi_{|f|\geq 1}|f|^p + \int \chi_{|f|<1}|f|^p \qquad \int |f|^r = \int \chi_{|f|\geq 1}|f|^r + \int \chi_{|f|<1}|f|^r$$

Then we get:

$$\chi_{|f|\geq 1}|f|^p \leq \chi_{|f|\geq 1}|f|^q < \infty \qquad \chi_{|f|<1}|f|^r \leq \chi_{|f|<1}|f|^q < \infty$$

Therefore, $\chi_{|f|\geq 1}f \in L^p$ and $\chi_{|f|<1}f \in L^r$. If $r = \infty$, then clearly $\|\chi_{|f|\leq 1}f\|_\infty \leq 1$, showing the result still holds, as we sought to show.

Proposition 3.1.13: L^p **superset**

Let $0 < p < q < r \leq \infty$. Then

$$L^p \cap L^r \subseteq L^q$$

In particular, if if we find that f is bounded in growth from above and bellow by p and r, $\|f\|_p < \infty$ and $\|f\|_r < \infty$, then f is bounded in growth by q.

***Proof*:**
Let $\|f\|_r < \infty$ and $\|f\|_q < \infty$. We need to show that $\|f\|_p < \infty$. Notice that it is equivalent to show that $\|f\|_r^n < \infty$ and $\|f\|_q^m < \infty$ then $\|f\|_p^\ell < \infty$ for $n, m, \ell \in \mathbb{R}_{\geq 0}$ (since if it's finite, then all powers will be finite). We'll prove it in the case where $r = \infty$ and $r < \infty$. In the infinite case, we simply take advantage of bounding properties of the essential supremum, and in the other case we use Hölder's inequality.

Starting with the $r = \infty$ case, notice that

$$|f|^q = |f|^{q-p+p} = |f|^q|f|^{q-p} \leq \|f\|_\infty^{q-p}|f|^p$$

and so, taking the integral and the qth root, we get:

$$\begin{aligned}
\|f\|_q &= \left(\int |f|^q\right)^{1/q} \\
&\leq \left(\int \|f\|_\infty^{q-p}|f|^p\right)^{1/q} \\
&= \|f\|_\infty^{\frac{q-p}{q}} \left(\int |f|^p\right)^{1/q} \\
&= \|f\|_\infty^{\frac{q-p}{q}} \left(\int |f|^p\right)^{p/pq} \\
&= \|f\|_\infty^{1-\frac{p}{q}} \|f\|_p^{p/q} < \infty
\end{aligned}$$

The fact that the last inequality is less than infinity comes from the that $\|f\|_p < \infty$ and $\|f\|_\infty < \infty$, so any power is less than infinity, and so $\|f\|_q < \infty$. Furthermore, notice that the sum of the power's is 1; this fact will come back for another intuition of $\|f\|_\infty$.

For the $r < \infty$ case the key is to find a way to integrate p and r using conjugate exponents. Since

$$p < q < r$$

then

$$\frac{1}{p} > \frac{1}{q} > \frac{1}{r}$$

therefore, define the continuous function $\lambda \mapsto \frac{1-\lambda}{p} + \frac{\lambda}{r}$. Since $(0, 1)$ is open, and $1/p > 1/r$, there exists a λ_0 such that

$$\frac{1-\lambda_0}{p} + \frac{\lambda_0}{r} = \frac{1}{q}$$

manipulating the equation around to get conjugate exponents, notice that:

$$\frac{(1-\lambda_0)q}{p} + \frac{\lambda_0 q}{r} = 1 \quad \Leftrightarrow \quad \frac{1}{\frac{p}{(1-\lambda_0)q}} + \frac{1}{\frac{r}{\lambda_0 q}} = 1$$

Thus, we have that $\frac{p}{(1-\lambda_0)q}$ and $\frac{r}{\lambda_0 p}$. With conjugate exponents established, we proceed with trying to bound $\|f\|_q$. As we commented, it is equivalent to bound $\|f\|_q^m$, so let $m = q$ (this will be a common trick to apply Hölder's inequality). Also, since $1 - \lambda_0 + \lambda_0 = 1$, we have that

$q = \lambda_0 q + (1-\lambda_0)q$, giving us the ability to split up our integral:

$$\begin{aligned}
\|f\|_q^q &= \int |f|^q \\
&= \int |f|^{\lambda_0 q + (1-\lambda_0)q} \\
&= \int |f|^{\lambda_0 q}|f|^{(1-\lambda_0)q} \\
&= \left\| |f|^{\lambda_0 q}|f|^{(1-\lambda_0)q} \right\|_1 \\
&\le \left\| |f|^{\lambda_0 q} \right\|_{\frac{p}{(1-\lambda_0)q}} \left\| |f|^{(1-\lambda_0)q} \right\|_{\frac{r}{\lambda_0 q}} && \text{Hölder's inequality} \\
&= \left(\int |f|^p\right)^{\frac{(1-\lambda_0)q}{p}} \left(\int |f|^r\right)^{\frac{\lambda_0 q}{r}} \\
&= \|f\|_p^{(1-\lambda_0)q} \|f\|_r^{\lambda_0 q} < \infty
\end{aligned}$$

Since the last two terms are finite, any power of them are also finite, and so we have bounded $\|f\|_q^q < \infty$, and so $\|f\|_q < \infty$, as we sought to show.

Proposition 3.1.14: ℓ^p Covariant Inclusion

If $0 < p < q \le \infty$, then $\ell^p(A) \subseteq \ell^q(A)$

***Proof*:**
They key part is to take advantage of the properties of the counting measure. Since the only set of measure 0 in the counting measure is the empty-set, we have that the essential supremum is actually the supremum!

Let's say $f \in \ell^p$, so that $\sum_{a\in A} |f(a)|^p = \|f\|_p < \infty$. We'll start by showing that $\|f\|_\infty < \infty$. Since the $\|f\|_\infty$ is in fact the supremum, we have that:

$$\|f\|_\infty^p = \sup_{a\in A} |f(a)|^p \le \sum_{a\in A} |f(a)|^p = \|f\|_p < \infty$$

Therefore, $\|f\|_\infty < \infty$.

For the $q < \infty$ case, using proposition 3.1.13 with $\lambda = p/q$, we get:

$$\|f\|_q \le \|f\|_p^\lambda \|f\|_\infty^{1-\lambda} \le \|f\|_p$$

therefore, $\|f\|_q < \infty$, completing the proof.

Proposition 3.1.15: L^p Contravariant Inclusion in finite measure

Let $(X, \mathcal{M}, \mu)$ be a measure space where $\mu(X) < \infty$. If $0 < p < q \le \infty$, then $L^p \supseteq L^q$

***Proof*:**
We need to show that if $f \in L^q$ then $f \in L^p$. Since $f \in L^q$, $\|f\|_q < \infty$, so $\|f\|_q^n < \infty$ for all $n \in \mathbb{N}$. If we can show that $\|f\|_p^m \leq \|f\|_q^n$ for some $m, n \in \mathbb{N}$, then the proof is complete.

We'll apply Hölder's inequality in the caes of $q = \infty$ and $q < \infty$. If $q = \infty$, then:

$$\|f\|_p^p = \int |f|^p = \int |f|^p \cdot 1 \overset{\text{H.}}{\leq} \|f\|_\infty^p \int 1 = \|f\|_\infty^p \mu(X) < \infty$$

since $\int 1 = \mu(X)$. If $q < \infty$, then we do a similar argument. First, since $p < q$, notice that:

$$\frac{1}{\frac{q}{p}} + \frac{1}{\frac{q}{q-p}} = \frac{p}{q} + \frac{q-p}{q} = \frac{q}{q} = 1$$

and so $\frac{q}{p}$ and $\frac{q}{p-q}$ are conjugate exponents. Therefore, by Hölder's inequality:

$$\begin{aligned}
\|f\|_p^p &= \int |f|^p \\
&= \int |f|^p \cdot 1 \\
&= \||f|^p \cdot 1\|_1 \\
&\leq \||f|^p\|_{\frac{q}{p}} \|1\|_{\frac{q}{p-q}} && \text{Hölder's Inequality} \\
&= \left(\int |f|^{p\frac{q}{p}}\right)^{\frac{p}{q}} \left(\int 1\right)^{\frac{q-p}{p}} \\
&= \|f\|_q^p \, \mu(X)^{\frac{q-p}{p}} < \infty
\end{aligned}$$

therefore, $\|f\|_p^p < \infty$, so $\|f\|_p < \infty$, showing that $f \in L^p$, as we sought to show.

Proposition 3.1.16: Relating L^p and L^∞

Let $p \in [1, \infty)$ and assume $\|f\|_p < \infty$. Show that:

$$\lim_{p \to \infty} \|f\|_p = \|f\|_\infty$$

***Proof*:**
For ease of typing, we simplify notation: $\|f\|_p := \|f\|_{L^p(\mu)}$

We'll show that $\lim_{p\to\infty} \|f\|_p \leq \|f\|_\infty$ and $\lim_{p\to\infty} \|f\|_p \geq \|f\|_\infty$. For the $\leq$ direction, note that $|f| \leq \|f\|_\infty$ a.e., and since x^n is a monotone function on $[0, \infty)$, we have $|f|^n \leq \|f\|_\infty^n$. Then for some fixed λ_0:

$$\begin{aligned}\|f\|_p &= \left(\int |f|^p\right)^{1/p} \\ &= \left(\int |f|^{p+\lambda_0-\lambda_0}\right)^{1/p} \\ &= \left(\int |f|^{p-\lambda_0}|f|^{\lambda_0}\right)^{1/p} \\ &\leq \|f\|_\infty^{\frac{p-\lambda_0}{p}} \|f\|_{\lambda_0}^{\frac{\lambda_0}{p}}\end{aligned}$$

where the last inequality is the same trick as part (a) of this question. Since λ_0 is fixed and this inequality holds true for all p, as $p \to \infty$:

$$\lim_{p\to\infty} \|f\|_p \leq \lim_{p\to\infty} \|f\|_\infty^{\frac{p-\lambda_0}{p}} \|f\|_{\lambda_0}^{\frac{\lambda_0}{p}} = \|f\|_\infty$$

For the $\geq$ direction, we have to bound $\|f\|_p$ from bellow by something that includes $\|f\|_\infty$, but ultimately the inequality will be independent of the choices we make. In particular, let $\epsilon > 0$. Take all points which are epsilon close to the essential supremum (which exists by the infimum property). Let M_ϵ be our set, in particular $M_\epsilon = \{x \mid |f(x)| \geq \|f\|_\infty - \epsilon\}$. Since $\|f\|_\infty = k$, then $\mu(M_\epsilon) > 0$. Thus:

$$\|f\|_p = \left(\int |f|^p\right)^{1/p} \geq \left(\int_{M_\epsilon} (\|f\|_\infty - \epsilon)^p\right)^{1/p} \stackrel{!}{=} (\|f\|_\infty - \epsilon)\mu(M_\epsilon)^{1/p}$$

where $\stackrel{!}{=}$ comes from the fact that we are integrating over a constant region. Since $\mu(M_\epsilon) > 0$, we have that $\mu(M_\epsilon)^{1/p} > 0$ for all p, and $\mu(M_\epsilon)^{1/p} \xrightarrow{p\to\infty} 1$. Furthermore, notice our inequalities are independent of choice of ϵ, and so since we can let ϵ be as small as we want:

$$\lim_{p\to\infty} \|f\|_p \geq \|f\|_\infty$$

and since we've shown the $\leq$ direction, we have:

$$\lim_{p\to\infty} \|f\|_p = \|f\|_\infty$$

as we sought to show

Now that we have examined L^p spaces in more detail, we finish off this seciton with some comments on the usefulness these spaces:

1. L^1 is natural for integration; we will essentially work with variation's of L^1 when we integrate. From an algebraic perspective, it can be thought of as a generalizatgion of the notion of "counting" all points in a space, and working in spaces where the count is finite.
2. L^∞ is useful as a broadening of the space of uniform-bound functions. It is a generlization of the idea of working with functions that are bounded.

3. L^p interpolates in between L^1 and L^∞. Many times, L^1 or L^∞ spaces have pathological behaviors (examples of which I yet do not know). L^p spaces behave more nicely, and can sometimes be used to go from L^1 to L^∞ step by step using the inequalities we constructed (ex. we got a function is L^1, we want to show it's L^∞, and we use the proceeding inequalities to build our way up)

4. L^2 is particularly nice since it is a Hilbert space, and so many euclidean intuitions work on L^2 and help us construct many nice properties. A lot of Fourier analysis happens on L^2, though sometimes we do use L^p.

   Another way of thinking of L^2 is that it is a generlization of the notion of a space that *can* have an inner product (this will be made rigorous with a particular universal property, see theorem 5.1.24). In ℓ^2, we would simply get a natural generalization of an inner product on a vector-space, while in L^2, we get the continuous equivalent.

3.2 The dual of Lebesgue Space

By Hölder's inequality, we have that if $f \in L^p$, then for all $g \in L^q$, $fg \in L^1$. We can thus define a function:

$$\varphi_g : L^p \to \mathbb{R} \qquad \varphi_g(f) = \int fg$$

and the operator norm of φ_g is always at most $\|g\|_q$[3]. With the opeartor norm, the mapping $g \mapsto \varphi_g$ will almost always be an isometry:

Proposition 3.2.1: Operator Norm And Isometry

Let p and q be conjugate exponents, $1 \leq q < \infty$. If $g \in l^q$, then:

$$\|g\|_q = \|\varphi_g\| = \left\{ \left| \int fg \right| \;\middle|\; \|f\|_p = 1 \right\}$$

if μ is semifinite, this results alos holds for $q = \infty$

***Proof*:**
Folland p. 188

Conversely, if $f \mapsto \int fg$ is a bounded linaer functional on L^p, then $q \in L^q$ in almost all cases:

[3] if $p = 2$, it is usually defined as mapping to $\int f\overline{g}$. This can also be done to $p \neq 2$, there it is just easier to work by conjugating for complex numbers

Theorem 3.2.2: L^p Linear Functional Implies G In L^q

Let p and q be conjugate exponents. Suppose that g is a measurable function on X such that $fg \in l^1$ for all f inteh space Σ of simple functions that vanish outside a set of finite measure, and the quanitty

$$M_q(g) = \{|fg| \mid f \in \Sigma, \ \|f\|_p = 1\}$$

is finite. Also, suppose either that $S_g = \{x \mid g(x) \neq 0\}$ is σ-finite or tha tμ is semi-finite. Then $g \in L^q$ and $M_q(g) = \|g\|_q$.

***Proof*:**
Folland p. 189

Finally, perhaps the most stricking thing is that the maps $g \mapsto \int fg$ are almost always a surjection of $(L^p)^*$!

Theorem 3.2.3: Description Of Dual Of Lp

Let p and q be conjugate exponents. If $1 < p < \infty$, for each $\varphi \in (L^p)^*$ there exits $g \in L^q$ such that $\varphi(f) = \int fg$ for all $f \in L^p$, and hence L^q is isometrically isomorphic to $(L^p)^*$. The same conclusion holds for $p = 1$ provided μ is $1s$-finite

***Proof*:**
Folland p. 190

Corollary 3.2.4: Lp Is Reflective

If $1 < p < \infty$, L^p is reflective, that is $(L^p)^{**} \cong L^p$

(Some comment in the case of $p = 1$ and $p = \infty$)

3.3 Some Useful Inequalities

In this section, we explore more inequalties, since they are at the heart of working with L^p spaces. We first present a common inequality that shows up often in statistics:

Theorem 3.3.1: Chebychev's Inequality

Let $f \in L^p$ $(0 < p < \infty)$. Then for any $\alpha > 0$

$$\mu(\{x \mid |f(x)| > \alpha\}) \leq \left(\frac{\|f\|_p}{\alpha}\right)^p$$

***Proof*:**
Let $E_\alpha = m(\{x \mid |f(x)| > \alpha\})$. Then:

$$\|f\|_p^p = \int_\Omega |f|^p \geq \int_{E_\alpha} |f|^p \geq \int_{E_\alpha} \alpha^p = \alpha^p m(E_\alpha)$$

re-arranging gives us:

$$\mu(\{x \mid |f(x)| > \alpha\}) \leq \left(\frac{\|f\|_p}{\alpha}\right)^p$$

as we sought to show

The next inequality gives us a general statement on being abel to get a function from $L^p(\mu)$ "transformed" into a function of $L^p(\nu)$:

Theorem 3.3.2: Transitioning L^p Spaces

Let$(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite measures spaces, K be a $M \otimes N$-measurable function on $X \times Y$, and let's say there exists a $C > 0$ such that

$$\int |K(x,y)| d\mu(x) \leq C \qquad \int |K(x,y)| d\nu(y) \leq C$$

for a.e. x and y. Then if $f \in L^p(\nu)$ $(1 \leq p \leq \infty)$, the integral:

$$Tf(x) = \int K(x,y) f(y) d\nu(y)$$

converges absolutely for a.e. $x \in X$, $Tf \in L^p(\mu)$, and $\|Tf\|_p \leq C\|f\|_p$

***Proof*:**
Let $p \in [1, \infty]$, and let q be it's conjugate exponent. Then:

$$|K(x,y)f(y)| = |K(x,y)||f(y)| = |K(x,y)|^{\frac{1}{p}+\frac{1}{q}}|f(y)| = (|K(x,y)|^{1/p})(|K(x,y)|^{1/q} f(y)|)$$

and applying Hölder to the functions in parenthesis, we get:

$$\begin{aligned}\int |K(x,y)f(y)| d\nu(y) &\leq \left(\int |K(x,y)| d\nu(y)\right)^{1/q} \left(\int |K(x,y)||f(y)|^p d\nu(y)\right)^{1/p} \\ &\leq C^{1/q} \left(\int |K(x,y)| f(y)|^p d\nu(y)\right)^{1/p}\end{aligned}$$

for a.e. $x \in X$. Hence, taking the integral over $X \times Y$, using Tonelli's theorem, and substituting the result we just found, we get:

$$\begin{aligned}\int\left[\int |K(x,y)f(y)d\nu(y)\right]^p d\mu(x) &\leq \int\int C^{p/q}|K(x,y)||f(y)|^p d\nu(y)d\mu(x)\\ &= C^{p/q}\int\int |K(x,y)||f(y)|^p d\mu(x)d\nu(y) \qquad \text{Tonelli}\\ &\leq C^{p/q}\int |f(y)|^p \int K(x,u)d\mu(x)d\nu(y)\\ &= C^{p/q+1}\int |f(y)|^p d\nu(y)\end{aligned}$$

Since $f \in L^p(\nu)$, the last integral is finite. Since this is true for a.e. x, $K(x,\cdot)f \in L^1(\nu)$ for a.e. x. Thus, Tf is well-defined a.e., meaning we can integrate it, and by the inequality we just showed:

$$\int |Tf(x)|^p d\mu(x) = \|Tf\|_p^p \leq C^{p/q+1}\|f\|_p^p$$

taking the pth root of each side get's us the desired inequality

The next result we work towards is a generalization of Minkowski: instead of summing we will be taking integrals. You can also interpret part (a) as you can move the $(|\cdot|^p)^{1/p}$ "down" a layer of integrals, and part (b) as the generalization of monotonicity ($|\int f| \leq \int |f|$).

Theorem 3.3.3: Minkowski's Inequality For Integrals

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite meaure spaces and let f be a [real?] $M \otimes N$-measurable function on $X \times Y$. Then:

1. if $f \geq 0$ and $1 \leq p \leq \infty$, then:

$$\left[\int\left(\int f(x,y)d\nu(y)\right)^p d\nu(x)\right]^{1/p} \leq \int\left[\int f(x,y)^p d\mu(x)\right]^{1/p} d\nu(y)$$

2. If $1 \leq p \leq \infty$, $f(\cdot,y) \in L^p(\mu)$ for a.e. y and the function $y \mapsto \|f(\cdot,y)\|_p$ is in $L^1(\nu)$, then $f(x,\cdot) \in L^1(\nu)$ for a.e. x, the function $x \mapsto \int f(x,y)d\nu(y)$ is in $L^p(\mu)$ and

$$\left\|\int f(\cdot,y)d\nu(y)\right\|_p \leq \int \|f(\cdot,y)\|_p \, d\nu(y)$$

***Proof*:**
If $p = 1$, then this becomes Tonelli's Theorem, so consider $1 < p < \infty$ (the case for $p = \infty$ comes later). Let q be it's conjugate exponent so that $1/p + 1/q = 1$. Suppose $\mu, \nu < \infty$ and $\|f\|_\infty < \infty$ (i.e. it's essential supremum is finite). For notational simplicity, define $H(x) = \int_Y f(x,y)d\nu(y)$.

Since the measure is finite and f is esentially bounded, we have that $\|H\|_p < \infty$. Then:

$$\begin{aligned}
\|H\|_p^p &= \int_X \left(\int_Y f(x,y)d\nu(y)\right)^p d\nu(x) \\
&= \int_X \left(\int_Y f(x,y)d\nu(y)\right)(H(x))^{p-1}d\mu(x) && x \text{ indep. of } y \\
&= \int_X \int_Y f(x,y)H(x)^{p-1}d\mu(x)d\nu(y) && \text{Fubini} \\
&\overset{!}{\leq} \int_X \|f(x,\cdot)\|_p \|H\|_p^{p-1} d\nu(y) && \text{Hölder, } q = p/(p-1) \\
&\leq \|H\|_p^{p-1} \int_X \|f(x,\cdot)\|_p d\nu(y)
\end{aligned}$$

where $\overset{!}{\leq}$ I'm not sure how Hölder's inequality was used. Finally, dividng by $\|H(x)\|_p^{p-1}$ get's us:

$$\|H\|_p \leq \int_X \|f(x,\cdot)\|_p d\nu(y)$$

For the general case, Yakov left it as an exercise to do $X = \bigcup_n X_n$, $Y = \bigcup Y_n$, $E_n = \{x \mid f(x) < n\}$, and

$$\tilde{X}_n = X_n \cap E_n \qquad \tilde{Y}_n = Y_n \cap E_n$$

and apply the result to each of these, and the monotone convergence theorem at the end (feels similar to how it was done for Fubini Tonnelli).

For $p = \infty$, This comes down to monotonicity of integrals:

$$\left\|\int f(x,y)d\nu(y)\right\|_\infty \leq \left|\int f(x,y)d\nu(y)\right| \leq \int |f(x,y)d\nu(y) = \int \|f(x,y)\|_\infty d\nu(y)$$

where the last inequality comes from the fact that the integral ignores zero-measure sets.

There are a few more results in the book, but Yakov decided to prove the generalized Hölder's inequality instead for now:

Theorem 3.3.4: Generalized Hölder's Inequality

Let $1 \leq p_i \leq \infty$ for $1 \leq i \leq n$ and $\sum_i^n p_i^{-1} = r^{-1} \leq 1$. If $f_i \in L^{p_i}$, then $\prod_i^n f_i \in L^r$ and

$$\left\|\prod_i^n f_i\right\|_r \leq \prod_i^n \|f_i\|_{p_i}$$

***Proof*:**

If $r = \infty$, then $r^{-1} = 0$, so all the $p_i^{-1} = 0$, so all the $p_i = \infty$, in which case we simply take:

$$\left|\prod_i^n f_i\right| \leq \prod_i^n |f_i|$$

For $r < \infty$, just like for Hölder's inequality, it suffice to assume $r = 1$, since we can do:

$$\sum \frac{1}{\frac{p_i}{r}} = 1$$

In that case, notice that:

$$\begin{aligned} \left\| \prod_i^n f_i \right\|_r^r &= \int |f_1 f_2 \cdots f_n|^r \\ &= \int f_1^r f_2^r \cdots f_n^r \\ &\leq \|f_1\|_{p_1}^r \|f_2\|_{p_2}^r \cdots \|f_n\|_{p_n}^r \end{aligned}$$

Notice that each f_i is raised to the power of p_i/r. (TBD)

Now, we do induction on i. Let the induction hypothesis be:

$$\left(\sum_i^{n-1} p_i^{-1} \right) + p_n^{-1} = r^{-1} + p_n^{-1} = 1$$

Then by Hölder's inequality:

$$\int f_1 \cdots f_n| \leq \|f_1 \cdots f_{n-1}\|_r \|f\|_{p_n} \overset{\text{I.H.}}{\leq} \|f_1\|_{p_1} \cdots \|f_n\|_{p_n}$$

giving our desired result

The final property we'll present a way to represent functions in terms of the integral:

Proposition 3.3.5: Layer Cake Representation

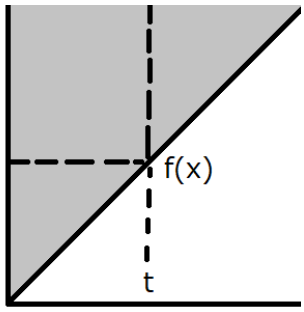
Let $\varphi(t) : [0, \infty] \to [0, \infty]$ be an increasing function which is C^1 on $(0, \infty)$, satisfies $\varphi(0) = 0$ and satisfies $\lim_{t\to\infty} \varphi(t) = \varphi(\infty)$. Let (X, M, μ) be a σ-finite measure space and $f : X \to [0, \infty]$ be a non-negative measurable function. Then:

$$\int_X \varphi(f(x))\, d\mu(x) = \int_0^\infty \mu(\{x \mid f(x) > t\})\varphi'(t)dt$$

One consequence of this equation is that with $\varphi(t) = t^p$ for $1 < p < \infty$ is the following formula for the L^p norm of f:

$$\|f\|_p = \left(p \int_0^\infty \mu(\{x \mid f(x) > t)\}\, t^{p-1} \right)^{1/p} dt$$

***Proof*:**
The proof can be captured pictorially as integrating over the following area:



Let $S \subseteq X \times [0,1]$, $S = \{(x,t) \mid f(x) \geq t\}$. With this set, define $F : X \times [0,\infty] \to [0,\infty]$, $F(x,t) = \varphi'(t)\chi_S$. That is, we are going to integrate the "upper triangle". Since φ is an increasing C^1 function and $\varphi'(t) > 0$, we have $F \in L^+(X,[0,\infty])$. Therefore, we can apply Tonneli's Theorem and see that it's equivalent to integrate over the iterated integrals to compute integrals.

To compute it, we need to be clever with the bounds we choose which depend on which iterated integral we choose. The different direction of the iterated integral will ultimately lead to the desired result. If we integrate with respect to $d\mu(x)dt$, then we take slices across t: $(\varphi'(t)\chi_S)_t = \varphi'(t)\chi_{S_t}$, where $(\chi_S)_t = \chi_S(x,t) = \chi_{S_t}$, with $S_t = \{x \mid f(x) \geq t\}$. And so we get:

$$\int_0^\infty \int_X \chi_{S_t}\varphi'(t) = \int_0^\infty \int_{\{x \mid f(x) \geq t\}} \varphi'(t)d\mu(x)dt = \int_0^\infty \mu(\{x \mid f(x) \geq t\})\varphi'(t)dt$$

Now, integrating with respect to $dtd\mu(x)$, we take x slices: $(\varphi'(t)\chi_S)_x = \varphi'(t)\chi_{S_x}$ where $(\chi_S)_x = \chi_S(x,t) = \chi_{S_x}$, with $S_x = \{x \mid 0 \leq f(x)\}$. And so we get:

$$\int_X \int_0^\infty \chi_{S_x}\varphi'(t)dtd\mu = \int_X \int_0^{f(x)} \varphi'(t)dtd\mu(x)$$

and since $[0,f(x)]$ is bounded and $\varphi(0) = 0$, the Lebesgue integral matches the Riemann integral, and so we get that:

$$\int_X \varphi(f(x))d\mu(x)$$

and by Tonnelli's theorem:

$$\int_X \varphi(f(x))d\mu(x) = \int_0^\infty \mu(\{x \mid f(x) \geq t\})\varphi'(t)dt$$

As we sought to show. Using this, if we let $\varphi(t) = t^p$, then we see that φ is C^1 on $(0,\infty)$ and $\varphi(0) = 0$ for $1 \leq p < \infty$. Thus, the value of $\|f\|_p$ (where we take f^p instead of $|f|^p$ since $f \geq 0$) is:

$$\begin{aligned} \|f\|_{L^p} = \left(\int |f|^p\right)^{1/p} = \left(\int f^p\right)^{1/p} &= \left(\int_X \varphi(f(x))d\mu(x)\right)^{1/p} \\ &= \left(p\int_0^\infty \mu(\{x \mid f(x) > t)\}\, t^{p-1}\right)^{1/p} dt \end{aligned}$$

Another use of this formula is by taking $\varphi(t) = t$. Then we will get that:

$$\int f(x)d\mu(x) = \int_0^\infty \mu(\{x \mid f(x) \geq t\})dt$$

This can be useful if we have information of what is happening to the growth of f as f get's larger, in particular if we know that somehow, the amount of points such that $\mu(\{x \mid f(x) \geq t\})$ is shrinking rapidly enough say that $\mu(\{x \mid f(x) \geq t\}) \leq \frac{C}{t^r}$ for some constant C).

3.4 Interpolation: The Riesz-Thorin Theorem

As we saw in the previous section, if a function is in L^p and L^r, it is also in L^q for all $p < q < r$. This is a type of "interpolation" of spaces. A natural question arises: if we have a linear operator T that is bounded on L^p and bounded on L^r, is it also bounded on L^q? The answer is yes, and the Riesz-Thorin theorem provides the precise quantitative relationship between the operator norms.

The proof relies on a powerful complex analysis technique known as the Phragmén-Lindelöf principle, specifically a version called the Three Lines Lemma.

Lemma 3.4.1: Three Lines Lemma

Let φ be a bounded continuous function on the strip $S = \{z \in \mathbb{C} \mid 0 \leq \text{Re}(z) \leq 1\}$ that is holomorphic on the interior of S. If $|\varphi(z)| \leq M_0$ for $\text{Re}(z) = 0$ and $|\varphi(z)| \leq M_1$ for $\text{Re}(z) = 1$, then for all $t \in [0, 1]$ and z with $\text{Re}(z) = t$:

$$|\varphi(z)| \leq M_0^{1-t} M_1^t$$

***Proof*:**

For $\epsilon > 0$, define $\varphi_\epsilon(z) = \varphi(z) M_0^{z-1} M_1^{-z} e^{\epsilon(z^2-1)}$. The term $e^{\epsilon(z^2-1)}$ helps handle the boundedness at infinity within the strip. By the maximum modulus principle applied to rectangles in the strip as the height goes to infinity, one can show $|\varphi_\epsilon(z)| \leq 1$ in the strip. Letting $\epsilon \to 0$ yields the result.

Theorem 3.4.2: Riesz-Thorin Interpolation Theorem

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be measure spaces. Let $p_0, p_1, q_0, q_1 \in [1, \infty]$. Define p_t and q_t for $t \in (0, 1)$ by:

$$\frac{1}{p_t} = \frac{1-t}{p_0} + \frac{t}{p_1}, \qquad \frac{1}{q_t} = \frac{1-t}{q_0} + \frac{t}{q_1}$$

Suppose T is a linear operator defined on the set of simple functions Σ such that for all $f \in \Sigma$:

$$\|Tf\|_{q_0} \leq M_0 \|f\|_{p_0} \quad \text{and} \quad \|Tf\|_{q_1} \leq M_1 \|f\|_{p_1}$$

Then for all $t \in (0, 1)$, T extends to a bounded operator from $L^{p_t}(\mu)$ to $L^{q_t}(\nu)$ satisfying:

$$\|Tf\|_{q_t} \leq M_0^{1-t} M_1^t \|f\|_{p_t}$$

***Proof*:**
Let $t \in (0,1)$ be fixed. By duality (Proposition 3.2.1), it suffices to estimate $|\int(Tf)g\,d\nu|$ where f and g are simple functions with $\|f\|_{p_t} = 1$ and $\|g\|_{q'_t} = 1$ (where q'_t is the conjugate exponent of q_t).

We construct a family of functions to apply the Three Lines Lemma. Let $f = |f|e^{i\theta}$ and $g = |g|e^{i\psi}$ be the polar decompositions. For any complex number z in the strip $0 \leq \mathrm{Re}(z) \leq 1$, define:

$$\alpha(z) = (1-z)\frac{1}{p_0} + z\frac{1}{p_1}, \qquad \beta(z) = (1-z)\frac{1}{q'_0} + z\frac{1}{q'_1}$$

Note that for $z = t$, $\alpha(t) = 1/p_t$ and $\beta(t) = 1/q'_t$. Now define deformations of f and g:

$$f_z = |f|^{p_t\alpha(z)}e^{i\theta}, \qquad g_z = |g|^{q'_t\beta(z)}e^{i\psi}$$

(If $f = 0$ or $g = 0$, set the function to 0). Note that $f_t = f$ and $g_t = g$.

Define the function $\Phi(z)$ on the strip $0 \leq \mathrm{Re}(z) \leq 1$:

$$\Phi(z) = \int_Y (Tf_z)g_z\,d\nu$$

Since f and g are simple, f_z is a finite linear combination of terms like $c_k\alpha^z$, making $\Phi(z)$ holomorphic and bounded.

We check the boundary conditions. For $\mathrm{Re}(z) = 0$, $f_z \in L^{p_0}$ and $g_z \in L^{q'_0}$. Specifically:

$$\|f_{iy}\|_{p_0}^{p_0} = \int ||f|^{p_t(\frac{1}{p_0}+i\mathrm{Im}(\alpha))}|^{p_0} = \int |f|^{p_t} = \|f\|_{p_t}^{p_t} = 1$$

Similarly $\|g_{iy}\|_{q'_0} = 1$. Thus by Hölder's inequality and the hypothesis on T:

$$|\Phi(iy)| = \left|\int T(f_{iy})g_{iy}\right| \leq \|Tf_{iy}\|_{q_0}\|g_{iy}\|_{q'_0} \leq M_0\|f_{iy}\|_{p_0} \cdot 1 = M_0$$

Similarly, for $\mathrm{Re}(z) = 1$, we find $|\Phi(1+iy)| \leq M_1$.

By the Three Lines Lemma, $|\Phi(t)| \leq M_0^{1-t}M_1^t$. Since $\Phi(t) = \int(Tf)g$, and this holds for all unit simple functions f, g, we conclude:

$$\|T\|_{L^{p_t}\to L^{q_t}} \leq M_0^{1-t}M_1^t$$

Counter-Example: The Linear Transport Equation

Consider the simple advection equation in one dimension:

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$

where $c \neq 0$ is a real constant velocity.

1. **The Operator:** The solution is a translation of the initial state $f(x)$:

$$u(x,t) = f(x - ct)$$

Therefore, the time-shift operator S_t acts as:

$$(S_t f)(x) = f(x - ct)$$

2. **The Adjoint:** To check if S_t is self-adjoint, we compute its adjoint S_t^* using the standard L^2 inner product $\langle f, g \rangle = \int_{-\infty}^{\infty} f(x)\overline{g(x)}\,dx$.

$$\langle S_t f, g \rangle = \int_{-\infty}^{\infty} f(x - ct)\overline{g(x)}\,dx$$

Perform a change of variables $y = x - ct$, so $x = y + ct$ and $dx = dy$:

$$\langle S_t f, g \rangle = \int_{-\infty}^{\infty} f(y)\overline{g(y + ct)}\,dy$$

$$\langle S_t f, g \rangle = \langle f, S_{-t} g \rangle$$

This shows that the adjoint is the inverse shift:

$$S_t^* = S_{-t}$$

3. **Conclusion:** For S_t to be self-adjoint, we require $S_t = S_t^*$. This would imply $S_t = S_{-t}$, or $f(x - ct) = f(x + ct)$ for all functions f. This is clearly false for $c \neq 0$ and $t \neq 0$. Thus, S_t **is not self-adjoint.**

General Context

- **Unitary vs. Self-Adjoint:** In conservative systems (like Quantum Mechanics or the Wave Equation), the evolution operator S_t is typically **unitary**, meaning $S_t^* = S_t^{-1} = S_{-t}$. A unitary operator is rarely self-adjoint (unless $S_t^2 = I$, which is not true for general time evolution).
- **When is it Self-Adjoint?** S_t *is* self-adjoint in specific dissipative systems, such as the **Heat Equation** ($u_t = \Delta u$). In this case, $S_t = e^{t\Delta}$. Since the Laplacian Δ is a self-adjoint operator (with appropriate boundary conditions), the operator exponential $e^{t\Delta}$ is also self-adjoint for real t.

Zak Katz: I see. Does that mean that S_t is not always "diagonalizable"?

Raycast AI: No, that is a common misconception! **"Not self-adjoint" does not mean "not diagonalizable."**

The time-shift operator S_t for the Transport Equation (which I used as a counter-example previously) **is** diagonalizable, even though it is not self-adjoint.

Here is the breakdown of why this works and when it fails.

1. The Key Concept: Normal Operators

In linear algebra and functional analysis, a deeper property governs diagonalizability: **Normality**.
An operator A is **Normal** if it commutes with its adjoint:

$$AA^* = A^*A$$

The **Spectral Theorem** states that **any Normal operator is diagonalizable** (specifically, unitarily diagonalizable).

- **Self-Adjoint Operators** are Normal ($A = A^*$, so $AA = AA$).
- **Unitary Operators** are Normal ($A^* = A^{-1}$, so $AA^{-1} = A^{-1}A = I$).

2. Revisiting the Transport Equation (The "Yes" Case)

Let's look at the Transport Equation again ($u_t + u_x = 0$), where S_t shifts the function: $f(x) \to f(x-t)$.

- We found that $S_t^* = S_{-t}$ (the inverse shift).
- Does it commute?
$$S_t S_t^* = S_t S_{-t} = S_0 = I$$
$$S_t^* S_t = S_{-t} S_t = S_0 = I$$
- Because $S_t S_t^* = S_t^* S_t$, the operator is **Normal**.

How do we "diagonalize" it? We use the **Fourier Transform**. In the frequency domain (Fourier space), the shift operator becomes multiplication:

$$\mathcal{F}(S_t f)(k) = e^{-ikt}\hat{f}(k)$$

Multiplication by a scalar (or function) is the infinite-dimensional equivalent of a diagonal matrix. The Fourier basis functions e^{ikx} are the "eigenvectors" of the time-shift operator.

Conclusion: The Transport operator is not self-adjoint, but it is normal, so it **is** diagonalizable.

---

3. When is S_t NOT Diagonalizable? (The "No" Case)

For S_t to fail to be diagonalizable (to be "defective"), the system must be **Non-Normal** in a specific way that creates "Jordan Blocks." This happens when eigenvectors coalesce and you lose a complete basis (this is called an **Exceptional Point** in physics).

Example: A "Defective" System of PDEs Consider a coupled system of two variables $u(x,t)$ and $v(x,t)$:

$$\frac{\partial}{\partial t}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}$$

Here, the change in u depends on v, but v is constant in time.

1. **Solve it:**
   - $v(x,t) = v(x,0)$ (constant)
   - $u_t = v(x,0) \implies u(x,t) = u(x,0) + t \cdot v(x,0)$
2. **The Time-Evolution Operator S_t:** We can write S_t as a matrix acting on the vector $\begin{pmatrix} f \\ g \end{pmatrix}$:

$$S_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

3. **Is it Diagonalizable?** Check the matrix $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$.
   - The eigenvalues are $\lambda = 1$ (with algebraic multiplicity 2).
   - However, there is only **one** independent eigenvector: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
   - Because we are missing a second eigenvector, this operator is **not diagonalizable** for any $t \neq 0$.

Summary Table

Type of PDE	**Operator Property**	**Self-Adjoint?**	**Diagonalizable?**
Heat Eq. ($u_t = \Delta u$)	Self-Adjoint	**Yes**	**Yes** (Real eigenvalues)
Transport / Wave	Unitary	**No**	**Yes** (Complex eigenvalues)
Defective System	Non-Normal	**No**	**No** (Jordan Block)

Bottom Line: Most standard textbook PDEs (Heat, Wave, Schrödinger) have diagonalizable time-evolution operators because they are typically governed by Normal operators. You usually have to construct specific coupled systems or non-Hermitian boundary conditions to break diagonalizability.

Zak Katz: Let me be clear: S_t shifts *in time* not in space. So to be clear: let a PDE $F(x_1, \ldots, x_n, t) = 0$ PDE w.r.t time t, if $u(x,t)$ is a solution and $f_t(x) = u(x,t)$ is a "state" at time t, and Let S_t be the "time-shift" operator so that $S_t(f_a) = f_{a+t}$ for some state f_a at time a. (NOT the space shift operator $f(x) \to f(x+t)$). Is S_t always diagonalizable?

Raycast AI: Thank you for the clarification. I understand the distinction you are making: you are asking about the **time-evolution operator** (often called the Propagator, $U(t)$ or S_t) that maps the initial state $u(\cdot, 0)$ to the state at time t, $u(\cdot, t)$.

The answer remains: **No, S_t is not always diagonalizable.**

While many standard textbook PDEs (Heat, Wave, Schrödinger) have diagonalizable time-evolution operators because their generators are Normal (Self-Adjoint or Skew-Adjoint), it is very easy to construct physical systems where S_t is **defective** (non-diagonalizable).

This happens specifically when the system exhibits **Linear Instability** or **Critical Damping** (often associated with "Exceptional Points" in modern physics).

The Mathematical Reason: Jordan Blocks

If a PDE can be written in the form $\frac{\partial}{\partial t}\psi = L\psi$, the time-evolution operator is formally $S_t = e^{Lt}$.

For S_t to be diagonalizable, the generator L must possess a complete basis of eigenfunctions. If L has a **Jordan Block** (i.e., it has generalized eigenvectors but not enough true eigenvectors), then the exponential $S_t = e^{Lt}$ will also possess a Jordan block structure for any $t \neq 0$, making it non-diagonalizable.

Here are two concrete counter-examples.

Counter-Example 1: The "Resonant" System (Algebraic Growth)

Consider a system of two fields, $u(x,t)$ and $v(x,t)$, on a domain Ω. The PDE is:

$$\frac{\partial}{\partial t}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}$$

where λ is a constant. This is a standard linear PDE system.

1. **Find the Solution:** This is a coupled system.
   - $v_t = \lambda v \implies v(x,t) = e^{\lambda t}v(x,0)$.
   - $u_t = \lambda u + v$. Substituting the solution for v: $u_t = \lambda u + e^{\lambda t}v(x,0)$.
   - The solution for u involves a secular term (a linear t):
   $$u(x,t) = e^{\lambda t}u(x,0) + \mathbf{te^{\lambda t}}v(x,0)$$
2. **The Time-Evolution Operator S_t:** Writing this as an operator matrix acting on the initial vector $\psi_0 = \begin{pmatrix} u_0 \\ v_0 \end{pmatrix}$:
   $$S_t = e^{\lambda t}\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$
3. **Why it is NOT Diagonalizable:** Look at the matrix part $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$.
   - **Eigenvalues:** It has a single eigenvalue $\mu = 1$ (repeated).
   - **Eigenvectors:** It has only **one** linearly independent eigenvector: $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
   - Because we need two independent eigenvectors to span the 2D space of fields at every point x, and we only have one, S_t is **not diagonalizable** for any $t \neq 0$.

Physical Interpretation: The term $te^{\lambda t}$ represents "algebraic growth" (or transient growth). Whenever you see a solution of the form $t \times$ (exponential), it is the signature of a non-diagonalizable operator. A diagonalizable operator would only have pure sums of exponentials.

Counter-Example 2: Critical Damping (Classical Mechanics / PDE)

Consider the Damped Harmonic Oscillator equation for a field $u(x,t)$, which is a second-order PDE in time:

$$u_{tt} + 2\gamma u_t + \gamma^2 u = 0$$

(Here $\gamma > 0$ is the damping coefficient).

To analyze S_t, we convert this to a first-order system (state-space form) by defining $v = u_t$:

$$\frac{\partial}{\partial t}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\gamma^2 & -2\gamma \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}$$

The Critical Case: The eigenvalues of the matrix $L = \begin{pmatrix} 0 & 1 \\ -\gamma^2 & -2\gamma \end{pmatrix}$ are found by $\det(L - \omega I) = 0$.

$$\omega^2 + 2\gamma\omega + \gamma^2 = (\omega + \gamma)^2 = 0$$

We have a repeated eigenvalue $\omega = -\gamma$.

Because the matrix L is not the identity matrix but has a repeated eigenvalue, it is a non-trivial Jordan block. The solution is:

$$u(t) = C_1 e^{-\gamma t} + C_2 \mathbf{t}\mathbf{e}^{-\gamma \mathbf{t}}$$

The presence of the $te^{-\gamma t}$ term confirms that the propagator S_t cannot be diagonalized. If it were diagonalizable, we could write the solution purely as $u(t) = Ae^{\omega_1 t} + Be^{\omega_2 t}$, which implies strictly exponential decay without the linear polynomial pre-factor.

Summary

- **Is S_t always diagonalizable? No.**
- **When does it fail?** It fails at **Exceptional Points**, where two or more eigenmodes of the system coalesce (become parallel), making the set of eigenvectors incomplete. This results in solutions that grow or decay with polynomial factors (like t, t^2) multiplying the exponentials.

Zak Katz: Okay. Now, when it *is* diagonalizable, is it *always* the case that we can use Fourier transforms to find an eigenbasis?

Raycast AI: No. Absolutely not.

Just because an operator is diagonalizable does not mean the Fourier Transform provides the basis.

The **Fourier Transform** is a very specific tool that only diagonalizes operators that are **Translation Invariant** (i.e., operators with **Constant Coefficients** defined on the entire space $\mathbb{R}^n$).

If your PDE has **variable coefficients** or is defined on a **bounded domain**, the Fourier basis (e^{ikx}) will generally fail to diagonalize the operator, yet a different eigenbasis might still exist perfectly fine.

Here is the hierarchy of logic:

1. **Diagonalizable:** Means *there exists* a basis $\{\varphi_n(x)\}$ such that the operator acts as multiplication on these functions.
2. **Fourier Diagonalizable:** Means that specific basis is $\{\varphi_k(x) = e^{ikx}\}$.

When does Fourier work?

Fourier analysis works **only** if your operator L commutes with the translation operator T_a (where $T_a f(x) = f(x+a)$).

- **Example:** The Heat Equation $u_t = u_{xx}$.
  - The second derivative behaves the same at $x = 0$ as it does at $x = 100$. Space is homogeneous.
  - Therefore, the eigenfunctions are plane waves e^{ikx}.

When does Fourier fail (but the operator is still diagonalizable)?

Fourier fails whenever **spatial symmetry is broken**. This happens in three common scenarios:

1. Variable Coefficients (Space is not homogeneous) Consider the **Quantum Harmonic Oscillator** (one of the most famous diagonalizable systems in physics).

$$i\frac{\partial \psi}{\partial t} = -\frac{\partial^2 \psi}{\partial x^2} + x^2\psi$$

- **Is it Diagonalizable?** Yes. The Hamiltonian is Self-Adjoint. It has a discrete, perfect set of eigenfunctions.
- **Are they Fourier Modes? No.** The term x^2 breaks translation invariance (physics looks different at $x = 0$ vs $x = 100$).
- **The Actual Basis:** The eigenfunctions are **Hermite Functions** (a Gaussian multiplied by a polynomial):
$$\varphi_n(x) = H_n(x)e^{-x^2/2}$$
  If you tried to apply a Fourier Transform to this equation, the x^2 multiplication term turns into a Laplacian in k-space ($-\partial_k^2$), so you effectively just swap the PDE for an identical PDE in frequency space without solving it.

2. Bounded Domains (The "Boundary Condition" Problem) Consider the heat equation $u_t = u_{xx}$ on a finite interval $x \in [0, L]$ with Dirichlet boundary conditions $u(0) = u(L) = 0$.

- **Is it Diagonalizable?** Yes.
- **Is it the Fourier Transform?** No. The Continuous Fourier Transform uses e^{ikx} for any real k.
- **The Actual Basis:** The basis is strictly **Sine functions**: $\sin(\frac{n\pi x}{L})$.

– While related to Fourier *Series*, the Continuous Fourier *Transform* basis (e^{ikx}) is invalid here because e^{ikx} does not satisfy the boundary conditions (it doesn't vanish at 0 and L). You are forced onto a discrete subset of the frequency spectrum.

3. Different Geometries If your PDE is on a sphere (like weather patterns on Earth):

$$\frac{\partial u}{\partial t} = \Delta_{\text{sphere}} u$$

- **Is it Diagonalizable?** Yes.
- **The Actual Basis:** The eigenfunctions are **Spherical Harmonics** $Y_l^m(\theta, \varphi)$.
- Standard planar Fourier modes (e^{ikx}) make no sense here.

Summary Table

PDE Feature	**Is S_t Diag.?**	**Fourier?**	**The Actual Eigenbasis**
Constant Coeffs on $\mathbb{R}$ ($u_t = u_{xx}$)	Yes	**Yes**	Plane Waves (e^{ikx})
Variable Coeffs ($u_t = xu_x$)	Yes	**No**	Power functions (x^λ)
Potential Well ($u_t = i\Delta u + iV(x)u$)	Yes	**No**	Hermite / Legendre / Bessel
Bounded Domain ($x \in [0, 1]$)	Yes	**No**	Fourier *Series* (Sines/Cosines)

Conclusion: Diagonalizability guarantees that *some* coordinate system exists where the operator is simple. It does not guarantee that the coordinate system is the one given by Fourier.

Zak Katz: Okay. Is it a necesssary and sufficient condition that S_t is diagonalizable by the Fourier basis iff the diagonalizable operator is translation invariant? Just to make sure I understand what translation invariant means here, we are talking about *space* translation, and we mean that if S_t is our operator and τ_a is our space-translation, then:

$$S_t(\tau_a(f)) = \tau_a(S_t(f))$$

Raycast AI: Yes, precisely.

For a linear operator S_t acting on functions defined on the entire space $\mathbb{R}^n$ (or a periodic domain), the condition is **necessary and sufficient**:

> **Theorem:** A bounded linear operator S_t is diagonalized by the Fourier Transform if and only if it is **Translation Invariant**.

Your definition of translation invariance is exactly correct:

$$[S_t, \tau_a] = S_t\tau_a - \tau_a S_t = 0 \quad \text{for all } a \in \mathbb{R}^n$$

where τ_a is the spatial shift operator.

Here is the logic for why this is an "if and only if" relationship.

1. Why is it Sufficient? (Translation Invariant $\implies$ Fourier Diagonal)

This relies on a fundamental principle of linear algebra: **Commuting operators share a common eigenbasis.**

1. Consider the family of translation operators $\{\tau_a\}_{a\in\mathbb{R}}$.
2. What are the eigenfunctions of a translation? Let's verify that the plane waves e^{ikx} are eigenfunctions of τ_a:
$$\tau_a(e^{ikx}) = e^{ik(x+a)} = e^{ika} \cdot e^{ikx}$$
The eigenvalue is the constant number e^{ika}, and the eigenfunction is e^{ikx}.
3. Since your time-evolution operator S_t commutes with **every** translation operator τ_a (for all shifts a), it must share the same set of eigenfunctions.
4. Therefore, the eigenfunctions of S_t must also be e^{ikx}.
5. If the eigenfunctions are e^{ikx}, then by definition, the operator is diagonalized by the Fourier Transform.

2. Why is it Necessary? (Fourier Diagonal $\implies$ Translation Invariant)

If an operator is diagonal in Fourier space, it acts as a simple multiplication. Let's call the Fourier transform $\mathcal{F}$.

1. Assume S_t is a Fourier Multiplier: $\widehat{S_t u}(k) = m(k)\widehat{u}(k)$ for some multiplier function $m(k)$.
2. The translation operator τ_a in Fourier space is also just a multiplication (by the phase factor e^{ika}).
3. Since regular multiplication of scalars is commutative ($a \cdot b = b \cdot a$), the multiplication by $m(k)$ commutes with the multiplication by e^{ika}.
4. Therefore, S_t and τ_a commute.

Summary

This concept is the bedrock of signal processing and linear systems theory (where these are called **Linear Time-Invariant (LTI)** systems, or in your PDE context, "Linear Space-Invariant" systems).

- **Does S_t commute with space shifts?**
  - **Yes:** You can use Fourier Transforms (e.g., Heat Eq, Wave Eq, Schrödinger with constant potential).

  - **No:** You cannot use Fourier Transforms to find the eigenbasis (e.g., Schrödinger with Harmonic Oscillator potential, variable coefficient Wave Eq).

Zak Katz: Right. Is ther a "theory" for diagonalizable operators S_t?

Raycast AI: Yes, there is a vast and rich "theory" for this, but it doesn't have a single name like "The Theory of Diagonalizable Operators." Instead, it is split into different fields of **Functional Analysis** depending on *why* the operator is diagonalizable.

If you are looking for the generalized version of "Fourier Analysis" for operators that are **not** translation invariant (i.e., not solvable by Fourier Transform), here is the map of the theory.

1. The "Gold Standard": The Spectral Theorem

This is the master theory. It applies if your operator S_t (or its generator L) is **Normal** ($LL^* = L^*L$).

- **The Theory: Spectral Theory of Normal Operators**.
- **What it says:** Any normal operator on a Hilbert space is unitarily equivalent to a multiplication operator.
- **Physical Meaning:** There *always* exists an orthonormal basis of eigenfunctions (or a "continuous basis" / spectral measure) that diagonalizes the operator.
- **How to find the basis:** This is the hard part. It usually requires solving the **Sturm-Liouville Problem** (for 1D) or using **Scattering Theory** (for unbounded domains).

2. The "Variable Coefficient" Theory: Sturm-Liouville Theory

If your PDE has variable coefficients but is in 1D (or separable), this is the standard replacement for Fourier analysis.

- **The Setup:** Your generator looks like $L = \frac{d}{dx}[p(x)\frac{d}{dx}] + q(x)$.
- **The Result:** Just like Fourier series uses $\sin(nx)$ and $\cos(nx)$, Sturm-Liouville theory guarantees a set of orthogonal eigenfunctions $\varphi_n(x)$ weighted by a density function.
- **Example:** Legendre polynomials, Bessel functions, Hermite polynomials (Quantum Harmonic Oscillator). These are the "custom Fourier bases" for specific geometries or potentials.

3. The "Abstract" Theory: Semigroup Theory (C_0-Semigroups)

This is the modern mathematical framework for time-evolution operators S_t.

- **The Setup:** You treat S_t as a "one-parameter semigroup" $T(t)$ acting on a Banach space.
- **The Theorem:** The **Hille-Yosida Theorem**.

- **What it does:** It characterizes exactly when a differential operator L generates a stable time-evolution S_t. While Hille-Yosida doesn't guarantee *diagonalizability* (it only guarantees existence and uniqueness of the flow), it is the starting point for analyzing the spectrum of S_t.
- **Spectral Mapping Theorem:** This theorem relates the eigenvalues of the generator L to the eigenvalues of the propagator S_t (basically $\sigma(S_t) = e^{t\sigma(L)}$). If you know the spectrum of L, you know the spectrum of S_t.

4. The "Non-Normal" Theory: Riesz Bases and Spectral Operators

What if S_t is diagonalizable but the eigenvectors are **not orthogonal**? (This happens in non-conservative systems, like convection-diffusion with non-standard boundary conditions).

- **The Theory: Non-Self-Adjoint Spectral Theory** (or the theory of **Riesz Bases**).
- **What it says:** You can still expand solutions in a basis of eigenfunctions, but because the basis is not orthogonal ($\langle \varphi_n, \varphi_m \rangle \neq \delta_{nm}$), you need a "dual basis" $\{\psi_n\}$ to find the coefficients.
  - Expansion: $f = \sum a_n \varphi_n$
  - Coefficients: $a_n = \langle f, \psi_n \rangle$ (instead of $\langle f, \varphi_n \rangle$).
- **Why it matters:** This is currently a very hot topic in fluid dynamics and laser physics, where systems are often "almost" normal but mathematically non-self-adjoint.

Summary: Which "Theory" do you need?

If your operator S_t is...	**The Theory is called...**	**The Basis is...**
Translation Invariant	Harmonic Analysis	Fourier Transform
Self-Adjoint / Normal	Spectral Theory	Orthonormal Eigenbasis
1D Variable Coeffs	Sturm-Liouville Theory	Special Functions
Not Normal (but diag.)	Non-Self-Adjoint Theory	Riesz Basis (Biorthogonal)
Not Diagonalizable	Jordan Form	Generalized Eigenvectors

4

Radon Measures and Dual Theory

(I really like the results of this chapter and am adamant to write it down here, especially the duality results. This chapter is a place-holder for now)

(intuition here once I have my own)

4.0.1 Convention In this chapter X will be a *locally compact Hausdorff space* (LCH space).

4.1 Positive Linear Functionals on Compact Support Continuous Functions

(I think there is a reason we start with $C_c(X)$ which I will eventually write down either here or in the chapter intro)

Definition 4.1.1: Positive Functional

Let $C_c(X)$ denote the real continuous functions with compact support. A linear functional $I : C_c(X) \to \mathbb{R}$ is called *positive* if $I(f) \geq 0$ whenever $f \geq 0$.

A positive functional was not defined to necessarily be continuous, however the positivity condition gives a sort of continuity that with $C_c(X)$ "having" compact-open topology, however this topology is in fact too coarse since as we'll see we shall want $\text{supp}(f) \subseteq K$ for compact $K \subseteq X$ which is not necessarily the case for the compact-open topology. Instead, we can form a Banach space $C_K(X)$ with the sup-norm $\|f\|_{u,K} = \sup_{x\ inK} |f(x)|$ and then take:

$$C_c(X) = \cup_{K \subseteq X} C_K(X)$$

and put the usual *filtered colimit topology* on it (in classical literature known as inductive limit topology). By the universal property of the topology, a functional is continuous if it is continuous on each $C_K(X)$.

Proposition 4.1.2: Continuity of Positivity

Let I be a positive linear functional on $C_c(X)$. Then for each compact $K \subseteq X$ there exists a C_K such that

$$|I(f)| \leq C_K \|f\|_u \qquad \forall f \in C_c(X) \text{ where supp} f \subseteq K$$

***Proof*:**
Let $f \geq 0$. Given $K \subseteq X$, by Urysohn's lemma choose $\varphi \in C - c(X, [0,1]0$ such that $\varphi = 1$ on K. Then if $\text{supp}(f) \subseteq K$,

$$|f(x)| \leq \|f\|_u \varphi(x) \qquad \text{i.e.} \qquad |f| \leq \|f\|_u \varphi$$

Hence $\|f\|_u \varphi - f \geq 0$. As f is positive $\|f\|_u \varphi + f \geq 0$. Thus by positivity of I:

$$\|f\|_u I(\varphi) - I(f) \geq 0 \qquad \text{and} \qquad \|f\|_u I(\varphi) + I(f) \geq 0$$

Hence:

$$\|I(f)| \leq I(\varphi)\|f\|_u$$

as we sought to show.

Now, given μ is a Bore measure on X such that $\mu(K) < \infty$ for all compact $K \subseteq X$, then clearly $C_c(X) \subseteq L^1(\mu)$ So the map:

$$f \mapsto \int f d\mu$$

is a positive linear functional on $C_c(X)$. What we shall show is that *every* positive linear functional on $C_c(X)$ is of this form, and with a little more regularity μ can be made unique in some way feels like this should be said earlier.

Definition 4.1.3: Regular Borel Measure

Let μ be a Borel measure on X and $E \subseteq X$ a Borel subset. Then μ is *outer-regular* on E if:

$$\mu(E = \inf \{\mu(U) \mid U \supseteq E, U \text{ open}\}$$

inner regular E if:

$$\mu(E = \inf \sup \mu(K) K \subseteq E, K \text{ compact}$$

and regular on E if E is inner- and outer-regular. The Borel μ is called outer-regular, inner-regular, and regular if all Borel sets are outer-regular, inner-regular, and regular respectively.

In most spaces, a regular measure shall work. However, we shall have some edge-cases of spaces that are not σ-compact on which we cannot define a regular measure (see ref:HERE for where this comes up)

Example 4.1.4: Too Big For Regular Measure
(you can make one on the LCH space $X = [0, \omega_1)$ which is the collection of all countable ordinals, which is not σ-compact, and take the Borel σ-algebra given the Dieudonné Measure)

Definition 4.1.5: Radon Measure

Let μ be a measure on X. Then if μ is a Borel measure that is:

1. finite on all compact sets
2. outer-regular on all Borel sets
3. inner-regular on all open sets

then μ is called a *Radon Measure*.

Proposition 4.1.8 shall show that all Radon measure are also inner-regular on all σ-finite sets.

Now, for the following theorem define the following notation: if $U \subseteq X$ is open and $f \in C_c(X)$ then:

$$f \prec U \qquad \text{iff} \qquad 0 \leq f \leq 1 \text{ and } \operatorname{supp}(f) \subseteq U$$

Note this is note quite the same as $0 \leq f \leq \chi_U$ as we this would imply $\operatorname{supp}(f) \subseteq \overline{U}$.

Theorem 4.1.6: Riesz Representation Theorem

If I is a positive linear functional on $C_c(X)$, there is a unique Radon measure μ on X such that $I(f) = \int f d\mu$ for all $f \in C_c(X)$. Moreover, μ satisfies

$$\mu(U) = \sup \{I(f) \mid f \in C_c(X), f \prec U\} \text{ for all open } U \subseteq X \tag{4.1}$$

and

$$\mu(K) = \inf \{I(f) \mid f \in C_c(X), f \geq \chi_K\} \text{ for all compact } K \subseteq X \tag{4.2}$$

***Proof*:**
We will first tackle the uniqueness of such a measure, which will naturally guide us toward the correct construction for existence.

Suppose such a Radon measure μ exists. Then for any open $U \subseteq X$, if $f \prec U$ (meaning $f \in C_c(X)$, $0 \leq f \leq 1$, and $\operatorname{supp}(f) \subseteq U$), we have:

$$I(f) = \int f d\mu \leq \int \chi_U d\mu = \mu(U)$$

Conversely, for any compact $K \subseteq U$, Urysohn's Lemma guarantees a function $f \in C_c(X)$ such that $f \prec U$ and $f = 1$ on K. Thus:

$$\mu(K) = \int \chi_K d\mu \leq \int f d\mu = I(f)$$

Since μ is a Radon measure, it is inner regular on open sets, meaning $\mu(U) = \sup_{K \subseteq U} \mu(K)$. Combining this with the inequalities above shows that $\mu(U)$ must be exactly the supremum of

$I(f)$ for $f \prec U$, satisfying (4.1). Since a Radon measure is determined by its values on open sets (due to outer regularity), μ is unique.

To prove existence, we simply turn this uniqueness property into a definition. For any open U, define:

$$\mu(U) = \sup \{I(f) \mid f \in C_c(X),\ f \prec U\}$$

and for an arbitrary set $E \subseteq X$, we define the outer measure μ^* standardly as:

$$\mu^*(E) = \inf \{\mu(U) \mid U \supseteq E,\ U \text{ open}\}$$

Note that for open sets, $\mu^*(U) = \mu(U)$ since $U \subseteq V \implies \mu(U) \leq \mu(V)$. To complete the theorem, we proceed in four steps:

1. Show μ^* is an outer measure.
2. Show every open set is μ^*-measurable.

Once these two are established, Carathéodory's Theorem (**??**) implies that all Borel sets are measurable and $\mu = \mu^*|_{\mathcal{B}_X}$ is a Borel measure. We then must show:

3. μ satisfies the inner regularity condition (4.2).
4. The functional is recovered by the integral: $I(f) = \int f d\mu$.

1. *μ^* is an outer measure:* The only non-trivial property to check is countable subadditivity on open sets (since the extension to arbitrary sets follows from **??**). Let $U = \bigcup_{j=1}^{\infty} U_j$. We want to show $\mu(U) \leq \sum \mu(U_j)$. Pick any $f \prec U$. Then $K = \text{supp}(f)$ is a compact subset of U, so it is covered by finitely many U_j, say N. Using a Partition of Unity ref:HERE (Proposition 4.41 in folland), we can find $g_1, \ldots, g_N \in C_c(X)$ such that $g_j \prec U_j$ and $\sum g_j = 1$ on K. Then $f = \sum_{j=1}^{N} f g_j$, and since each $f g_j \prec U_j$:

$$I(f) = \sum_{j=1}^{N} I(f g_j) \leq \sum_{j=1}^{N} \mu(U_j) \leq \sum_{j=1}^{\infty} \mu(U_j)$$

   Taking the supremum over all such f, we get $\mu(U) \leq \sum \mu(U_j)$, as desired.

2. *Open sets are measurable:* we need to show that for any open U and any E with finite outer measure, $\mu^*(E) \geq \mu^*(E \cap U) + \mu^*(E \setminus U)$. First, assume E is open. Given $\epsilon > 0$, we can choose an $f \prec E \cap U$ such that $I(f)$ is close to $\mu(E \cap U)$, specifically $I(f) > \mu(E \cap U) - \epsilon$. Similarly, since $E \setminus \text{supp}(f)$ is open, we choose $g \prec E \setminus \text{supp}(f)$ with $I(g) > \mu(E \setminus \text{supp}(f)) - \epsilon$. Since f and g have disjoint supports, $f + g \prec E$. Thus:

$$\begin{aligned} \mu(E) \geq I(f+g) &= I(f) + I(g) \\ &> \mu(E \cap U) + \mu(E \setminus \text{supp}(f)) - 2\epsilon \\ &\geq \mu^*(E \cap U) + \mu^*(E \setminus U) - 2\epsilon \end{aligned}$$

   Letting $\epsilon \to 0$ gives the result for open E. For a general E, we approximate it from the outside by an open $V \supset E$ such that $\mu(V) < \mu^*(E) + \epsilon$ and apply the previous logic to V.

3. *Inner Regularity:* We verified outer regularity by construction; now we check (4.2) for compact K. If $f \in C_c(X)$ and $f \geq \chi_K$, let $U_\epsilon = \{x \mid f(x) > 1 - \epsilon\}$. This is an open set containing K. If we take any $g \prec U_\epsilon$, then $f \geq (1-\epsilon)g$, so $I(g) \leq (1-\epsilon)^{-1} I(f)$. Taking the supremum over g, we get $\mu(K) \leq \mu(U_\epsilon) \leq (1-\epsilon)^{-1} I(f)$. As $\epsilon \to 0$, $\mu(K) \leq I(f)$. Conversely, for any open $U \supset K$, Urysohn's Lemma gives us an f where $\chi_K \leq f \prec U$, so $I(f) \leq \mu(U)$. Taking the infimum over U confirms the equality.

4. *Recovering the Integral:* it suffices to prove $I(f) = \int f d\mu$ for functions with range $[0,1]$ (by linearity). The strategy is to slice the range of f into N strips of height $1/N$, similar to constructing a Lebesgue integral. Let $K_j = \{x \mid f(x) \geq j/N\}$ for $j = 0, \ldots, N$. We define a "layer" function f_j that lives between K_j and K_{j-1}:

$$f_j = \min\left\{\max\left\{f - \frac{j-1}{N}, 0\right\}, \frac{1}{N}\right\}$$

   Notice that $f = \sum_{j=1}^N f_j$. We can bound f_j by characteristic functions: $\frac{1}{N}\chi_{K_j} \leq f_j \leq \frac{1}{N}\chi_{K_{j-1}}$. Integrating this gives:

$$\frac{1}{N}\mu(K_j) \leq \int f_j d\mu \leq \frac{1}{N}\mu(K_{j-1})$$

   Applying the functional I (and using the fact that $N f_j \prec U$ for any open $U \supset K_{j-1}$) yields the same bounds:

$$\frac{1}{N}\mu(K_j) \leq I(f_j) \leq \frac{1}{N}\mu(K_{j-1})$$

   Summing these up for all j, both $\int f d\mu$ and $I(f)$ are trapped in the same interval of size $\frac{1}{N}(\mu(K_0) - \mu(K_N))$. Since $\mu(K_0)$ is finite (it's the support of f) and N is arbitrary, the difference must be zero, so $I(f) = \int f d\mu$, completing the proof.

It is worth noting that the construction in this proof actually yields something stronger than just the statement of the theorem. We obtain not just a Borel measure μ, but an extension $\overline{\mu}$ of μ to the σ-algebra of all μ^*-measurable sets. Recall that μ is outer regular. Because of this, for any $E \subseteq X$, the outer measure is effectively approximated by Borel sets:

$$\mu^*(E) = \inf\{\mu(B) \mid B \in \mathcal{B}_X,\ B \supseteq E\}$$

This means that μ^* corresponds exactly to the outer measure induced by μ. Consequently, the extension $\overline{\mu}$ relates to the standard completion of a measure: if μ is σ-finite, $\overline{\mu}$ is precisely the completion of μ. In the general case, it is the saturation of the completion.

Finally, while we have focused on $\mathcal{B}_X$, some authors prefer to work with a slightly smaller σ-algebra, specifically the one generated by the continuous functions themselves. For ease of indexing, we box the definition:

Definition 4.1.7: Baire Sets

The σ-algebra generated by $C_c(X)$ (i.e., the smallest σ-algebra with respect to which every $f \in C_c(X)$ is measurable) is denoted by $\mathcal{B}_X^0$. The elements of $\mathcal{B}_X^0$ are called *Baire sets*.

Proposition 4.1.8: Inner Regularity on σ-finite Sets

Every Radon measure is inner regular on all of its σ-finite sets.

***Proof*:**
Let μ be a Radon measure and E be a σ-finite set. We proceed by cases.

1. Assume $\mu(E) < \infty$. The strategy here is to approximate E from the outside with an open set, and then approximate that open set from the inside with a compact set, ensuring we stay "mostly" inside E. For any $\epsilon > 0$, since μ is outer regular on Borel sets, we can choose an open $U \supseteq E$ such that $\mu(U) < \mu(E) + \epsilon$. Since μ is inner regular on open sets, we can find a compact $F \subseteq U$ such that $\mu(F) > \mu(U) - \epsilon$.

   However, F is not necessarily a subset of E. The part of F that "spills over" is contained in $U \setminus E$. Notice that:
   $$\mu(U \setminus E) = \mu(U) - \mu(E) < \epsilon$$
   Since this "error set" $U \setminus E$ is Borel, we can use outer regularity again to cover it with an open set $V \supseteq U \setminus E$ such that $\mu(V) < \epsilon$. Now, define $K = F \setminus V$. Since V is open, K is a closed subset of the compact set F, and thus K is compact. Furthermore, since we removed the part covering $U \setminus E$, we have $K \subseteq E$. Finally, we check the measure:
   $$\mu(K) = \mu(F) - \mu(F \cap V) > (\mu(E) - \epsilon) - \mu(V) > \mu(E) - 2\epsilon$$
   Since ϵ is arbitrary, μ is inner regular on E.

2. Assume $\mu(E) = \infty$. Since E is σ-finite, it is an increasing union of sets E_j with $\mu(E_j) < \infty$ and $\mu(E_j) \to \infty$. For any large number N, there exists some j such that $\mu(E_j) > N$. Since E_j has finite measure, by the argument in Case 1, we can find a compact $K \subseteq E_j \subseteq E$ such that $\mu(K)$ is arbitrarily close to $\mu(E_j)$, and specifically $\mu(K) > N$. Thus, the supremum of the measures of compact subsets is infinite, satisfying inner regularity.

Corollary 4.1.9: Regularity of Radon Measures

Every σ-finite Radon measure is regular. Furthermore, if X is σ-compact, every Radon measure on X is regular.

4.2 Dual of $C_0(X)$

In the previous section, we established the correspondence between positive linear functionals and Radon measures. However, $C_c(X)$ is generally not a Banach space under the uniform norm $\|\cdot\|_u$ (unless X is compact). We need a complete space. The completion of $C_c(X)$ under the uniform norm is $C_0(X)$.

Definition 4.2.1: Functions Vanishing at Infinity

A continuous function $f : X \to \mathbb{R}$ is said to *vanish at infinity* if for every $\epsilon > 0$, the set $\{x \in X \mid |f(x)| \geq \epsilon\}$ is compact.
The space of all continuous functions vanishing at infinity is denoted $C_0(X)$. Equipped with the uniform norm $\|f\|_u = \sup_{x \in X} |f(x)|$, $C_0(X)$ is a Banach space, and $C_c(X)$ is dense in it.

Now, instead of restricting ourselves to *positive* functionals, we want to look at the continuous dual space $C_0(X)^*$, which consists of *all* bounded linear functionals on $C_0(X)$. Since these functionals can be negative, they will naturally correspond to signed measures.

Definition 4.2.2: Space of Finite Radon Measures $M(X)$

Let $M(X)$ denote the space of all finite signed regular Borel measures on X. A signed measure μ is regular if its positive and negative variations (from the Hahn Decomposition), μ^+ and μ^-, are finite Radon measures.
We equip $M(X)$ with the *total variation norm*, defined as $\|\mu\| = |\mu|(X) = \mu^+(X) + \mu^-(X)$.

We are now ready to state the full Riesz Representation Theorem. This is the exact realization of the "dual theory" philosophy. Just as $(L^p)^* \cong L^q$, we now identify the dual of $C_0(X)$.

Theorem 4.2.3: Riesz Representation for $C_0(X)^*$

Let X be an LCH space. For every bounded linear functional $I \in C_0(X)^*$, there exists a unique finite signed regular Borel measure $\mu \in M(X)$ such that

$$I(f) = \int f d\mu \qquad \forall f \in C_0(X)$$

Moreover, the map $\mu \mapsto I$ is an isometric isomorphism, meaning $\|I\| = \|\mu\| = |\mu|(X)$.

***Proof*:**
The strategy here is to split the bounded functional I into a difference of two positive functionals, say $I = I^+ - I^-$. Once we do that, we can just apply our previous theorem 4.1.6 to each piece, get two positive Radon measures, and subtract them.

Let's construct I^+. For any non-negative $f \in C_c(X)$, we define:

$$I^+(f) = \sup\{I(g) \mid g \in C_c(X),\ 0 \leq g \leq f\}$$

Since I is bounded, $I(g) \leq \|I\|\|g\|_u \leq \|I\|\|f\|_u$, so this supremum is finite. It requires a standard but slightly tedious algebraic verification to show that I^+ is additive and positive on non-negative functions. Once we have I^+ defined for $f \geq 0$, we extend it linearly to all of $C_c(X)$ by writing $f = f^+ - f^-$.

Next, we simply define $I^- = I^+ - I$. Since $I^+(f) \geq I(f)$ for all $f \geq 0$, it immediately follows that I^- is also a positive functional.

By the Riesz Representation Theorem for positive functionals (theorem 4.1.6), there exist unique positive Radon measures μ^+ and μ^- such that:

$$I^+(f) = \int f d\mu^+ \qquad \text{and} \qquad I^-(f) = \int f d\mu^-$$

Let $\mu = \mu^+ - \mu^-$. Because I is bounded on $C_c(X)$ with respect to the uniform norm, one can show that μ^+ and μ^- must be finite measures. Thus $\mu \in M(X)$, and $I(f) = \int f d\mu$ for all $f \in C_c(X)$. Because $C_c(X)$ is dense in $C_0(X)$ and I is continuous, this integral representation uniquely extends to all of $C_0(X)$.

Finally, we must show the isometry $\|I\| = \|\mu\|$. First, we show $\|I\| \leq \|\mu\|$. For any $f \in C_0(X)$ with $\|f\|_u \leq 1$:

$$|I(f)| = \left| \int f d\mu \right| \leq \int |f| d|\mu| \leq \int 1 d|\mu| = |\mu|(X) = \|\mu\|$$

Taking the supremum over all such f gives $\|I\| \leq \|\mu\|$.

To get the reverse inequality, we use the Hahn decomposition $X = P \cup N$ where P is the positive set for μ and N is the negative set. If we could just plug the function $f = \chi_P - \chi_N$ into our functional, we would get exactly:

$$\int (\chi_P - \chi_N) d\mu = \mu(P) - \mu(N) = \mu^+(X) + \mu^-(X) = \|\mu\|$$

While $\chi_P - \chi_N$ is sadly not continuous (so it's not in $C_0(X)$), we can use the regularity of μ and Lusin's Theorem to approximate it from within with a sequence of continuous functions $f_n \in C_c(X)$ such that $\|f_n\|_u \leq 1$. Taking the limit as $n \to \infty$ yields $\|I\| \geq \|\mu\|$, locking in the exact isometry.

This theorem fully characterizes the geometry of $M(X)$: it tells us that regular Borel measures are the natural algebraic dual to continuous functions. In the specific case where X is compact, $C_0(X) = C(X)$, giving us $C(X)^* \cong M(X)$.

4.2.4　Better Than Smooth Functions　When covering distributions in **??**, we shall see that it is a good to replace evaluation with integrating against a smooth function with compact support. The results of this section show that for distributions represented by continuous functions, it is natural to replace $C_c^\infty(X)$ with $M(X)$, and in fact this would work even for distributions represented by continuous functions as this would recover evaluation; see ref:HERE.

5

Hilbert Spaces

There are many reasons why Hilbert spaces are interesting. We shall start with a motivation stemming from looking for the ideal space where the product of two functions is still integrable, and then take a moment to elaborate on some more advanced intuitions of Hilbert spaces for readers with more background.

We are naturally interested in integrating products of functions:

$$\int_X fg$$

The problem we may encounter when working over L^1 is that $\int_X fg$ need not be finite. We thus may say "let us thus work work the functions in which this *is* well-defined". If $L'(X)$ is the space of all functions measurable functions where for any choice $f, g \in L'(X)$:

$$\int_X f\overline{g} < \infty \qquad \text{i.e.} \qquad f\overline{g} \in L^1$$

then we have just made a circular-reasoning mistake, for we are quantifying existence of elements in $L'(X)$ using elements of $L'(X)$. Instead, we may ask: given a set S, what is the subset of measurable functions T such that:

$$T = \left\{ f \in \mathcal{M}(X) \;\middle|\; \int_X f\overline{g} < \infty,\ \forall g \in S \right\}$$

The two extremes of this are if $S = \{0\}$, then T is all measurable functions, and if S is all measurable functions, then $T = \{0\}$ only contains the zero function. The ideal is that there is a middle ground choice of S where $S = T$ so that we don't need to keep track of the two spaces we are choosing our functions from, namely we would not like to have $(S, T)(X)$. Fortunately, as will be proven in section 3.2, this is *exactly* the space L^2. Namely, if $S = L^2$, then:

$$T = \left\{ f \in \mathcal{M}(X) \;\middle|\; \int_X f\overline{g} < \infty\ \forall g \in L^2 \right\} = L^2$$

Though we can't prove this here, the high level is that if $f, g \in L^2$, then by the Cauchy-Schwartz inequality[1]

$$\int_X |fg| \le \left(\int_X |f|^2\right)^{1/2} \left(\int_X |g|^2\right)^{1/2} = \|f\|_2 \|g\|_2$$

showing that $L^2 \subseteq T$. Conversely, by Hölder's inequality, we have that if $f \in L^p$, then for all $g \in L^q$, $fg \in L^1$. If $p = 2$, then $q = 2$, showing that all functionals $(L^2)^*$ are characterized by L^2 functions:

$$\varphi_g : L^p \to \mathbb{R} \qquad \varphi_g(f) = \int fg$$

showing that $T \subseteq L^2$, hence $T = L^2$. From this perspective, L^2 is the ideal space in which to multiply functions and they remain integrable (i.e. $f, g \in L^2$, $fg \in L^1$)! We shall call such multiplication the *inner product*:

$$\langle f, g \rangle = \int_X f\overline{g}$$

The conjugation in the definition is used so that $f\overline{f} = |f|^2$. Noe that we are not looking for the product of two functions to remain within L^1. In section **??**, we shall see that *convolution* will have the desired property so that $f, g \in L^1(G)$ then $f * g \in L^1(G)$ for G a locally compact group, in particular that shall be a *group algebra*.

5.0.1 post-Hoc Motivation A lot of the following motivation came *a-posteriori*, that is, I felt much more comfortable with Hilbert spaces after leaning about L^p spaces, and generalizing some result from Riemannian Geometry to Banach spaces to notice that L^2 is the only space with flat geometry, hence really is the "correct" generalization of euclidean space/geometry in infinite dimensions.

Banach spaces provide a robust framework for measuring the magnitude of elements, allowing us to quantify convergence and bound operations via the triangle and Minkowski inequalities. However, a fundamental deficiency in general Banach spaces is the lack of a linear measurement theory to quantify *similarity*. While we can measure the total size of a sum $\|f + g\|$, we often lack the granular information required to isolate the contribution of f from g. For instance, determining how much a vector projects onto a specific basis element is straightforward in Euclidean space but algebraically obstructed in general norms.

To make this deficiency precise, consider how the inner product arises in Euclidean geometry. Define the *energy* of a vector as $E(x) = \frac{1}{2}\|x\|_2^2$. Then the directional derivative of this energy at x in the direction of a perturbation y is:

$$\frac{d}{dt} E(x + ty)\Big|_{t=0} = \frac{d}{dt} \frac{1}{2} \langle x + ty, x + ty \rangle \Big|_{t=0} = \langle x, y \rangle$$

Thus, the inner product $\langle x, y \rangle$ is not just an algebraic artifact; it physically represents the *sensitivity* of the energy of x to a perturbation by y. In particular, the energy satisfies the clean decomposition:

$$E(x + y) = E(x) + E(y) + \langle x, y \rangle$$

[1] which shall be proven in this section, but the reader most likely has already encountered it and hence it is used here to give intuition

The interaction term $\langle x, y\rangle$ is *bilinear*—it decomposes the interference between two vectors according to the simple rules of superposition. In general Banach spaces, however, such a clean, linear decomposition of the interaction is impossible due to the non-flat geometry of the unit ball.

The natural question, then, is: *what replaces the inner product in L^p?* The correct generalization is to replace the quadratic energy with the *p-energy*:

$$F(f) = \frac{1}{p}\|f\|_p^p = \frac{1}{p}\int |f|^p$$

and ask: how does $F(f)$ change if we perturb f by a function g? To answer this, we must compute the gradient ∇F, which requires first understanding the natural duality of L^p spaces.

The Duality Map. What does it mean to go in the "same direction" in a function space? (For simplicity we work throughout with L^p spaces for $1 < p < \infty$.[2]) Rigorously, we are asking for the functional φ_f of unit norm that *maximizes* the evaluation at f. We can interpret this φ_f as the *optimal filter* or sensor for f—it captures the full "energy" of f without loss ($\varphi_f(f) = \|f\|_p$ and $\|\varphi_f\|_p = 1$).

By theorem 3.2.3 (and the equality condition of Hölder's inequality), this functional corresponds to $g \in L^q$ such that:

$$g(x) = |f(x)|^{p-1}\operatorname{sgn}(f(x))$$

Note that $g \in L^q$ since:

$$|g|^q = |f|^{q(p-1)} = |f|^{qp-q} \overset{!}{=} |f|^p$$

where $\overset{!}{=}$ holds since $1/p + 1/q = 1$ iff $p + q = pq$ iff $p = q(p-1)$. Hence as $f \in L^p$:

$$\int |g|^q = \int |f|^p < \infty$$

Thus, we have a map $f \mapsto \varphi_f$. We define the *duality map*:

$$J_p : L^p \to L^q, \qquad f \mapsto \varphi_f$$

As $(L^q)^* \cong L^p$, we can equivalently view this as a map $J_p : L^p \to (L^p)^*$.

The Central Geometric Fact. The duality map J_p is not an algebraic accident—it is the *geometric normal* to the level sets of the p-energy. To see this, consider the function:

$$F(x) = \frac{1}{p}\|x\|_p^p = \frac{1}{p}\sum |x_i|^p$$

(where the ℓ^p norm is used for visualization, but the logic holds for integrals). From multivariable calculus, the unit sphere is the level set $F(x) = \frac{1}{p}$, and the gradient $\nabla F(x)$ is the vector strictly perpendicular (normal) to the level set at point x. Computing the gradient (the Fréchet derivative):

$$\frac{\partial}{\partial x_i}\left(\frac{1}{p}|x_i|^p\right) = |x_i|^{p-1}\operatorname{sgn}(x_i)$$

[2] The reasoning can extend to nuclear Fréchet spaces, which admit descriptions as projective limits of Hilbert spaces. For general metric spaces, a different approach is needed via CAT(k) geometry; see [5] for details.

So, the gradient vector is:

$$\nabla F(x) = \big(|x_1|^{p-1}\mathrm{sgn}(x_1), \ldots, |x_n|^{p-1}\mathrm{sgn}(x_n)\big)$$

This is exactly our duality map:

$$\boxed{\nabla F(x) = J_p(x)}$$

This identification is the key to the entire theory. It tells us that the optimal functional for f—the element of $(L^p)^*$ that best "measures" f—is precisely the gradient of the energy at f: the direction of steepest increase in p-energy. The duality map is a *geometric* object, not merely an algebraic one.

The p-Pairing and Its Interpretation. With the gradient in hand, we can now answer our original question: how does the p-energy $F(f)$ change under a perturbation by g? The first-order variation is[3]:

$$dF(f; g) = \langle \nabla F(f), g\rangle_{\mathrm{dual}} = \int \nabla F(f) \cdot g$$

Substituting $\nabla F = J_p(f)$, this defines a natural pairing:

$$\langle f, g\rangle_p := \int J_p(f) \cdot g$$

Note that it is important to pass through $J_p(f)$ since $\int fg$ need not be finite in general. However, by Hölder's inequality:

$$\int |J_p(f) \cdot g| < \infty$$

This pairing has a precise geometric interpretation. The gradient $J_p(f)$ is a cotangent vector—a 1-form $dF|_f \in T_f^* L^p$—and the perturbation g is a tangent vector $g \in T_f L^p$. Thus the pairing is the evaluation of a 1-form on a tangent vector:

$$\langle f, g\rangle_p = dF|_f(g) \;\in\; T_f^* L^p \times T_f L^p \to \mathbb{R}$$

This canonical evaluation pairing between the tangent and cotangent spaces exists in *every* geometric setting without any choice of metric. What J_p adds is the ability to *produce* a specific cotangent vector from a tangent vector, turning the pairing from a bilinear form on $T_f L^p \times T_f^* L^p$ into a (generally nonlinear) form on $T_f L^p \times T_f L^p$—that is, internalizing the measurement within the space itself.

The Infinitesimal Meaning of V^*. We are now in a position to make a precise statement about the geometric content of the dual space. In a bare vector space without additional structure, the dual V^* is merely an algebraic object—a collection of linear maps with no geometric significance beyond probing linearly. However, in L^p equipped with its norm, the identity $J_p = \nabla F$ reveals that *every element of* $(L^p)^*$ *arises as the gradient of the energy functional*—a differential $dF|_f$. This endows $(L^p)^*$ with genuine infinitesimal meaning: its elements are not just linear maps, but *cotangent vectors* that encode the first-order sensitivity of the geometry.

Crucially, cotangent vectors are a different species of infinitesimal object than tangent vectors. A tangent vector $g \in T_f L^p \cong L^p$ represents an *infinitesimal direction of motion*—it answers "which way?". A cotangent vector $\varphi \in T_f^* L^p \cong L^q$ represents an *infinitesimal measurement*—it answers "how fast is this quantity changing?". These are related by the evaluation pairing $\varphi(g) = dF|_f(g)$, but they

[3]The dF here can be thought of as the Gâteaux derivative, commonly defined for functions between locally convex topological vector spaces.

transform in opposite ways: tangent vectors push forward under maps, while cotangent vectors pull back. The duality map $J_p : T_f L^p \to T_f^* L^p$ provides a (nonlinear) identification between these two types of infinitesimal data, allowing us to transfer the directional interpretation onto the dual space. However, this transfer *depends essentially on the norm geometry*—it is canonical only for $p = 2$.

Geometrically, if we view the identity map $f \mapsto f$ as the radial vector field (a *section* of the tangent bundle TL^p), then the duality map J_p transforms this into a covector field (a *section* of the cotangent bundle T^*L^p). In the language of Riemannian geometry, this transformation is the *musical isomorphism* $\flat$ (flat), which maps a vector field X to a 1-form $X^\flat$ by lowering indices via the metric tensor. The four corollaries that follow are all consequences of this single geometric fact.

Corollary 1: Nonlinearity of the Musical Isomorphism. The duality map is *not* linear:

$$J_p(2f) = |2f|^{p-1}\mathrm{sgn}(2f) = 2^{p-1}|f|^{p-1}\mathrm{sgn}(f) = 2^{p-1}J_p(f)$$

This is only linear in the special case of $p = 2$, since:

$$J_2(nf) = |nf|^{2-1}\mathrm{sgn}(nf) = nJ_2(f)$$

This distinction is profound. In a general Banach space ($p \neq 2$), the musical isomorphism is *nonlinear*: the operation of lowering an index depends on the magnitude of the vector itself, meaning the "metric" changes as we move through the space. However, when $p = 2$, this map is linear. This implies a linear bundle isomorphism $TL^2 \cong T^*L^2$, effectively trivializing the distinction between vectors and covectors. Only in L^2 is the musical isomorphism a linear operator, meaning the "force" required to correct an error is linearly proportional to the error itself.

Corollary 2: Non-Symmetry of the Pairing. The pairing $\langle f, g\rangle_p$ is *not* symmetric for $p \neq 2$. This is an immediate consequence of the nonlinearity of J_p: the 1-form $dF|_f$ evaluated on g has no reason to equal the 1-form $dF|_g$ evaluated on f, since the gradient depends nonlinearly on its base point. If however $p = 2$, then $J_p(x)$ maps identically to itself and we recover the symmetric form $\int fg$. This asymmetry is the first signal that the geometry of L^p is *not flat* for $p \neq 2$.

Corollary 3: Failure of Rotational Invariance. This lack of symmetry in the pairing $\langle f, g\rangle_p$ is directly linked to the fact that L^2 is the unique L^p space that is *rotationally invariant.* Geometrically, rotational invariance means that the properties of a vector depend only on its length, not on its alignment with the coordinate axes. Mathematically, this requires that the duality map commutes with rotations. If R is a rotation operator (an isometry), we expect the "normal vector" of the rotated point to be the rotated "normal vector" of the original point:

$$J_p(Rf) \stackrel{?}{=} R(J_p(f))$$

In L^2, since J_2 is linear (essentially the identity), this holds trivially: $J_2(Rf) = Rf = R(J_2(f))$. However, for $p \neq 2$, the map $f \mapsto |f|^{p-1}\mathrm{sgn}(f)$ applies a non-linear distortion component-wise. Rotating the vector mixes the components, and applying the power law *after* mixing is not the same as applying it *before.*

To see this concretely, consider ℓ^3 ($p = 3$) in $\mathbb{R}^2$ and let $f = (1, 0)$. If we first compute the functional and then rotate by $\pi/4$ (45°), we get:

$$R(J_3(f)) = R(1^2, 0^2) = R(1, 0) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

However, if we rotate the vector first, the components become $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$. Applying the duality map J_3 (squaring components) yields:

$$J_3(Rf) = J_3\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

Since $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) \neq (\frac{1}{2}, \frac{1}{2})$, we have $J_p(Rf) \neq R(J_p(f))$. This confirms that for $p \neq 2$, the geometry is *anisotropic*: the shape of the unit ball looks different depending on the angle from which you view it.

Corollary 4: Birkhoff-James Orthogonality. With this understanding of the position-dependent metric, we can clarify the notion of orthogonality. In Euclidean geometry, orthogonality is symmetric ($x \perp y \iff y \perp x$) and additive. In general Banach spaces, without a global bilinear form, we must rely on *Birkhoff-James orthogonality*: we say $x \perp y$ if moving in the direction of y does not decrease the length of x ($\|x + \lambda y\| \geq \|x\|$). Geometrically, this means the line passing through x in the direction of y is tangent to the unit sphere at x.

Analytically, utilizing the duality map J_p, this condition is equivalent to requiring that the tangent vector y be annihilated by the cotangent vector generated by x:

$$\langle J_p(x), y \rangle = 0$$

This is precisely the statement that y lies in the kernel of the 1-form $dF|_x$—a natural geometric condition. However, it reveals the fundamental asymmetry in non-Hilbert geometry. In L^2, because J_2 is linear, the condition is symmetric. However, in L^p ($p \neq 2$), the non-linearity of J_p breaks reciprocity. The set of vectors orthogonal to x (kernel of $J_p(x)$) does not align symmetrically with the set of vectors orthogonal to y. Consequently, it is common to have $x \perp y$ but $y \not\perp x$. This lack of symmetry illustrates why Hilbert spaces are the unique setting where Euclidean geometric intuition remains valid.

The Riemannian Perspective. The corollaries above all concern the first derivative of the energy (the gradient J_p). The full geometry of the space, however, is determined by how this gradient *changes*—the second derivative, which acts as the metric tensor. In this framework, the metric tensor $g_{ij}(x)$ at a point x dictates how to measure the length of tangent vectors. In our context, this metric is generated by the Hessian of the energy function $F(x) = \frac{1}{p}\|x\|_p^p$. For L^2, this Hessian is proportional to the identity operator everywhere; the metric is *constant* (flat). However, for L^p ($p \neq 2$), the second derivative depends on position x, specifically scaling as $|x|^{p-2}$. This implies that the "metric" changes as we move around the space.

It is worth noting why the theory breaks down at the boundary cases. For L^1, the unit ball is a diamond with corners; at these corners, the gradient is not unique (it is a set-valued subgradient). For L^∞, the unit ball has flat faces; on these faces, the gradient is non-unique or vanishes in certain directions. This lack of smoothness (L^1) or rotundity (L^∞) prevents a well-defined duality map, leaving L^p ($1 < p < \infty$) as the sweet spot for smooth geometry.

Topological vs. Geometric Structure. To visualize the practical consequences of this non-flat geometry, it is helpful to consider the "shape" of the unit ball in infinite dimensions. In finite dimensions, all norms are equivalent, meaning the unit ball of one norm can always be fit inside a scaled unit ball of another. In infinite dimensions, this equivalence breaks. The curvature of the norm dictates how "energy" is allowed to concentrate without bound. The L^∞ unit ball acts like a strict box, while the L^1 unit ball allows for infinitely long thin spikes.

Ideally, we would like to know if these distortions are topological or "merely" geometric. This is answered by the *Mazur Map* $M_{p,q} : L^p(B) \to L^q(B)$ where B is the unit ball. The map:

$$M_{p,q}(f) = \operatorname{sgn}(f) \cdot |f|^{p/q}$$

shows that all $L^p(B)$ spaces are *uniformly homeomorphic* to each other (not just homeomorphic as theorem 2.5.5 shows). This means that we can smoothly deform the "spikes" of L^1 into the "sphere" of L^2. However, while the Mazur map preserves the topology, it destroys the linear structure (additivity). The Hilbert space L^2 is the unique fixed point where this topological structure aligns perfectly with the linear algebraic structure (where the Mazur map would be the identity).

This geometric disparity explains why convergence depends on the norm. Convergence $f_n \to 0$ is not about movement, but about entering arbitrarily small ϵ-balls. Consider the sequence $f_n(x) = x^n$ on $[0, 1]$. In the L^1 geometry, this sequence travels down one of the "spikes" of the unit ball: the functions get thinner and thinner, their area shrinking to zero despite their height remaining constant. In the L^∞ geometry, however, these functions are trapped on the surface of the unit sphere; they cannot enter the ϵ-ball because the L^∞ "box" does not extend down the "spikes" of concentration.

Consequences for Hilbert Spaces. The unique geometry of Hilbert spaces—specifically self-duality, isotropy, and the inner product—unlocks powerful results unavailable in general Banach spaces. The preceding analysis explains *why*: in L^2, the duality map J_2 is the identity, making the musical isomorphism linear, the pairing symmetric, and the geometry flat. The following are key consequences we will explore:

1. *The Projection Theorem (Best Approximation):* In general Banach spaces, finding the element in a subspace M closest to a point x is difficult (the closest point might not be unique, e.g., the flat edge of L^1). In Hilbert spaces, the strict convexity and "roundness" of the unit ball guarantee that for any closed subspace M, there exists a *unique* element $P_M x$ minimizing the distance $\|x - m\|$. Furthermore, the error vector $x - P_M x$ is orthogonal to the subspace.

2. *The Riesz Representation Theorem (Self-Duality):* The duality map $J : H \to H^*$ is linear and bijective. This means we can completely identify linear functionals with vectors. Every continuous linear functional φ acts exactly like an inner product: $\varphi(x) = \langle x, y\rangle$ for a unique y. This collapses the study of H^* to the study of H.

3. *Orthonormal Bases and Fourier Series:* In L^p, we may have a Schauder basis[4], but determining the coefficients is non-trivial. In Hilbert spaces, orthogonality allows us to decouple coefficients. This is key for the Fourier transform, where any element is reconstructed by projection: $x = \sum \langle x, e_n\rangle e_n$. Additionally, the *Parseval Identity* allows us to measure size by summing squared coefficients: $\|x\|^2 = \sum |\langle x, e_n\rangle|^2$.

4. *Existence of Adjoints:* In Banach spaces, the adjoint of $T : X \to Y$ maps $Y^* \to X^*$. Because Hilbert spaces are self-dual, we can pull this operator back to the original space. Thus, for every bounded operator T on a Hilbert space, there is a unique adjoint T^* on the *same space* satisfying $\langle Tx, y\rangle = \langle x, T^* y\rangle$. This is foundational for spectral theory.

In summary, while Banach spaces provide the necessary topological framework (completeness), Hilbert spaces add the rigid geometric structure required to generalize linear algebra to infinite dimensions.

[4] A sequence $\{b_n\}$ where every $v \in V$ has a unique expansion $v = \sum \alpha_n b_n$.

The duality map J_p, understood as the gradient of the p-energy, reveals the precise mechanism: it is the *cotangent structure* of the space—the infinitesimal measurement theory—that determines the geometry. The canonical evaluation pairing between tangent and cotangent vectors is the universal source of "derivative-like" phenomena; J_p is the geometric data that internalizes this pairing within the space itself. In L^2 alone, this internalization is linear, making vectors and covectors—directions and measurements—two aspects of a single object.

5.1 Basic definition and properties

Definition 5.1.1: Inner Product Space

Let H be a complex vector space with an inner product $\langle \cdot, \cdot \rangle : H \times H \to \mathbb{C}$ where

1. $\langle x, y \rangle = \overline{\langle y, x \rangle}$
2. $\langle x + y, z \rangle = \langle x, z \rangle + \langle x, z \rangle$
3. $\langle ax, y \rangle = a\langle x, y \rangle$, $a \in \mathbb{C}$
4. $\langle x, x \rangle \in (0, \infty)$ for all nonzero $x \in H$
5. $\langle x, x \rangle = 0$ if and only if $x = 0$

We'll define $\|x\|_2 = \sqrt{\langle, x, x \rangle}$ where the 2 subscript hints that this will become the L^2-norm and is added to not hold ambiguity with the L_1 norm. This is naturally a norm since:

1. $\|x\|_2 = \sqrt{\langle x, x \rangle} = 0$ if and only if $x = 0$
2. $\|ax\|_2 = \sqrt{\langle ax, ax \rangle} = \sqrt{a\overline{a}\langle x, x \rangle} = \sqrt{a^2\langle x, x \rangle} = |a|\sqrt{\langle x, x \rangle} = |a|\|x\|_2$

Proving the triangle inequality is a bit more difficult: IF we square both sides, we it is equivalent to show:

$$\|x + y\|_2^2 \leq \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2$$

expanding out the left hand side, we get:

$$\|x + y\|_2^2 = \|x\|_2^2 + 2\text{Re}\langle x, y \rangle + \|y\|_2^2$$

so we have to show that $\text{Re}\langle x, y \rangle \leq \|x\|_2\|y\|_2$. In fact, a bit more is true: $|\langle x, y \rangle| \leq \|x\|_2\|y\|_2$. Though we are using this inequality to prove the triangle inequality, notice that this in fact gives us a general bound on the value of the inner product! This inequality is one of the central tools used in the study of Hilbert spaces:

Theorem 5.1.2: Cauchy-Schwartz Inequality

$$|\langle x, y \rangle| \leq \|x\|_2\|y\|_2$$

***Proof*:**
If $\|y\|_2 = 0$ then $y = 0$, and then $|\langle x, 0\rangle| = 0$, so assume $\|y\|_2 \neq 0$. Then for all $t \in \mathbb{R}$

$$\begin{aligned} 0 &\leq \langle x - ty, x - ty\rangle \\ &= \|x\|_2^2 + 2t\mathrm{Re}\langle x, y\rangle + t^2\|y\|_2^2 \\ &= \|x\|_2^2 + \frac{2\mathrm{Re}\langle x, y\rangle^2}{\|y\|_2^2} + \frac{\mathrm{Re}\langle x, y\rangle^2}{\|y\|_2^2} \qquad\qquad t = \frac{\mathrm{Re}\langle x, y\rangle}{\|y\|_2^2} \\ \Rightarrow |\mathrm{Re}\langle x, y\rangle| &\leq \|x\|_2\|y\|_2 \end{aligned}$$

To finish the proof, pick $\alpha \in \mathbb{C}$ with $|\alpha| = 1$ such that

$$|\mathrm{Re}(\alpha\langle x, y\rangle)| = |\langle x, y\rangle|$$

Notice this does not break the inequality, and so we can repeat the entire process but with α taken into account with t, and so:

$$|\langle x, y\rangle \leq \|x\|_2\|y\|_2$$

as we sought to show

With this, the rest of the proof that $\| - \|_2$ induced by $\langle -, -\rangle$ is a norm follows

Theorem 5.1.3: Triangle Inequality For Induced Norm

$$\|x + y\|_2 \leq \|x\|_2 + \|y\|_2$$

***Proof*:**

$$\begin{aligned} \|x + y\|_2^2 &= \langle x + y, x + y\rangle \\ &\leq \|x\|_2^2 + 2\mathrm{Re}\langle x, y\rangle + \|y\|_2^2 \\ &\leq \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2 \\ &= (\|x\|_2 + \|y\|_2)^2 \end{aligned}$$

The Cauchy-Shwartz inequality is the tool we use to bring in geometry as we know it into Hilbert space. For example, in real inner product spaces, we will define the angle between two vectors to be θ where θ satisfies:

$$\langle u, v\rangle = \cos(\theta)\|u\|_2\|v\|_2$$

Many other cool properties of the Cauchy-Shwartz inequality can be found here: (commented)

Now, notice we can define a metric by doing $\rho(x, y) = \|x - y\|_2$:

1. $\rho(x, x) = \|x - x\|_2 = 0$ if and only if $x - x = 0$ so $x = x$
2. $\rho(x, y) = \|x - y\|_2 \geq 0$
3. $\rho(x, y) = \|x - y\|_2 = \|x - y + y - z\|_2 \leq \|x - y\|_2 + \|y - z\|_2 = \rho(x, y) = \rho(y, z)$

Thus, we can define a natural topology on any pre-Hilbert space. If that topology is complete, we give it a name:

Definition 5.1.4: Hilbert Space

Let H be an inner product space. Then H is a Hilbert space if it is *complete* with respect to the induced metric.

Example 5.1.5: Hilbert Spaces

1. $\mathbb{R}^n$ is a (real) Hilbert space with the inner product being:

$$\langle x, y\rangle = \sum_i x_i y_i$$

   verifying this is a Hilbert space is left as an exercise.

2. $L^2_\mu(X)$ for a complex measure μ is a Hilbert space with the inner product being:

$$\langle f, g\rangle_{L^2} = \int_X f(x)\overline{g(x)}d\mu(x)$$

   To show it's well-defined, recall that if $a, b \geq 0$:

$$2|ab| = 2|a||b| \leq |a|^2 + |b|^2$$

   Thus:

$$|f(x)\overline{g(x)}| = |f(x)||g(x)| \leq |f(x)|^2 + |g(x)|^2$$

   As integrating keeps inequality when integrating positive functions:

$$\|f\overline{g}\|_1 \leq \int_X |f(x)|^2 + \int_X |g(x)|^2 < \infty$$

   showing it is well-defined. Alternatively, using Cauchy-Schwartz:

$$\|f\overline{g}\|_1 \leq \|f\|_2\|g\|_2 < \infty$$

3. As an exercise, show that $\ell^2(\mathbb{Z})$ is a Hilbert Space.

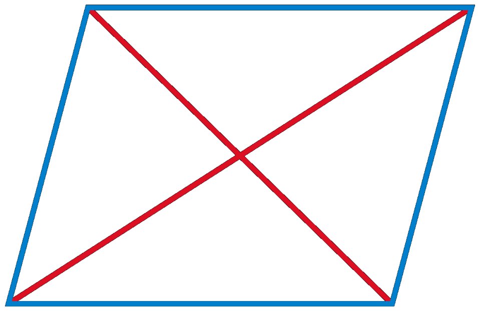
Hilbert spaces are very euclidean in nature, for example:

Theorem 5.1.6: Parallelogram Law

Let $\|-\|$ be a norm. Then $\|-\|$ is induced by an inner product if and only if:

$$\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

Visually, this means that the "area" of a parallelogram is can be measured in two ways:



***Proof*:**

$$
\begin{aligned}
\|x+y\|^2 + \|x-y\|^2 &= \langle x+y, x+y\rangle + \langle x-y, x-y\rangle \\
&= 2\langle x,x\rangle + \langle x,y\rangle - \langle x,y\rangle + \langle y,x\rangle - \langle y,x\rangle + 2\langle y,y\rangle \\
&= 2\langle x,x\rangle + 2\langle y,y\rangle \\
&= 2\|x\|^2 + 2\|y\|^2
\end{aligned}
$$

It turns out that a norm satisfying the parallelogram law will in fact define an inner product. This shows how norms define geometries that are more general than euclidean:

Example 5.1.7: Exploring Parallelogram Law

1. (The polarization Identity)$\mathcal{H}$ be a Hilbert space. For any $x, y \in \mathcal{H}$,

$$\langle x,y\rangle = \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2)$$

Proof. This is simply a matter of computation:

$$
\begin{aligned}
&\frac{1}{4}(\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2) \\
&= \frac{1}{4}(\langle x+y, x+y\rangle - \langle x-y, x-y\rangle + i\langle x+iy\rangle - i\langle x-iy, x-iy\rangle) \\
&= \frac{1}{4}(\langle x,x\rangle + \langle x,y\rangle + \langle y,x\rangle + \langle y,y\rangle - (\langle x,x\rangle - \langle x,y\rangle - \langle y,x\rangle + \langle y,y\rangle) \\
&\quad + i(\langle x,x\rangle + \langle x,iy\rangle + \langle iy,x\rangle + \langle iy,iy\rangle - (\langle x,x\rangle - \langle x,iy\rangle - \langle iy,x\rangle + \langle iy,iy\rangle)) \\
&= \frac{1}{4}(2\langle x,y\rangle + 2\langle y,x\rangle + i(2\langle x,iy\rangle + 2\langle iy,x\rangle)) \\
&= \frac{1}{4}(2\langle x,y\rangle + 2\langle y,x\rangle + i(-i)2\langle x,y\rangle + 2ii\langle y,x\rangle) \\
&= \frac{1}{4}(2\langle x,y\rangle + 2\langle y,x\rangle + 2\langle x,y\rangle - 2\langle y,x\rangle) \\
&= \frac{1}{4}4\langle x,y\rangle \\
&= \langle x,y\rangle
\end{aligned}
$$

$\square$

2. If $\mathcal{H}'$ is another Hilbert space, a linear map from $\mathcal{H}$ to $\mathcal{H}'$ is unitary if and only if it is isometric and surjective

Proof. Let U be unitary so that $\langle Ux, Uy\rangle_2 = \langle x, y\rangle_1$ for any $x, y \in \mathcal{H}$. Letting $x = y$ gives us $\|Ux\|_2 = \|x\|_1$, giving us that it's an isometry. Since U is invertible, it is surjective, and so U is a surjective isometry.

Conversely, let U be a surjective isometry. We need to show its invertible (it suffices to show it's injective), an isometry, and that U and U^{-1} are bounded. Clearly, U is bounded since $\|U(x)\| \leq 1\|x\|$. For injectivity, let $Ux = Uy$, so $Ux - Uy = 0$. Taking the norm on both sides, we get:

$$0 = \|0\|_2 = \|Ux - Uy\|_2 = \|U(x-y)\|_2 = \|x-y\|_1$$

where the last equality is due to U being an isometry. Since $\|z\| = 0$ if and only if $z = 0$, we have $x - y = 0$, i.e. $x = y$, proving that U is injective. Along with surjectivity, that makes U invertible. With this, we can also show that U^{-1} is bounded: for any $y \in \mathcal{H}'$, there exists an $x \in \mathcal{H}$ such that $U(x) = y$. Thus:

$$\|U^{-1}(y)\| = \|x\| \stackrel{!}{=} \|U(x)\| = \|y\|$$

where $\stackrel{!}{=}$ comes from U being an isometry, showing that $\|U^{-1}(y)\| \leq 1 \cdot \|y\|$, making it also bounded.

Finally, to show that $\langle Ux, Uy\rangle_2 = \langle x, y\rangle_1$, we will use part (a):

$$\begin{aligned}
\langle x, y\rangle &= \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2) \\
&\stackrel{!}{=} \frac{1}{4}(\|U(x+y)\|^2 - \|U(x-y)\|^2 + i\|U(x+iy)\|^2 - i\|U(x-iy)\|^2) \\
&= \frac{1}{4}(\|U(x)+U(y)\|^2 - \|U(x)-U(y)\|^2 + i\|U(x)+iU(y))\|^2 - i\|U(x)-iU(y)\|^2) \\
&= \langle U(x), U(y)\rangle
\end{aligned}$$

where $\stackrel{!}{=}$ comes from U being an isometry. Thus:

$$\langle Ux, Uy\rangle = \langle x, y\rangle$$

As we sought to show. $\square$

Proposition 5.1.8: Inner Product is Continuous

Let V be a Hilbert space. Then $\langle -, -\rangle : V \times V \to \mathbb{R}$ (or $\mathbb{C}$) is continuous

***Proof*:**
As $V \times V$ has the product topology, it is enough to show that if $x_n \to x$ and $y_n \to y$ in V then

$$\langle x_n, y_n\rangle \to \langle x, y\rangle$$

Take:

$$\begin{aligned}
|\langle x,y\rangle - \langle x_n,y_n\rangle| &= |\langle x,y\rangle - \langle x_n,y\rangle + \langle x_n,y\rangle - \langle x_n,y_n\rangle| \\
&= |\langle x-x_n,y\rangle + \langle x_n,y-y_n\rangle| \\
&\leq |\langle x-x_n,y\rangle| + |\langle x_n,y-y_n\rangle| \qquad \leq \|x-x_n\|\|y\| + \|x_n\|\|y-y_n\| \\
&\to 0
\end{aligned}$$

where the last line comes since $x_n \to x$ and $y_n \to y$, completing the proof.

The next concept we consider is a way of defining two vectors being somehow "unrelated" or going in some notion of "direction" that doesn't give information for the other vector:

Definition 5.1.9: Orthogonal

We say that x is *orthogonal* to y if $\langle x,y\rangle = 0$. We will write $x \perp y$

If two vectors are orthogonal, the triangle inequality becomes instant:

Theorem 5.1.10: Pythagoras

$$x \perp y \Rightarrow \|x+y\|^2 = \|x\|^2 + \|y\|^2$$

***Proof*:**

$$\|x+y\|^2 = \langle x+y, x+y\rangle = \|x\|^2 + \|y\|^2$$

We might wonder if we take $E \subseteq H$, can we find all vectors in H that are somehow "dual" to E? Define it's *orthogonal compliment* to be:

$$E^\perp = \{x \in H \mid \langle x,y\rangle = 0 \ \forall y \in E\}$$

Notice that E is *always* a closed subspace:

Proof. The fact it's a subspace follows from linearity of the second component.t

for closed, suppose there is a sequecne $\{x_n\} \subseteq E^\perp$ where $x_n \to x \in H$. Then:

$$|\langle x,y\rangle| \leq |\langle x-x_n,y\rangle| \leq \|x-x_n\|\|y\| \to 0 \quad \text{as } n \to \infty$$

□

We can also find closed (and not closed) subspace of hilbert spaces:

Example 5.1.11: Subspaces Examples

1. If $\mathcal{H}$ is a finite-dimensional Hilbert space, then all subspace are closed
2. If $\mathcal{H}$ is infinite-dimensional, consider $\mathcal{H} = L^1([0,1])$ with the L^2 inner product. and take $P([0,1]) \subseteq L^1([0,1])$ to be the subspace of all polynomial functions. This is is certainly not equal to $L^1([0,1])$, however polynomial functions are dense in $L^1([0,1])$ (they can arbitraily

approximate continuous bump characteristic functions, and hence simple functions). However, their closure is $L^1([0,1])$, showing that $P([0,1])$ cannot be closed.

3. Assume $E \subseteq \mathcal{H}$ is an open subspace. Then since $0 \in E$, there exists a $r > 0$ such that

$$B_r(0) \subseteq E$$

However, for any nonzero element $x \in \mathcal{H}$,

$$\frac{r}{2\|x\|} x \in B_r(0) \subseteq E$$

Since E is closed under scalar multiplication, this shows that $x \in E$, so $\mathcal{H} \subseteq E$, so $\mathcal{H} = E$, showing that the only open subspace of $\mathcal{H}$ is $\mathcal{H}$ itself.

Thus, we are interested in closed subspace since are closed under limits, meaning we can use our tools from analysis to get the following decomposition result:

Theorem 5.1.12: Hilbert Space Decomposition

Let $E \subseteq \mathcal{H}$ be a closed subspace. Then:

$$\mathcal{H} = E \oplus E^\perp$$

In particular, for any $x \in H$, there is a unique $y \in E$ and $z \in E^\perp$ such that $x = y + z$.

***Proof*:**
Let $x \in \mathcal{H}$, and let $\delta = \inf\{\|x-y\| = \rho(x,y) \mid y \in E\}$ (which exists by completeness of $\mathbb{R}^n$ and $\mathcal{H}$). let $\{y_n\} \subseteq E$ be a sequence such that $\|x - y_n\| \to \delta$. We will show that $\{y_n\}$ is a cauchy sequence, and then find a limit point for it. By the parralelogram law:

$$2(\|x-y_n\| + \|x-y_m\|) = \|y_n - y_m\|^2 + \|(y_n+y_n) - 2x\|^2$$

Since $y_n + y_m \in E$, for sufficinetly large N so that $n, m > N$:

$$\begin{aligned}\|y_n - y_m\|^2 &= 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\|\frac{1}{2}(y_n+y_n) - x\|^2 \\ &\leq 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\delta^2\end{aligned}$$

and as $n, m \to \infty$, this sequence goes to $4\delta^2 - 4\delta^2 = 0$, and so $\|y_n - y_m\| \rho(y_n, y_m) \to 0$, meaning $\{y_i\}_i$ is a cauchy sequence. Take $y \lim y_n$. Then since E is closed, $y \in E$. By construction $\|x-y\| = \delta$. Now let $z = x - y$.

I claim that $z \in E^\perp$, that is $\langle z, u\rangle = 0$ for all $u \in E$. So, consider $\langle u, z\rangle$ for any $u \in E$. If $\langle u, z\rangle$ is complex, multiply it by a [complex] unit-length scalar to make it real ($\langle u, z\rangle = 0$ if and only if $a\langle u, z\rangle = 0$). With the fact that it's real, we will do a sneaky trick where we take advantage of optimization of differentiable functions. In particular, let

$$f : \mathbb{R} \to \mathbb{R} \quad f(t) = \|z + tu\|^2 = \|z\|^2 + 2t\langle z, u\rangle + \|u\|^2$$

f is certainly a differentiable function, being a polynomial with a square-root on $\mathbb{R}$. Furthermore, $z = x - y$ is the minimal value of $\|x-y\|$, and so the minimal value of $\|x-y\|^2$. It follows that $z + ut$ is not minimal, and so this equation is has a minimum when $t = 0$. Thus, we get $0 = f'(0) = 2\langle u, z\rangle$, implying that $\langle u, z\rangle = 0$. Since u was arbitrary, we get that $z \in E^\perp$.

To show that z is the minimal distance from x, let's say there was another $z' \in E^\perp$ that satisfied the equation. Then by the Pythagorean theorem:

$$\begin{aligned} \|x - z'\| &= \|(x - z) + (z - z')\|^2 \\ &= \|x - z\|^2 + \|z - z'\|^2 && x - y \in E, z - z' \in E^\perp \\ &\geq \|x - z\|^2 \end{aligned}$$

So equality holds if and only if $z = z'$, meaning z is of minimal distance to x. A similar reasoning shows the same for y.

To show uniqueness, let's say $y + z = x = y' + z'$. Then $y - y' = z - z' \in E \cap E^\perp = \{0\}$, so $y - y' = z - z' = 0$, which can be used to show that $y = y'$ and $z = z'$, completing the proof.

We said that orthogonal vector are somehow dual to each other. However, there is the more classical notion of "dual" vector spaces. Let $\mathcal{H}^*$ be set of all linear functions from $\mathcal{H}$ to the underlying field (usually $\mathbb{C}$, but sometimes $\mathbb{R}$); such linear functions are called functionals. It turns out that all linear functionals can be written in the form $f_y(x) = \langle x, y\rangle$, and we can define a norm on $\mathbb{H}^*$ such that $\|f_y\| = \|y\|$. Thus, $y \mapsto f_y$ will define a conjugate-linear isometry from $\mathcal{H}$ to $\mathcal{H}^*$. The following theorem shows tha this map is surjective:

Theorem 5.1.13: Natural Isomorphism To $\mathcal{H}^*$ (Riesz Representation)

Let $f \in \mathcal{H}^*$. Then there exists a $y \in \mathcal{H}$ such that $f(x) = \langle x, y\rangle$ for all $x \in X$.

***Proof*:**
It easy easy to check uniqueness: if $\langle x, y\rangle = \langle x, y'\rangle$ for all x, then let $x = y - y'$. Manipulating the equation, we get that $\|y - y'\| = 0$, so $y = y'$.

If f is the zero functional, then take $y = 0$. If otherwise, then let $\mathcal{M} = \{x \mid f(x) = 0\}$ be the kernel of f. It is clearly a subspace, and any squence will also necesarily converge within $\mathcal{M}$, and so it is also closed. Furthermore, $\mathcal{M}$ is a proper subspace, since $f \neq 0$. Thus, $\mathcal{M}^\perp \neq \{0\}$ by the previous proof.

Now, pick any $z \in \mathcal{M}^\perp$ with $\|z\| = 1$. Then if we set $u = f(x)z - f(z)x$, we have that $u \in \mathcal{M}$. Hence:

$$0 = \langle u, z\rangle = f(x)\|z\|^2 - f(z)\langle x, z\rangle = f(x) - \langle x, \overline{f(z)}z\rangle$$

re-arrangine and setting $y = \overline{f(z)}z$, we get

$$f(x) = \langle y, z\rangle$$

as we sought to show.

The uniqueness makes this map to be injective, and by linearity of f and the first component of the inner product makes it is an isomorphism. Furthermore, since $\|f_y\| = \|y\|$, limits are also preserved, and so this in fact defines an isomorphism between $\mathcal{H} \cong \mathcal{H}^*$. Notice how this differs from vector-spaces where there is no natural isomorphism between V and V^* (there is one between V and V^{**}).

Since we are working over a vector-space, we should be able to define a basis. Since we are studying infinite dimensional vector-spaces, we often care more about "countable linear combination" rather than "finite linear combination" as we have for vector-spaces. To distinguish the two, we will call *Hamel basis* a set of linearly independent elements for which all *finite linear combinations* can represent every element of the vector space. We will call a *Schauder basis* a set of linearly independent elements for which all *countable inear combinations* can represent every element of the vector space. If every element in a Schauder basis has norm 1 and is orthogonal, we call it an *orthonormal basis.* Since we have an inner-product that is complete, it would be nicest if this vector-space has an orthonormal basis. We strive to this end in the following build-up:

Theorem 5.1.14: Bessel's Inequality

If $\{u_\alpha\}_{\alpha \in A}$ is an orthonormal set in $\mathcal{H}$, then for any $x \in \mathcal{H}$:

$$\sum_{\alpha \in A} |\langle x, u_\alpha \rangle|^2 \leq \|x\|^2$$

As a consequence, $\{\alpha \mid \langle x, u_\alpha \rangle \neq 0\}$ is countable.

***Proof*:**
It suffices to show that $\sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2 \leq \|x\|^2$ for any finite subset $F \subseteq A$, since if it fails for some countable amount, it will fail for some finite amount. Then we see that:

$$\begin{aligned} 0 &\leq \left\| x - \sum_{\alpha \in F} \langle x, u_\alpha \rangle u_\alpha \right\| \\ &= \|x\|^2 - 2\mathrm{Re} \left\langle x, \sum_{\alpha \in F} \langle x, u_\alpha \rangle u_\alpha \right\rangle + \left\| \sum_{\alpha \in F} \langle x, u_\alpha \rangle u_\alpha \right\| \\ &= \|x\|^2 - 2 \sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2 + \sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2 && \text{Pythagorean Theorem} \\ &= \|x\|^2 - \sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2 \end{aligned}$$

and manipulating the final equation we get:

$$\sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2 \leq \|x\|^2$$

as we sought to show

This in particular is important for uncountable sets. In the following we establish when equality holds, and thus when an orthonormal set is an orthonormal basis

Theorem 5.1.15: Orthonormal Basis Condition

Let $\{u_\alpha\}_{\alpha \in A}$ be an orthonormal set in $\mathcal{H}$. Then the following are equivalent:

1. Finite linear combinations of $\{u_\alpha\}$ are dense in $\mathcal{H}$.
2. (Completeness) If $\langle x, u_\alpha \rangle = 0$ for all α, then $x = 0$
3. For each $x \in \mathcal{H}$, $x = \sum_{\alpha \in A} \langle x, u_\alpha \rangle u_\alpha$ where the right hand side sum has only countably many nonzero terms and converges in the norm topology no matter the order of the terms
4. (Parseval's Identity) $\|x\|^2 = \sum_{\alpha \in A} |\langle x, u_\alpha \rangle|^2$ for all $x \in \mathcal{H}$

***Proof*:**

(1) ⇒ (2) Assume $\langle f, u_\alpha \rangle = 0$. As finite linear combinations of $\{u_\alpha\}$ are dense in $\mathcal{H}$, for any $f \in \mathcal{H}$ there is a sequence g_n such that $\|f - g_n\| \xrightarrow{n \to \infty} 0$. As $\langle f, u_\alpha \rangle = 0$, then $\langle f, -g_n \rangle = 0$. By bi-linearity and Cauchy-Schwartz:

$$\|f\|^2 = \langle f, f \rangle = \langle f, f - g_n \rangle \leq \|f\| \|f - g_n\| \xrightarrow{x \to \infty} 0 \tag{5.1}$$

hence $f = 0$

(2) ⇒ (3) As the nonzero portion of the sum is countable, we can reduce to the countable case. For any $f \in \mathcal{H}$, let:

$$S_n(f) = \sum_{k=1}^{n} a_k u_k \qquad a_k = \langle f, u_k \rangle$$

Let's first show $S_n(f) \to g$ for some $g \in \mathcal{H}$, then show $g = f$. First, we show an alternative proof of Bessel's identity with a satisfying trick. Notice that certainly $S_n(f) \perp (f - S_n(f))$. Thus:

$$\|f\|^2 = \|S_n(f)\|^2 + \|f - S_n(f)\|^2 = \sum_{k=1}^{n} a_k u_k + \|f\|^2$$

Taking $n \to \infty$ we get:

$$\|f\| \geq \sum_{k=1}^{\infty} a_k u_k$$

Importantly, $\sum_{k=1}^{\infty}$ *converges*, and hence $S_n(f)$ forms a Cauchy sequence in $\mathcal{H}$, so by completeness $S_n(f) \to g$. To show $f = g$, notice that for any k, for sufficiently large n:

$$\langle f - S_n(f), u_k \rangle = a_k - a_k = 0$$

Thus:

$$\langle f - g, u_k \rangle = 0 \qquad \forall k$$

so by (2), $f - g = 0$, so $f = g$.

(3) ⇒ (4) Assuming (3) so that $= \sum_k^\infty \langle f, u_k\rangle$, then $\|f - S_n(f)\| \to 0$ so plugging this into equation (5.1):

$$\|f\|^2 = \sum_k^\infty |a_k|^2$$

(4) ⇒ (1) Notice that $S_n(f)$ is a finite linear combinations elements u_k, hence using equation (5.1) again we get the desired implication.

Any orthonormal set satisfying these conditions is given a name:

Definition 5.1.16: Orthonormal Basis

Let $\{u_\alpha\}_{\alpha\in A}$ be an orthonormal set. Then if the set satisfies any of the 3 conditions in theorem 5.1.15, it is called an *orthonormal basis*.

An argument similar to the one for vector-spaces gives that every Hilbert spaces has an orthonormal basis, essentially an orthonormal basis is an orthonormal set that is maximal:

Theorem 5.1.17: Hilbert Space Basis

Every Hilbert space has an orthonormal basis.

***Proof*:**
Application of Zorn's lemma: The collection of ordered orthonormal sets ordered by inclusion has a maximal element, and maximality is equivalent to the first property of theorem 5.1.15.

Example 5.1.18: Orthonormal Basis

Given a Hilbert space $L^2(G)$ where $G = \mathbb{R}^n$ or $\mathbb{T}^n$, we would want to find an orthonormal basis to get decomposition of functions. go over all of these!

1. Legendre basis (can be thought of as "global" Taylor approximation) (must add integrate image. Also a good example of orthonormal basis that is not the Fourier one)

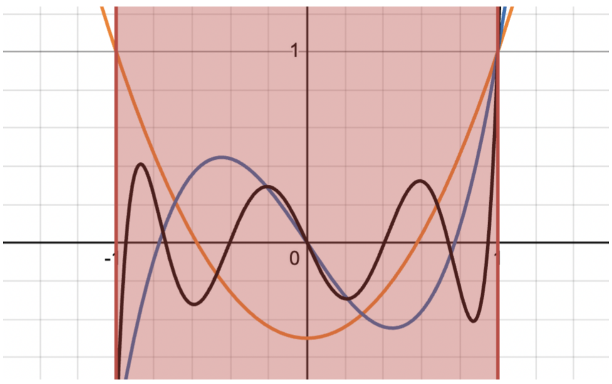


Figure 5.1: Legendre polynomials integrate to 0

2. Hermite basis
3. Jacobi Basis
4. Chebyshev

5.1.19 Basis On Banach Space Without an inner product, we cannot define an orthonormal basis on L^p. We can ask if we can at least define a *Schauder basis*, i..e any $f \in L^p$ is a (potentially) linear of a *single* sequence $\{b_n\}$ with different weights:

$$f = \sum a_n b_n$$

This was actually a very difficult problem and is related to the following question: is every compact operator on a Banach space the limit of finite operators. It isn't too hard to check this is the case in Hilbert spaces (exercise ref:HERE). However, this was proven in the negative by Per Enflo [4] (for a refinement of the proof see [2]). A consequence of this is that not every Banach space has a Schauder basis [1]. Another consequence of this paper for Banach spaces is that not all Banach space embed into $\ell^p(\mathbb{Z})$, $L^p(\mathbb{R}^d)$, or $L^p(K)$ for a compact set K; the family of Banach spaces is in fact larger (ex. $C(K)$ Is a Banach space not belonging to any of these spaces) .

Now that we have established a basis for Hilbert spaces, we will separate them into 3 categories: Those with finite, countable, and uncountable basis. In most cases, we will be working with countable basis due to the following property:

Proposition 5.1.20: Separable Hilbert Spaces

A Hilbert space $\mathcal{H}$ is separable if and only if it has a countable orthonormal basis, in which case every orthonormal basis for $\mathcal{H}$ is countable.

***Proof*:**
p. 176

Though all separable Hilbert spaces are "equivalent" (isometric), it is often the specific manifestation of the Hilbert space that matters as it frames how operators on the space transform functions (ex. the choice of basis, the choice of inner product).

Theorem 5.1.21: Enflo's Rigidity Theorem

A Banach space is uniformly homeomorphic to a Hilbert space if and only if it is linearly isomorphic to a Hilbert space.

***Proof*:**
This was a seminal result by Per Enflo (1969) [3]. It settled the question of whether the "uniform" structure of a space determines its linear structure in the context of Hilbert spaces. Since L^p (for $p \neq 2$) is not isomorphic to L^2, this theorem implies they are not uniformly homeomorphic either.

5.1.22 Hierarchy of Sameness It is useful to summarize how "same" two infinite-dimensional separable spaces can be. Let X and Y be infinite-dimensional separable Banach spaces (e.g., L^p and L^2).

1. Topological Homeomorphism : They are homeomorphic by the *Anderson-Kadec Theorem*, i.e. all such spaces are homeomorphic to $\mathbb{R}^{\mathbb{N}}$. They are topologically indistinguishable.
2. Uniform Homeomorphism: Generally no. By *Enflo's Rigidity Theorem*, if X is a Hilbert space and Y is not linearly isomorphic to it (like $L^p, p \neq 2$), they are *not* uniformly homeomorphic. *Note:* Their unit balls *are* uniformly homeomorphic via the **Mazur Map**, but this does not extend to the whole space.
3. Linear Isomorphism (The "Algebraic" level): Is there a linear bijection? Depends. L^p is not isomorphic to L^q for $p \neq q$.
4. Isometric Isomorphism (The "Geometry" level): They don't they preserve distance. The geometry of the unit ball (round vs. diamond vs. box) is strictly preserved here.

We next talk about the isomorphisms in the category of Hilbert spaces:

Definition 5.1.23: Unitary Map

Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two Hilbert spaces with inner products $\langle\cdot,\cdot\rangle_1$ and $\langle\cdot,\cdot\rangle_2$, and let $U : \mathcal{H}_1 \to \mathcal{H}_2$. Then U is called an *isometry* if if it is a linear map that preserves inner products:

$$\langle Ux, Uy\rangle_2 = \langle x, y\rangle_1$$

If it is also bijective, it is a *unitary map*.

Note that by taking $x = y$, we get that $\|x\|_1 = \|Ux\|_2$, i.e. it is an isometry. In fact, the converse is true too: every surjective isometry is unitary. With the definition of an isomorphism, we can see that all Hilbert spaces are isomorphic to $\ell^2(A)$ given basis A:

Theorem 5.1.24: Classifying Hilbert Spaces

Let $\{e_\alpha\}_{\alpha\in A}$ be an orthonormal basis for $\mathcal{H}$. Then the correspondence

$$x \mapsto \hat{x} \qquad \hat{x}(\alpha) = \langle x, u_\alpha\rangle$$

is a unitary map from $\mathcal{H}$ to $\ell^2(A)$.

***Proof*:**
We'll show it's a surjective isometry, which is equivalent to it being unitary.

First off, the map $x \to \hat{x}$ is linear since the inner product is linear in each component, and it is an isometry by Parseval identity: $\|x\|^2 = \sum |\hat{x}(x)|^2$.

For surjectivity, let $f \in \ell^2(A)$, so $\sum_{\alpha\in A} |f(\alpha)^2| < \infty$, so the Pythagorean shows that all partial sumns of the series $\sum f(\alpha)u_\alpha$ are Cauchy [a], so $x = \sum_\alpha f(\alpha)u_\alpha$ exists in $\mathcal{H}$ and $\hat{x} = f$.

[a]Note that only countably many terms are nonzero

As an application that ties together several threads from previous chapters, we close off with the proof a theorem of Grothendieck showing that the Hilbert space structure of L^2 imposes rigid constraints on subspaces of L^p. The result says, roughly, that a closed subspace of L^p cannot consist entirely of bounded functions unless it is finite-dimensional. The proof combines the closed graph theorem from the Baire category section (theorem 2.2.10), the L^p inclusion theory (proposition 3.1.13), and the Bessel inequality developed above.

Theorem 5.1.25: Grothendieck's Theorem on Closed Subspaces of L^p

Let $(X, \mathcal{F}, \mu)$ be a finite measure space, i.e. $\mu(X) < \infty$. Suppose that E is a closed subspace of $L^p(X, \mu)$ for some $1 \leq p < \infty$, and that $E \subseteq L^\infty(X, \mu)$. Then E is finite-dimensional.

***Proof*:**
Step 1: The L^∞ norm is controlled by the L^p norm on E. Consider the inclusion $\iota : (E, \|\cdot\|_p) \to L^\infty(X, \mu)$ defined by $\iota(f) = f$. We verify that its graph is closed. Suppose $f_n \to f$ in L^p with $f_n \in E$, and simultaneously $f_n \to g$ in L^∞. Since E is closed in L^p, $f \in E$. Convergence in L^p gives $f_{n_k} \to f$ a.e. along a subsequence, while convergence in L^∞ gives $f_n \to g$ a.e. Hence $f = g$ a.e., showing the graph is closed. By the closed graph theorem (theorem 2.2.10), ι is bounded: there exists $C > 0$ such that:

$$\|f\|_\infty \leq C\|f\|_p \qquad \text{for all } f \in E$$

Step 2: E carries a Hilbert space structure. Since $\mu(X) < \infty$ and $E \subseteq L^\infty$, the L^2 norm is finite on E. We show the L^p and L^2 norms are equivalent on E.

For one direction: $\|f\|_2^2 = \int |f|^2 \leq \|f\|_\infty^2 \mu(X) \leq C^2\|f\|_p^2 \mu(X)$, giving $\|f\|_2 \leq C\mu(X)^{1/2}\|f\|_p$.

For the reverse: if $p \leq 2$, then Hölder directly gives

$$\|f\|_p \leq \mu(X)^{1/p-1/2}\|f\|_2$$

If $p > 2$, then interpolate:

$$\|f\|_p^p \leq \|f\|_\infty^{p-2}\|f\|_2^2 \leq (C\|f\|_p)^{p-2}\|f\|_2^2$$

so that $\|f\|_p^2 \leq C^{p-2}\|f\|_2^2$, i.e. $\|f\|_p \leq C^{(p-2)/2}\|f\|_2$.

In either case, the L^p and L^2 norms are equivalent on E. Since E is complete in $\|\cdot\|_p$, it is complete in $\|\cdot\|_2$, so $(E, \langle\cdot,\cdot\rangle_{L^2})$ is a Hilbert space. Write C' for a constant satisfying $\|f\|_\infty \leq C'\|f\|_2$ for all $f \in E$ (obtained by combining Step 1 with the norm equivalence).

Step 3: Bounding the dimension via Bessel's inequality. Let $\{e_1, \ldots, e_N\}$ be any orthonormal set in $(E, \|\cdot\|_2)$. Note that N is independent of the choice of C'; this will be the key fact to bound $\dim(E)$. For almost every $x \in X$, the evaluation functional $\varphi_x : E \to \mathbb{C}$ defined by $\varphi_x(f) = f(x)$ satisfies:

$$|\varphi_x(f)| \leq \|f\|_\infty \leq C'\|f\|_2$$

so φ_x is a bounded linear functional on the Hilbert space E with $\|\varphi_x\| \leq C'$. By the Riesz representation theorem (theorem 5.1.13), there exists $K_x \in E$ with $\varphi_x(f) = \langle f, K_x\rangle$ and

$\|K_x\|_2 = \|\varphi_x\| \leq C'$. In particular,

$$\varphi_x(e_n) = e_n(x) = \langle e_n, K_x \rangle$$

for each n. Applying Bessel's inequality (theorem 5.1.14) to K_x:

$$\sum_{n=1}^{N} |e_n(x)|^2 = \sum_{n=1}^{N} |\langle K_x, e_n \rangle|^2 \leq \|K_x\|_2^2 \leq C'^2$$

Integrating over X and using $\|e_n\|_2 = 1$:

$$N = \sum_{n=1}^{N} \|e_n\|_2^2 = \int_X \sum_{n=1}^{N} |e_n(x)|^2 \, d\mu \leq C'^2 \mu(X)$$

Since N was the size of an arbitrary orthonormal set, $\dim(E) \leq C'^2 \mu(X) < \infty$.

The element K_x appearing in the proof is a *reproducing kernel*: it allows one to recover pointwise values of functions in E via the inner product. The argument shows that the existence of such a kernel, together with the finiteness of the measure, forces the space to be finite-dimensional. The bound $\dim(E) \leq C'^2 \mu(X)$ is quantitative: the larger the ratio $\|f\|_\infty / \|f\|_2$ can be on E, the fewer orthonormal directions the space can accommodate.

It is worth pausing to appreciate how three seemingly unrelated tools conspire here. The closed graph theorem (theorem 2.2.10) upgrades the set-theoretic inclusion $E \subseteq L^\infty$ into a norm inequality. The L^p interpolation inequalities (proposition 3.1.13) transfer this to equivalence of all L^q norms on E, revealing a hidden Hilbert space. And Bessel's inequality (theorem 5.1.14) converts the uniform pointwise bound into a dimension bound via integration.

5.2 Operators on Hilbert Spaces

In this section, we try to generalize many notion's we've seen before, including:

1. adjoints, self-adjoint, and normal operators
2. sign and absolute value and when can we decompose a function into this form
3. diagonalizable
4. polar decomposition (think of how for every $z \in \mathbb{C}$, $z = \arg(z)|z|$)

Adjoints

Hilbert spaces have very simple dual structure, since $\mathbb{H} \cong \mathbb{H}^*$ in a very natural way with the inner product of $\mathbb{H}$. Thus, the problem of finding the structure of dual spaces is fully finished for hilbert spaces. Due to this very simple dual structure, we will be able to properly define an "opposite" operator given any operator $T : \mathbb{H}_1 \to \mathbb{H}_2$. In essense, there is an interesting symmetry in the structure of any map: Given a map $T : \mathbb{H}_1 \to \mathbb{H}_2$ between two hilbert spaces, there is a natural dual map $T^* : \mathbb{H}_2^* \to \mathbb{H}_1^*$. Hence, we have

$$(-)^* : \mathrm{Hom}(\mathbb{H}_1, \mathbb{H}_2) \to \mathrm{Hom}(\mathbb{H}_2^*, \mathbb{H}_1^*)$$

and since in Hilbert spaces $\mathbb{H}_i \cong \mathbb{H}_i^*$, we may envision this map as "flipping" the direction of arrows in a natural way:

$$(-)^* : \mathrm{Hom}(\mathbb{H}_1, \mathbb{H}_2) \to \mathrm{Hom}(\mathbb{H}_2, \mathbb{H}_1)$$

Not only does this map exist, but it is an isometric isomrophism (in **Ban**)! We will start with the special case of $\mathrm{Hom}(\mathbb{H}, k) = \mathbb{H}^*$. We will find that $(-)^*$ is a map

$$(-)^* : \mathrm{Hom}(\mathbb{H}, \mathbb{H}) \to \mathrm{Hom}(\mathbb{H}, \mathbb{H})$$

The following shows how we can define such a dual and some consequences:

Lemma 5.2.1: Endomorphism and And Inner Products

Let $\mathbb{H}$ be a complex hilbert space. Then there is a bijective isometric correspondence between $\mathrm{End}_k(\mathbb{H})$ and bounded sesquilinear forms on $\mathbb{H}$ given by tne map $T \to B_T$ where

$$B_T(x, y) = \langle x, T(y) \rangle$$

***Proof*:**
Let $T \in \mathrm{End}_k(\mathbb{H})$. Then certainly B_T is a sequilinear form on $\mathbb{H}$ bounded by $\|T\|$ since

$$\begin{aligned} \|B_T\| &= \sup \{ |B_T(x, y)| \mid \|x\| \le 1,\ \|y\| \le 1 \} \\ &= \sup \{ |\langle x, T(y) \rangle| \mid \|x\| \le 1, \|y\| \le 1 \} \\ &\le \|T\| \end{aligned}$$

On the other hand

$$\begin{aligned} \|T(x)\|^2 &= \langle T(x), T(x) \rangle \\ &= B_T(T(x), x) \\ &\le \|B_T\| \|T(x)\| \|x\| \\ &\le \|B_T\| \|T\| \|x\|^2 \end{aligned}$$

showing that $\|T\| \le \|B_T\|$, and so $\|T\| = \|B_T\|$.

Theorem 5.2.2: Existence Of Adjoint

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then there exists a unique T^* in $\mathrm{End}_k(\mathbb{H})$ such that

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle$$

for all $x, y \in \mathbb{H}$. The map $T \mapsto T^*$ is conjugate linear, contravariant with respect to multiplication, an isometry of $\mathrm{End}_k(\mathbb{H})$ onto itself, an involution, and satisfies

$$\|T^* T\| = \|T\|^2$$

***Proof*:**
Fix $T \in \mathrm{End}_k(\mathbb{H})$, and let $B_T(x, y)$ be a continuous sesquilinear form defined as $(x, y) \mapsto \langle T(x), y\rangle$. Then by lemma 5.2.1, there exists a $T^* \in \mathrm{End}_k(\mathbb{H})$ such that $T^* \to B_T$. In particular

$$B_T(x, y) = \langle x, T^*(y)\rangle$$

Thus,

$$\langle T(x), y\rangle = \langle x, T^*(y)\rangle$$

giving us existence of such an operator. We now establish all the properties named in the theorem. By conjugate symmetry

$$\begin{aligned} \langle x, T^{**}(y)\rangle &= \langle T^*(x), y\rangle \\ &= \overline{\langle y, T^*(x)\rangle} \\ &= \overline{\langle T(y), x\rangle} \\ &= \langle x, T(y)\rangle \end{aligned}$$

Thus, T^{**} and T correspond to the same sesquilienar form, thus $T^{**} = T$. This also shows that $T \mapsto T^*$ is an involution. Furthermore:

$$\langle x(ST)^*y\rangle = \langle S(T(x)), y\rangle = \langle T(x), S^*(y)\rangle = \langle x, T^*(S^*(y))\rangle$$

thus $(S \circ T)^* = T^* \circ S^*$ showing $(-)^*$ is a contravariant endofunctor. Since the operator norm is submiltiplicative, we have $\|T^*T\| \leq \|T\|\|T^*\|$. On the other hand, applying $\angle T(x), y\rangle = \langle x, T^*(y)\rangle$ to T^* we get

$$\|T(x)\|^2 = \langle T(x), T(x)\rangle = \langle T^*(T(x)), x\rangle = \|T^*T\|\|x\|^2$$

showing that $\|T\|^2 \leq \|T^*T$
. Combining these we get $\|T\| \leq \|T^*\|$, hence $\|T\| = \|T^*\|$ (since $T^{**} = T$). This finishes the proof.

Note that if $\mathbb{H}$ is a finite dimensional complex Hilbert space, then each operator T has a matrix representation, say (α_{ij}). Then $T^* = (\overline{\alpha_{ji}})$, i.e. $T^* = \overline{T^t}$ where T^t is the transpose given the matrix representation.

Definition 5.2.3: Adjoint And Self-Adjoint

Let $T \in B(\mathbb{H})$ be a continuous linear operator between Hilbert spaces. Then T^* is called the *adjoint* of T. If $T = T^*$, then T is said to be *self adjoint* (or *Hermitian*).

Proposition 5.2.4: Adjoint Properties I

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then

$$\ker(T^*) = (T(\mathbb{H}))^\perp$$

***Proof*:**
Since $\langle T(x), y\rangle = \langle x, T^*(y)\rangle$ we have $y \in \ker(T^*)$ when $y \in (T(\mathbb{H}))^\perp$. Conversely, if $y \in (T(\mathbb{H}))^\perp$, then $T^*(y) \in \mathbb{H}^\perp = \{0\}$.

For ease of reference, we shall list some of the results we have proven in the theorem:

Proposition 5.2.5: Adjoint Properties II

Let $T, S \in B(\mathbb{H})$. Then:

1. $(T + S)^* = T^* + S^*$
2. $(cT)^* = \overline{c}T^*$
3. $(TS)^* = S^*T^*$
4. $(T^*)^* = T$
5. $I^* = I$

***Proof*:**
If you're not convinced of any of these, double check the theorem or try proving it again.

Proposition 5.2.6: Adjoint Properties III

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then the following are equivalent:

1. T is invertible (so $T^{-1} \in \mathrm{End}_k(\mathbb{H})$)
2. T^* is invertible
3. T and T^* are bounded away from zero.
4. T and T^* are injective, and $T(\mathbb{H})$ is closed
5. T is injective and $T(\mathbb{H}) = \mathbb{H}$

***Proof*:**
Since T is invertible, $T^{-1}T = TT^{-1} = I$, and so $(T^{-1})^*$ is the inverse of T^*, giving (1) $\Leftrightarrow$ (2).

For (1) $\Rightarrow$ (3), we say that an operator F is bounded away from zero if

$$\|x\| \le C\|F(x)\|$$

for some constant C. In our case:

$$\|x\| = \|T^{-1} \circ T(x)\| \le \|T^{-1}\|\|T(x)\|$$

Since (1) $\Leftrightarrow$ (2), the same holds for T^*.

For (3) $\Rightarrow$ (4), We have $\|T(x)\| \ge \epsilon\|x\|$ and $\|T^*(x)\| \ge \epsilon\|x\|$ for some $\epsilon > 0$ and all $x \in \mathbb{H}$. Thus certainly both T and T^* are injective: if $T(x) = T(y)$ then

$$0 = \|T(x) - T(y)\| = \|T(x - y)\| \ge \epsilon\|x - y\|$$

hence $\|x-y\|=0$, so $x=y$. Furthermore, $\|T(x)-T(y)\| \geq \|x-y\|$ imples that $T(\mathbb{H})$ is complete, hence a closed subspace of $\mathbb{H}$.

For (4) $\Rightarrow$ (5), by corollary **??** ($\overline{Y}=(Y^\perp)^\perp$) and proposition 5.2.4 we get

$$\overline{T(\mathbb{H})} = (T(\mathbb{H})^\perp)^\perp = (\ker(T^*))^\perp = \{0\}^\perp = \mathbb{H}$$

where the last equality come from T^* bein injective.

Finally, for (5) $\to$ (1), we use the open mapping theorem.

This allows us to generlize this result to the followin

Proposition 5.2.7: Inversing Maps

Let $\mathbb{H}_1$ and $\mathbb{H}_2$ be hilbert spaces and consider an operator $T \in \mathrm{Hom}_k(\mathbb{H}_1, \mathbb{H}_2)$. Then there exists a unique operator $T^* \in B(\mathbb{H}_2, \mathbb{H}_1)$ such that

$$\langle T(x), y\rangle = \langle x, T^*(y)\rangle$$

show that the map $T \to T^*$ is conjugate linear isometry of $\mathrm{Hom}_k(\mathbb{H}_1, \mathbb{H}_2)$ to $\mathrm{Hom}_k(\mathbb{H}_2, \mathbb{H}_1)$ and satisfies

$$\|T^*T\| = \|T^2 = \|TT^*\|$$

***Proof*:**
Let

$$\hat{T} = \begin{pmatrix} 0 & 0 \\ T & 0 \end{pmatrix} \in \mathrm{End}_k(\mathbb{H}_1 \oplus \mathbb{H}_2)$$

By theorem 5.2.2, there is a unique function $\hat{T}^*$ such that

$$\langle \widehat{T}(x_1, x_2), (y_1, y_2)\rangle = \langle (x_1, x_2), \widehat{T}^*(y_1, y_2)\rangle$$

By definition of $\widehat{T}$, we see

$$\widehat{T}^* = \begin{pmatrix} 0 & T^* \\ 0 & 0 \end{pmatrix}$$

Hence we have that T^* is a function with domain and codomain $T^* : \mathbb{H}_2 \to \mathbb{H}_1$. It follows that

$$\langle T(x), y\rangle = \langle x, T^*(y)\rangle$$

and by theorem 5.2.2, T and T^* respect all the desired properties, completing the proof.

For the rest of this seciton, we will focus on $T \in B(\mathbb{H})$ insead of $T \in B(\mathbb{H}_1, \mathbb{H}_2)$. The reason for this is to find suitable decompositino theorems of T generlizing the notion of eigenvalues.

Recall that over finite dimensional vector spaces, $TT^* = T^*T$ implied T was diagonalizable, with the usual special case of $T = T^*$ (self-adjoint) being looked into. We would like to know if this notion of diagonalizability for normal operators remains true for infinite dimensional Hilbert spaces. We shall see that this is not the case, some more conditions is required (for ex. a special case of digaonalizable operators will be comapct normal operators). Before we get into the technicalities, I want to put a word on how I think about $TT^* = T^*T$. In my mind, when two functions commute, it means that one function is somehow "invariant" with respect to the other. Think of PDE's with equations that commute with the differential operator that makes it invariant to the Lorenzian metric. In this case, I like to think of T being somehow "symmetric". Since $TT^*(x) = T^*T(x)$ for all $x \in \mathbb{H}$, and T^* is in some ways the dual (think of ref:HERE, the generalization), we see that T is "invariant" to the effect of the dual T^*. We shall see in a moment that the invariant to the dual is translated to T and T^* being metrically identical. In the simpleset case of $T = T^*$, it is evident why this is the case. If $T = T^*$, we can deepen our intuition with complex numbers: if we consider $T : \mathbb{C} \to \mathbb{C}$ where $T(z) = z$, then $T^*(z) = \overline{z}$. Hence, $T = T^*$ if and only if $T(\mathbb{C}) \subseteq \mathbb{R}$, since only the real number are invariant under conjugation.

Definition 5.2.8: Normal Operator

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then T is said to be *normal* if it commutes with its adjoint, i.e. if $T^*T = TT^*$

As an immedaite consequence, we have

$$\|T(x)\| = \langle T^* \circ T(x), x\rangle^{1/2} = \langle T \circ T^*(x), x\rangle^{1/2} = \|T^*x\|$$

showing that T and T^* are metrically identical. In fact, by the palalalogram law (also known as polarization identity) if $T \in \mathrm{End}_k(\mathbb{H})$ such that T and T^* ar emetrically identical, we can show $\langle T^* \circ T(x), y\rangle = \langle T \circ T^*(x), y\rangle$ for all x, y and so T is normal by lemma 5.2.1. By proposition 5.2.6, an operator is normal if and only if it is bounded away from zero.

Positive Operators

Definition 5.2.9: Positive Operator

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then T is said to be *positive* if $T = T^*$ and $\langle T(x), x\rangle \geq 0$ for all $x \in \mathbb{H}$. If $k = \mathbb{C}$, the positivey condition implies self-adjointness by the polarization identity.

Certainly, the sum of two positive operators is positive (geometrically, positive operatros form a cone in $\mathrm{End}_k(\mathbb{H})$). However the product is not positive, and need not be self-adjoint.

Example 5.2.10: Positive And Non-Positive Operators

here

However, if the opeartors commute (if the product is self-adjoint), we see that there is a product that is positive since ($ST = S^{1/2}TS^{1/2} \geq 0$ by what we'll show in the next proposition)

Let $B(\mathbb{H})_{sa} \subseteq \mathrm{End}_{\mathbb{R}}(\mathbb{H})$ be the collection of self-adjoint operators. Then this is a closed subspace, this subspace intersected with the conte of positive operators defines an ordder $S \leq T$ if $T - S \geq 0$. By taking the case of 2 dimensional real subspaces and $\mathrm{End}_{\mathbb{R}}(\mathbb{R}^2)$, this order is not total, not even a lattice order.

We require the following lemma to gain some important results (which will be an immedate consequence of the spectral theorem in ref:HERE)

Lemma 5.2.11: Square Root Lemma

If $S \leq T$ in $B(\mathbb{H})_{sa}$, then $A^*SA \leq A^*TA$ for every $A \in \text{End}_k(\mathbb{H})$. If morever $) \leq S$, then $\|S\| \leq \|T\|$

***Proof*:**
Since $S \leq T$, we know that $\langle (T-S)(y), y\rangle \geq 0$ for evey $y \in \mathbb{H}$. Replacing y with $A(x)$ for appropraite $x \in \mathbb{H}$, it follows that $A^*(T-S)A \geq 0$, so $A^*SA \leq A^*TA$.

If $0 \leq S \leq T$, by the Cauchy-Schwarz inequality on the positive sesquilinear form $\langle S(\cdot), \cdot\rangle$, we get that for every pair of unit vectors $x, y \in \mathbb{H}$

$$|\langle S(x), y\rangle|^2 \leq \langle S(x), x\rangle\langle S(y), y\rangle \leq \langle T(x), x\rangle\langle T(y), y\rangle \leq \|T\|^2$$

Hence, $\|S\|^2 \leq \|T\|^2$ by lemma 5.2.1.

Lemma 5.2.12: Approximation Lemma

There is a sequence $(p_n) \subseteq \mathbb{R}[x]$ of polynomials with positive coefficients such that $\sum p_n$ converges uniformally on $[0,1]$ to

$$t \mapsto 1-(1-t)^{1/2}$$

***Proof*:**
This is immediate from real analysis, but we can also construct it explicitly.If $q_0 = 0$ and $q_n(t) = \frac{1}{2}(t + q_{n-1}(t)^2)$, then we can see that each coefficent of q_n is positive and that $q_n(t) \leq 1$ for every $t \in [0,1]$. Morever, since

$$2(q_{n+1} - q_n) = q_n^2 - q_{n-1}^2 = (q_n - q_{n-1})(q_n + q_{n-1})$$

it follows by induction that each polynomial $p_n = q_n - q_{n-1}]$ has positive coefficients. Taking the q_n as elements of $C([0,1])$, we see they converge pointwise to $[0,1]$ to a function q such that $2q(t) = t + q(t)^2$. Thus, $q(t) = 1-(1-t)^{1/2}$. Sicne $q \in C([0,1])$ and $[0,1]$ is compact, by the monotone ocnvergence $q_n \nearrow q$ si in fact uniform by Dini's lemma, hence

$$\sum p_n = \lim q_n = q$$

completing the proof.

Proposition 5.2.13: Square-Rooting Positive Operator

Let $T \in \text{End}_k(\mathbb{H})$ be a positive operator. Then there is a unique positive operator $T^{1/2}$ such that $(T^{1/2})^2 = T$. Moreover, $T^{1/2}$ commutes with every operator commuting with T.

***Proof*:**
Analysis Now p.92

Proposition 5.2.14: Invertible Positive Operator

A positive operator T is invertible in $\text{End}_k(\mathbb{H})$ if and only if $T \geq \epsilon I$ for some $\epsilon > 0$. I that case, $T^-1 \geq$ and $T^{1/2}$ is invertib,e and $(T^{-1})^{1/2} = (T^{1/2})^{-1}$. If $T \leq S$, then $S^{-1} \leq T^{-1}$

***Proof*:**
Analayis now p.94

Projection

Let $W \subseteq \mathbb{H}$ be a closed subspace of a hilbert space H. Then we know that $\mathbb{H} = W + W^\perp$. In particular, for eah $y \in \mathbb{H}$, we have $y = w + w^\perp$, $w \in W$ and $w^\perp \in W^\perp$. Since

$$\alpha y_1 + \beta y_2 = (\alpha x_1 + \beta x_2) + (\alpha x_1^\perp + \beta x_2^\perp)$$

is the decomposition of a linear combination, it follows that $P : \mathbb{H} \to \mathbb{H}$ given by $P(y) = x$ is an operator in $\text{End}_k(\mathbb{H})$ with $\|P\| \leq 1$. Note P is idempotent ($P^2 = P$), self-adjoint, and positive since

$$\langle P(y_1), y_2 \rangle = \langle x_1 < x_2 + x_2^\perp \rangle = \langle x_1, x_2 \rangle = \langle x_1 + x_1^\perp, x_2 \rangle = \langle y_1, P(y_2) \rangle$$

and

$$\langle P(y), y \rangle = \langle x, x + x^\perp \rangle = \|x\|^2 \geq 0$$

Definition 5.2.15: (Orthogonal) Projection

The map P in the above is called the *(orthogonal) projection.*

If P is a self-adjoint and idempotent in $\text{End}_k(\mathbb{H})$, then

$$W = \{x \in \mathbb{H} \mid P(x) = x\} = P(\mathbb{H})$$

is a closed subsapce of $\mathbb{H}$. If $w^\perp \in W^\perp$, we have (since $P = P^*$) that

$$\|P(w^\perp)\|^2 = \langle w^\perp, p^2(w^\perp) \rangle = 0$$

Thus, P is an orthogonal projection of $\mathbb{H}$ onto W. Note that $I - P$ is the projection on $W^\perp$.

Projections are some of the east operators we may construct, and can be thought as the "characteristic function" from real analysis. The analogue of simple function is then obtained by decomposing $\mathbb{H}$ into a finite orthogonal sum

$$\mathbb{H} = W_1 \oplus W_2 \oplus \cdots \oplus W_n$$

corresponding to the projections $P_1, ..., P_n$ which are pairwise orthogonal ($P_i P_j = 0$ for $i \neq j$ satisfying $\sum P_i = I$). We may then consider

$$T = \sum \lambda_n P_n$$

for scalars $\lambda_1, \lambda_2, ..., \lambda_n \in k$. This may look familiar to the notion of *diagonalization.* Indeed, we may generalize the concept to infintie dimensions like so:

Definition 5.2.16: Diagonalizable

Let $T \in \mathrm{End}_k(\mathbb{H})$ be an operator. Then T is said to be *diagonalizable* if there is an orthonormal basis $\{e_i \mid i \in I\}$ for $\mathbb{H}$ and a bounded set $\{\lambda_i \mid i \in I\} \subseteq k$ such that

$$T(x) = \sum \lambda_i \langle x, e_i \rangle e_i$$

The number $\langle x, e_i \rangle$ are the coordiantes for x in the basis $\{e_i\}$ and each λ_i is an *eigenvalue* for T corresponding to the eignevector $e_i >$ Denotign P_i to be the projecti nofo $\mathbb{H}$ onto ke_i, we have $P_i(x) = \langle x, e_i \rangle e_i$, so we may write the operator as:

$$T = \sum \lambda_i P_i$$

Note that the above sum is not necessarily convergent in the norm topology on $\mathrm{End}_k(\mathbb{H})$, but only pointwise convergent (i.e. in convergent in the strong operator topology). If T is diagonalizable, then T^* is diagonalizable with

$$T^*(x) = \sum \overline{\lambda_i} \langle x, e_i \rangle e_i$$

with eigenvalues $\{\overline{\lambda_i}\}$. Furthermore, if T is diagonalizable, $T^*T = TT^*$ (with eigenvalues $|\lambda_i|^2$ with basis $\{e_i\}$), and hence T is normal. T is self-adjoint if and only if all λ_i are real and T is positive if and only if $\lambda_i \geq 0$ for all i. If $\mathbb{H}$ is finite-dimensional, every normal operator is diagonalizable. This is not generally true for infinite dimensions. In general, most “interesting” (normal) operators on infinite-dimensional hilbert sapces are nondiagonalizable, and must be handled with a continuous analogue of the concept of a basis (this will be the spectrum, see section 5.3 and chapter 6).

Unitary Operators

Recall $T : \mathbb{H} \to \mathbb{H}$ is unitary if T is an isometric isomorphism from $\mathbb{H}$ onto itself (if $k = \mathbb{R}$ we say it is an *orthogonal operator*). By the polarization identity, we see that $U^*U = UU^* = I$, so that U is normal and invertible with $U^{-1} = U^*$. Conversely, if $U \in \mathrm{End}_k(\mathbb{H})$ such that $U^{-1} = u^*$ then U is unitary since

$$\|U(x)\|^2 = \langle U^*U(x), x \rangle = \|x\|^2$$

So U is an isometry and is surjective since U is invertible. If $\{e_i \mid i \in I\}$ is an orthonormal basis for $\mathbb{H}$ and U is unitary, then $\{Ue_i \mid i \in I\}$ is a basis for $\mathbb{H}$. Conversely, we saw that every transition between orthonormal bases is given by a unitary operator. For future reference the denote the group of unitary operators $U(\mathbb{H}) \subseteq \mathrm{End}_k(\mathbb{H})$ (it should be evident it forms a group).

Definition 5.2.17: Unitary Equivalent

Let $S, T \in \mathrm{End}_k(\mathbb{H})$. Then we say that S and T are *unitarily equivalnet* of $S = UTU^*$ for some unitary operator U.

It should be clear that unitary equivalent preserves norm, self-adjointness, normality, diagonalizability, and unitarity.

Definition 5.2.18: Partial Isometry

Let $U \in \operatorname{End}_k(\mathbb{H})$. Then U is said to be a *partial isometry* if there is a closed subpspace $X \subseteq \mathbb{H}$ such that $U|_X$ is isometric and $U|_{X^\perp} = 0$

This immediately implies that $X^\perp = \ker(U)$, hence $X = \bar{(}U^*(\mathbb{H}))$. Setting $P = U^*U$, then

$$\langle P(x), x\rangle = \|U(x)^2 = \|x\|^2$$

for every $x \in X$, hence $P(x) = x$ by the Cauchy-Schwarz inequality. Furthermore, $P(x^\perp) = 0$ for every $x^\perp \in X^\perp$, so P is a projetion of $\mathbb{H}$ onto X. Conversely, if $U \in \operatorname{End}_k(\mathbb{H})$ such that $U^*U = P$ for some projeciotn P, taking $X = P(\mathbb{H})$ and see from the equation $\|U(x)$
$^2 = \langle p(x), x\rangle$ that U is isometric on X and 0 on $X^\perp$, and so U is a partial isometry. Using this, we can also show that U^* is a partial isometry. Since $U(I-P) = 0$,

$$(UU^*)^2 = UU^*UU^* = UPU^* = UU^*$$

so UU^* is self-adjoint idempodent, i.e. a projection. Note that U^* is the projectio nof $U(\mathbb{H})$ back onto $U^*(\mathbb{H})$ so that U^* is a *partial inverse* of U.

Be weary that there exist isometries that are not unitary (since they need not be surjective). For example, if $\mathbb{H}$ is a separable and has countable basis $\{e_n \mid n \in \mathbb{N}\}$, we have

$$S\left(\sum \alpha_n e_n\right) = \sum \alpha_n e_{n+1}$$

Then S (the unilateral shift operator) is an isometry of $\mathbb{H}$ ont othe subspace $\{e_1\}^{|}perp$ and consequently S^* is a patial isometry of $\{e_1\}^\perp$ onto $\mathbb{H}$. In particular, $S^*S = I$, wheas SS^* is the projection onto $\{e_1\}^\perp$.

Theorem 5.2.19: Polar Decomposition of T

Let $T \in \operatorname{End}_k(\mathbb{H})$. Then there is a unique positive operator $|T| \in \operatorname{End}_k(\mathbb{H})$ such that

$$\|T(x)\| = \||T|(x)\|$$

and $|T| = (T^*T)^{1/2}$. Moreover, there is a unique partial isometry U with kernel $\ker(U) = \ker(T)$ and $U \circ |T| = T$. In particular

$$U^*U|T| = |T| \qquad U^*T = |T| \qquad UU^*T = T$$

***Proof*:**
Analysis Now p.96

We may think of U as the generalized "sign" of T and $|T|$ as the generalized "absolute value" of T. By the formula

$$T^* = |T|U^* = U^*(U|T|U^*)$$

and the unique of polar decomposition, we see that U^* is the sign of T^*, whereas $U|T|U^*$ is the absolute value fo T^*. In general, the absolute value of a sum or product of noncommuting operators doens't have much relation to the sum or product of the absolute values. Similarly, the sign (in the finite dimensional case), can't always be chosen to be unitary (the unilateral shift is an immedaite counter example). For T to have the form $U|T|$ for some unitary operator U, it is necessary and sufficient that the cloesd subspaces $\ker(T)$ and $\ker(T^*)$ have the same dimension (finite or infinite). Easy cases of this general theorem are estalblisehd in the next result

Proposition 5.2.20: Polar Decomposition For Isomorphisms

Let $T \in \mathrm{End}_k(\mathbb{H})$ be invertible. Then the partial isometry in its polar decomposition is unitary

***Proof*:**
Analysis Now p.97

Proposition 5.2.21: Polar Decomposition For Normal Operators

Let $T \in \mathrm{End}_k(\mathbb{H})$ be normal. Then there is a unitary operator W commuting with T, T^*, and $|T|$ such that $T = W|T|$

***Proof*:**
Analysis Now p.97

I want to add one more thing, I just don't konw where: for any $T \in B(\mathbb{H})$, $T = T_1 + iT_2$ for self-adjoint operators T_1, T_2. This lets us generalize the spectral theorem for noraml operators, since we will konw it for self-adjoint operator.

Lemma 5.2.22: lemma I

If $T = T^*$ and $\|T\| \leq 1$, the operator $U = T + i(I - T^2)^{1/2}$ is unitary and $T = \frac{1}{2}(U + U^*)$

***Proof*:**
Analysis Now p.97

Lemma 5.2.23: Lemma II

Let $S \in \mathrm{End}_k(\mathbb{H})$ with $\|S\| < 1$. Then for every unitary U, there are unitaries U_1, V_1 such that

$$S + U = U_1 + V_1$$

***Proof*:**
Analysis Now p.98

Theorem 5.2.24: Russo-Dye-Gardner Theorem

Let $T \in \text{End}_k(\mathbb{H})$ with
$$\|T\| < 1 - \frac{2}{n} \qquad n > 2$$
Then there are unitary operators $U_1, U_2, ..., U_n$ such that
$$T = \frac{1}{n}(U_1 + U_2 + \cdots + U_n)$$

***Proof*:**
Analysis Now p.98

Since the self-adjoint elements span $\text{End}_k(\mathbb{H})$, we see from lemma 5.2.22 that every $T \in \text{End}_k(\mathbb{H})$ is the linear combination of (at least 4) unitary operators The estimate in the Russo-Dye-Gardner Theorem is much more precise with respect to the norm, and shows that every element in the open unit ball of $\text{End}_k(\mathbb{H})$ is a convex combination (indeed, a mean) of unitary operators. On the surface of the ball this need not be the case, in fact the unilateral shift cannot be expressed as a convex combination of unitaries (exercise 3.2.8 in Gert K. Pederson Analysis Now).

Proposition 5.2.25: Numerical Radius Of Operator

Let $\mathbb{H}$ be a complex hilbert space and $T \in \text{End}_{\mathbb{C}}(\mathbb{H})$. Then we deifn eth *numerical radius* of T as
$$|||T||| = \sup\{\langle T(x), x\rangle \mid x \in \mathbb{H}, \|x\| \leq 1\}$$
Then $\|\cdot\|$ is a norm on $\text{End}_{\mathbb{C}}(\mathbb{H})$ satisfying $\frac{1}{2}\|\cdot\| \leq |||\cdot||| \leq \|\cdot\|$. More over, $|||T^2||| \leq |||T|||^2$ for every T and $|||T||| = \|T$
if T is normal

***Proof*:**
Analysis Now p.99

5.3 Compact Operators

Throughout this section, let $\mathbb{H}$ be a hilbert space over $k \in \{\mathbb{R}, \mathbb{C}\}$ unless otherwise specified.

Definition 5.3.1: Finite Rank Operator

Let $T \in \text{End}_k(\mathbb{H})$. Then T is said to have *finite rank* if $T(\mathbb{H})$ is finite-dimensional subspace of $\mathbb{H}$. Denote by $B_f(\mathbb{H}) \subseteq \text{End}_k(\mathbb{H})$ the set of all finite rank opeartors.

Clearly, $B_f(\mathbb{H})$ is a subspace, even a subalgebra, even an ideal of $\text{End}_k(\mathbb{H})$. If $T \in B_f(\mathbb{H})$, by proposition 5.2.4, we get an orthogonal decomposition $\mathbb{H} = T(\mathbb{H}) \oplus \ker(T^*)$, which shows that $T^*(\mathbb{H}) - T^*T(\mathbb{H})$, so T^* has finite rank. Thus, $B_f(\mathbb{H})$ is *self-adjoint* ideal in $\text{End}_k(\mathbb{H})$ $((B_f(\mathbb{H}))^* = B_f(\mathbb{H})$ as a set). The subset $B_f(\mathbb{H}) \subseteq \text{End}_k(\mathbb{H})$ is similar in relation to $C_c(X) \subseteq C_b(X)$ when X is a locally compact Hausdorff space. In particular, these classes describe the local phenomena on $\mathbb{H}$ and on X. Passing to a limit in norm may destroy the "locality". but enogh structure is preservd to make these "quasilocal" operators and functions very useful and interesting. We shall stud the closure of $B_f(\mathbb{H})$ in this section as a noncommutative analogue of $C_0(X)$ in function theory.

Lemma 5.3.2: Net And Compact Operators

There is a net $(P_\lambda)_{\lambda\in\Lambda}$ of projection in $B_f(\mathbb{H})$ such that $\|P_\lambda(x) - x\| \to 0$ for each $x \in \mathbb{H}$.

that is, the projection are "dense" in $B_f(\mathbb{H})$.

***Proof*:**
We want to find a next P_n such that $\|P_n(x) - x\| \to 0$. Let $\{e_i \mid i \in I\}$ be an orthonormal basis for $\mathbb{H}$ and let N be a net of finite subsets of I ordered by inclusion. For each $n \in N$, let P_n denote the projection of $\mathbb{H}$ onto span $\{e_i \mid i \in n\}$ so that $(P_n)_{n\in N}$ is a net in $B_f(\mathbb{H})$.

Now, if $x \in \mathbb{H}$, then $x = \sum a_i e_i$, thus

$$\|P_n(x) - x\|^2 = \sum_{i\notin n} |a_i|^2$$

By Parseval's identity, this tends to zero, completing the proof.

Theorem 5.3.3: Closed Unit Balls In Hilbert Spaces

Let $B \subseteq \mathbb{H}$ be a closed unit ball in a hilbert space $\mathbb{H}$. Then the following conditions on an operator $T \in \mathrm{End}_k(\mathbb{H})$ are equivalent

1. $T \in \overline{B_f(\mathbb{H})}$
2. $T|_B$ is a weak-norm continuous function from B into $\mathbb{H}$
3. $T(B)$ is compact in $\mathbb{H}$
4. $\overline{T(B)}$ is compact in $\mathbb{H}$
5. each net in B has a subnet whose iamge under T converges in $\mathbb{H}$

***Proof*:**
Analysis Now p.105

Definition 5.3.4: Compact Operators

Let $T \in B(\mathbb{H})$. Then T is compact if $T(B)$ is compact. The set of compact operators is denoted by $B_0(\mathbb{H})$ to show that these operators "vanish at infinity".

Lemma 5.3.5: Diagonalizable Operator Is Compact

Let $T \in \mathrm{End}_k(\mathbb{H})$. Then T is comapct if and only if its eigenvalues $\{\lambda_i \mid i \in I\}$ belongs to $c_0(I)$

***Proof*:**
Let $T(x) = \sum \lambda_i \langle x, e_i \rangle e_i$. If $T \in B_0(\mathbb{H})$, then for any $\epsilon > 0$, define

$$I_\epsilon = \{i \in I \mid |\lambda_i| \geq \epsilon\}$$

If I_ϵ is infinite, the net $(e_i)_{i \in I_\epsilon}$ converges weakly to zero for any well-ordering of I_ϵ since $\langle e_i, x \rangle \to 0$ by Parsaval's identity. Sicne $\|T(e_i)\| = |\lambda_i| \geq \epsilon$ for $i \in I_\epsilon$< this contradicts theorem 5.3.3(3). Thus, I_ϵ must be finite for each $\epsilon > 0$, hence λ_i must vanish at infinity.

Conversely, if I_ϵ is finite for all $\epsilon > 0$. et

$$T_\epsilon = \sum_{i \in I_\epsilon} \lambda_i \langle \cdot, e_i \rangle e_i$$

Then T_ϵ has finite rank, and

$$\begin{aligned} \|(T - T_\epsilon)(x)\|^2 &= \left\| \sum_{i \notin I_\epsilon} \lambda_i \langle x, e_i \rangle e_i \right\|^2 \\ &= \sum_{i \notin I_\epsilon} |\lambda_i|^2 |\langle x, e_i \rangle|^2 \\ &\leq \epsilon^2 \|x\|^2 \end{aligned}$$

hence, $\|T - T_\epsilon\| \leq \epsilon$, thus $T \in B_0(\mathbb{H})$ by theorem 5.3.3(1), completing the proof.

Lemma 5.3.6: Eigen Vectors Between Adjoint

Let x be an eigenvector for a normal operator $T \in \mathrm{End}_k(\mathbb{H})$ corresponding to an eigenvalue λ. Then x is an eigenvector for T^* corresponding to the eigenvalue $\overline{\lambda}$. Eigenvectors for T correspondign to different eigenvalues are orthogonal.

***Proof*:**
Note that the operator $T - \lambda I$ is normal and it's adjoitn is $T^* - \overline{\lambda} I$. COnversely,

$$\|(T^* - \overline{\lambda} I)(x)\| = \|(T - \lambda I)(x)\| = 0$$

If $T(y) = \gamma y$ where $\gamma \neq \lambda$, then we may certainly assume $\lambda \neq 0$ so that

$$\begin{aligned} \langle x, y \rangle &= \lambda^{-1} \langle T(x), y \rangle \\ &= \lambda^{-1} \langle x, T^*(y) \rangle \\ &= \lambda^{-1} \langle x, \overline{\gamma} y \rangle \\ &= \lambda^{-1} \gamma \langle x, y \rangle \end{aligned}$$

Hence, $\langle x, y\rangle = 0$, showing they are different eigenvectors, completing the proof.

Lemma 5.3.7: Normal Compact Operator, then Eivenvalue Exists

Every normal, compact operator T on a complex Hilbert space $\mathbb{H}$ has an eigenvalue λ with $|\lambda| = \|T\|$

***Proof*:**
Analysis now p. 107

Theorem 5.3.8: Normal Compact, Then Diagonalizable

Every normal compact operator $T \in \mathrm{End}_k(\mathbb{H})$ is diagonalizable and its eigenvalues (counted-with multiplicity) vanish at infinite. Conversely, each such operator is normal and compact.

***Proof*:**
Analysis now p.108

It will be convenient for future sections to introduce a new notation: Let $x \odot y$ represent the rank one operator in $\mathrm{End}_k(\mathbb{H})$ determined by the vectors $x, y \in \mathbb{H}$ and the formula

$$(x \odot y)(z) = \langle z, y\rangle x$$

The map $(x, y) \mapsto x \odot y$ is sesquilinear map of $\mathbb{H} \times \mathbb{H}$ into $B_f(\mathbb{H})$. If $\|e\| = 1$ then $e \odot e$ is the one-dimensional projection of $\mathbb{H}$ on $\mathbb{C}e$. Every normal compact operator $\mathbb{H}$ can now be written by the above theorem in the form

$$T = \sum \lambda_i e_i \odot e_j$$

For suitable orthonormal basis $\{e_i \mid i \in I\}$. The sum will converge in norm, since either the set $I_0 = \{i \in I \mid \lambda_i \neq 0\}$ is finite (so that $T \in B_f(\mathbb{H})$) or else countably infinite in which case the sequence $\{\lambda_i \mid i \in I_0\}$ converges to zero.

Definition 5.3.9: Spectrum

We say that the compact set

$$^{(}T) = \{\lambda_i \mid i \in I_0\} \cup \{0\}$$

is the *spectrum* of T.

For every continuous function f on $^{(}T)$, define

$$f(T) = \sum f(\lambda_i) e_i \odot e_i$$

Then $f(T)$ is compact if and only if $f(0) = 0$ and the map $f \mapsto f(T)$ Is an isometric $*$-preserving homomorhpism of $C(^{(}T))$ into $\mathrm{End}_k(\mathbb{H})$. If $f(z) = \sum \alpha_{nm} z^n \overline{z}^m$ is a polynomial of two commuting variables z and $\overline{z}$, then $f(T) = \sum \alpha_{nm} T^n (T^*)^m$. This result is the *spectrial (mapping) theorem* for normal compact operators. In the next chapter, we shall generalize this to any normal operator.

Calkin Algebra and Index

Since $B_0(\mathbb{H}) \subseteq B(\mathbb{H})$ is a closed ideal, we may define $B(\mathbb{H})/B_0(\mathbb{H})$ with the quotient norm[5]. This quotient is given a name

Definition 5.3.10: Calkin Algebra

Let $B_0(\mathbb{H}) \subseteq B(\mathbb{H})$. Then the algebra

$$B(\mathbb{H})/B_0(\mathbb{H})$$

is called the *Calkin Algebra* of $\mathbb{H}$

We shall see that many properties of $B(\mathbb{H})$ are conveniently expressed in terms of the Calkin algebra.

Definition 5.3.11: Compact Pertubation

If S and T are elements in $B(\mathbb{H})$ and $S - T \in B_0(\mathbb{H})$, we say that S is a *compact pertubation* of T.

This is antoher way of saying that S and T hav the same image in the Calkin algebra. Such "local" pertubation happen frequently in application and properties of an operator that are invariatn under compact pertubations are highly used. The best known of these we shall study know is known as the *index*

Theorem 5.3.12: Atkinson's Theorem

Let $T \in B(\mathbb{H})$. Then the following are equivalent:

1. There is a unique operator $S \in B(\mathbb{H})$ such that ST and TS are the projections on $(\ker(T))^\perp$ and $\ker(T^*)^\perp$, respectively, and both projections have finite co-rank
2. For some operator $S \in B(\mathbb{H})$ both operators $ST - I$ and $TS - I$ are compact
3. the image of T is invertible in the Calkin algebra $B(\mathbb{H})/B_0(\mathbb{H})$
4. Both $\ker(T)$ and $\ker(T^*)$ are finite-dimensional subsapces and $T(\mathbb{H})$ is closed

***Proof*:**
Analysis now p.109

In essence, Fredholm operators are operators that are are almost invertible up to some finite-dimension beign ignored, in particular they are invertible modulo compact operators.

Definition 5.3.13: Fredholm Operator

The operator satisfying the condition's in Atkinson's theorem are called *Fredholm operators*, and the class of them is dneoted $F(\mathbb{H})$.

[5] This is even a C^*-algebra

Definition 5.3.14: Index

For each $T \in F(\mathbb{H})$, define the *index* of T to be

$$\operatorname{index}(T) = \dim(\ker(T)) - \dim(\ker(T^*))$$

If we shooe S for T as in (1) of Atkinson's theorem and write $suchthat = I - P$ and $TS = I - Q$ with P and Q projection of rank, then another way of writing this would be

$$\operatorname{index}(T) = \operatorname{rank}(P) - \operatorname{rank}(Q)$$

By (3) of Atkinson's Theorem, the product of Fredholm operators is again a Fredholm operator. In particular, every product $RT \in F(\mathbb{H})$ if $T \in F(\mathbb{H})$ and R is invertible in $B(\mathbb{H})$, and in this case

$$\operatorname{index}(RT) = \operatorname{index}(TR) = \operatorname{index}(T)$$

since R is bijective. It should also be clear that $T^* \in F(\mathbb{H})$ if $T \in F(\mathbb{H})$ with index $\operatorname{index}(T^*) = -\operatorname{index}(T)$. Finally, the partial inverse S to T in (1) of Atkinson's theorem is a Fredholm operator with index $S) = -T)$. F

Definition 5.3.15: Indexed Fredholm Operators

For each $n \in \mathbb{Z}$ define

$$F_n(\mathbb{H}) = \{T \in F(\mathbb{H}) \mid T) = n\}$$

All of these subsets are nonemtpy, since if S is the unilateral shift, then $S^n \in F_{-n}(\mathbb{H})$ and $(S^*)^n \in F_n(\mathbb{H})$ for every $n > 0$.

Lemma 5.3.16: Finite Rank And 0 Index

If $A \in B_f(\mathbb{H})$, then $I + A \in F_0(\mathbb{H})$

***Proof*:**
Analysis now p.110

Corollary 5.3.17: Fredholm Alternative

If $A \in B_0(\mathbb{H})$ and $\lambda \in \mathbb{C} \setminus \{9\}$, then either $\lambda I - A$ is invertible in $B(\mathbb{H})$ or λ is an eigenvalue of A with finite multiplicity, in which case $\overline{\lambda}$ is an eigenvalue for A^* with the same multiplicity

***Proof*:**
Let $T + I - \lambda^{-1}A$. Since $T \in F_0(\mathbb{H})$, by lemma 5.3.16 we know that $T(\mathbb{H})$ is closed and that

$$\dim(\ker(T)) = \dim(\ker(T^*)) < \infty$$

If these dimension are 0, then $T(\mathbb{H}) = (\ker(T^*))^\perp = \mathbb{H}$ and $\ker(T) = \{0\}$, hence T is invertible by proposition 5.2.6, completing the proof.

This lemma implies that the spectrum of a compact operator (i.e. those λ such that $\lambda I - T$ is not invertible, as we'll see in the next chapter) consists of $\{0\}$ and the eigenvalues for T. IN particualr, it is a countable subset of $\mathbb{C}$ with 0 as the only possible accumulatpion point.

Theorem 5.3.18: Index Invariant Under Compact Operator

Let $T \in F(\mathbb{H})$ and A be a compact operator. Then

$$T + A) = T)$$

***Proof*:**
Analysis now p.111

Proposition 5.3.19: Fredholm Class Is Open

Each Fredholm class $F_n(\mathbb{H})$ for each $n \in \mathbb{H}$ is open in $B(\mathbb{H})$

***Proof*:**
analysis now p.112

Proposition 5.3.20: Grading Structure Of Freidholm Operators

If $T_1 \in F_n(\mathbb{H})$ and $T_2 \in F_m(\mathbb{H})$ then $T_1T_2 \in F_{n+m}(\mathbb{H})$

***Proof*:**
Analysis now p. 112

We close with an interesting topological fact that brings some light to the index .Let G denote the group of invertible elements of the Calkin algebra $B(\mathbb{H})/B_0(\mathbb{H})$. and G_0 is the connected component of the identity (which is the group generated by the elements of the form $\exp(A)$ for some $A \in B(\mathbb{H})/B_0(\mathbb{H})$). Then G_0 is an open and closed subgroup of G and G/G_0 is a discrete group that labels the connected components in G. In this case, G/G_0 is isomorphic to $\mathbb{Z}$, and the index of a Fredholm operator T is the image of T under the compsoed quotinet maps from $F(\mathbb{H})$ to G and from G to G/G_0 $(= \mathbb{Z})$

5.4 Trace

5.5 *Intuition to add

This is a conversation I had with Gemini that I really want to incorporate somewhere as it gave some good insights. I will save it here for now:

Zak Katz: Let $F(x_1, \ldots, x_n, t)$ represent a PDE w.r.t time. Let's say $u(x, t)$ represent a solution and $f_t(x) = u(x,t)$ is a "state" at time 0. Let S_t be the "time-shift" operator so that $S_t(f_a) = f_{a+t}$ for some state f_a at time a.

My question is: Is S_t always self-adjoint? If yes, can you prove it, and if not, can you give a counter-example?

Raycast AI:

`Tool: web_search`

No, the time-shift operator S_t is not always self-adjoint.

In fact, for many common PDEs (like the wave equation, transport equation, or Schrödinger equation), S_t is almost never self-adjoint. It is usually unitary (energy-conserving) or a contraction semi-group (dissipative).

6

Spectral Theory

In short, Spectral theory generalizes the notion of eigen vectors and values to infinite dimensional operators. In the previous chapter, we saw how there is a notion of diagonalizability for compact normal operators. In this chapter, we extend this result to normal operators. We also along the way learn some important dualities such as the Gelfand duality which has some useful applications, and is an intersting motivator for classical algebraic geometry.

6.1 Banach Algebras

Recall the definition of a banach algebra

Definition 6.1.1: Banach Algebra Revisited

A *Banach Alagebra* is an algebra A with a submultiplicative norm (so that it is continuous) st the unerlying vector space is a banach space.

All the same applies for Banach Algebras as we expect. Note that for A/I to be a Banach algebra for an ideal I, I has to be closed, just like for subspaces. If A is unital and $A \neq \{0\}$, then it is certainly unique and $\|I\| \geq 1$ (why not =?? Analysis now p.128) If A has no units, we may try to isometrically embed it into a larger Banach Algebra $\widehat{A}$ such that $A \subseteq \widehat{A}$ becomes an ideal such that every nonzero ideal of $\widehat{A}$ has a nonzero intersection with A (such an ideal is called an *essential ideal*). This can be thought as the algebraic counterpart of the notion of compactification of a topological space. For classical Banach algebra, there is usually a natural way of *adjoining a unit*, but there is always a general method of doing so that is the counter-part of one-point compactification. Take the direct sum $A \oplus k = \widehat{A}$ along with the product

$$(x, \alpha)(y, \beta) = (xy + \alpha y + \beta x, \alpha\beta)$$

and the submultiplicative 1-norm. Then $A \oplus k$ is a unital Banach algebra with $I(0,1)$ and contains A as an ideal of co-dimension 1.

Definition 6.1.2: Regular Representation

Let A be a Banch Algebra. Then the (left) *regular representation* $\rho : A \to B(A)$ by $\rho(x)(y) = xy$ for $x, y \in A$.

It should be verified that ρ is a norm decreasing algebra homomorphism, and if A is unital, the inequalities

$$\|A\| \leq \|\rho(A)\|\|I\| \leq \|A\|\|I\|$$

show that ρ is a homeomrophism. Renorming A by using the equivalent norm from $B(A)$, we may assume that $\|I\| = 1$ (if $A \neq \{0\}$), which we do.

Definition 6.1.3: Approximate Unit

A net $(E_\lambda)_{\lambda \in \Lambda}$ in the unit ball of a Banah algebra A is an *approximate unit* if

$$\lim E_\lambda x = \lim x E_\lambda = x$$

for all $x \in A$.

Example 6.1.4: Approximate Units

Can you think of approximate units for the non-unital Banach algebra $C_0(X)$, $B_0(\mathbb{H})$ and $L^1(\mathbb{R}^n)$? Show that the existence of an approximate unit guarantees that the regular representation ρ is an isometry

We shall denote $A^\times = \mathrm{GL}(A)$ the set of all invertible elements of A (in algebra, we'd usually call these elements unital, but the term unit in functional analysis is more commonly reserved for the identity)

Lemma 6.1.5: Almost Invertible Elements

Let $x \in A$ where A is a unital banach algebra with $\|A\| < 1$. Then $I - A \in \mathrm{GL}(A)$ and

$$(I - A)^{-1} = \sum_{n=0}^{\infty} A^n$$

***Proof*:**

Since the norm is submultiplicative $\|x^n\| \leq \|x\|^n$, thus the series $\sum x^n$ converges in x to an element y. Sicne $xy = yx = y - I$ by manipulating the above power series, it follwos that $y = (I - x)^{-1}$, completing the proof.

Proposition 6.1.6: Properties Of $\mathrm{GL}(A)$

Let A be a unital algebra and $\mathrm{GL}(A)$ be it's multiplicative group. Then $\mathrm{GL}(A)$ is an open subset of A, an the map $x \mapsto x^{-1}$ is a homeomorphism of $\mathrm{GL}(A)$

***Proof*:**
Let $x \in \mathrm{GL}(A)$ and $y \in A$, Then by lemma 6.1.5

$$y = x - (x - y) = x(I - x^{-1}(x - y)) \in \mathrm{GL}(A)$$

provided that $\|A^{-1}(A - B)\| < 1$. In particular, the ball $B_\epsilon(x)$ is contained in $\mathrm{GL}(A)$ for $\epsilon < \|x^{-1}\|^{-1}$, which proves that $\mathrm{GL}(A)$ is open. If $y \to a$, we see from the above equation and lemma 6.1.5 that $y^{-1} \mapsto a^{-1}$, thus inverting is a continuous process and therefore a homeomorphism since $(A^{-1})^{-1} = A$, completing the proof.

From ehre on out, all Banach algebra will be over $\mathbb{C}$, i.e. all Banach algebras will be *complex*. See exercise ref:HERE on the complexicication of general real Banach algebras (or E.2.1.13 in Analysis Now).

Definition 6.1.7: Spectrum

Let A be a complex unital Banach algebra. Then for every $x \in A$, define the *spectrum* of x t obe the set

$$^{(}x) = \{\lambda \in \mathbb{C} \mid \lambda I - x \notin \mathrm{GL}(A)\}$$

The comliment of $^{(}x)$ is the *resolvent set*.

On the resolvent set, we can define the resolvent function

$$R(x, \lambda) = (\lambda I - A)^{-1}$$

Definition 6.1.8: Spectral Radius

Let A be a complex unital Banach algera, $x \in A$, and $^{(}x)$ be its spectrum. Then the smalleest $r \geq 0$ such that $^{(}x) \subseteq B_r(0)$ is called the *spectral radius* of x, and is denoted $r(x)$. THus

$$r(x) = \sup \left\{ |\lambda| \;\middle|\; \lambda \in^{(} x) \right\}$$

Lemma 6.1.9: Build-Up Lemma I

Let $x \in A$ and $f(z) = \sum \alpha_n z^n$ be a holomorphic function in a region contined in the closed disk $B_r(0)$ with $\|x\| \leq r$. then we can define $f(x) = \sum \alpha_n x^n$ in A. Morever, if $\lambda \in^{(} x)$ and $|\lambda| \leq r$, then $f(\lambda) \in^{(} f(x))$

***Proof*:**
The series $f(x)$ is absolutely convergence ($\sum |\alpha_n| \|x\|^n < \infty$), so certainly $f(x) \in A$. Moerover

$$\begin{aligned} f(\lambda)I - f(x) &= \sum \alpha_n(\lambda^n I - x^n) \\ &= (\lambda I - x) \sum \alpha P_{n-1}(\lambda, x) \\ &= (\lambda I - x)y \end{aligned}$$

where $P_{n-1}(\lambda, x) = \sum_{k=0}^{n-1} \lambda^k A^{n-k-1}$ so that $\|P_{n-1}(\lambda, x)\| \le nr^{n-1}$, which implies the series of polynomials converges in A to an element y commuting with x. We see form this computation that if $f(\lambda)I - f(x) \in \mathrm{GL}(A)$ with inverse v, then yv will be the inverse of $\lambda I - x$, so tha $\lambda I - x \in \mathrm{GL}(A)$

Lemma 6.1.10: Build-Up Lemma II

For every $x\ inA$, we have

$$r(x) \le \inf \|A^n\|^{1/n}$$

***Proof*:**
If $|\lambda| > \|A\|$, thn by lemma 6.1.5

$$(\lambda I - x)^{-1} = \lambda^{-1}(I - \lambda^{-1}x)^{-1} = \sum \lambda^{-n-1}x^n$$

which shows that $r(x) \le \|x\|$. Now if $\lambda \in^(x)$, then $\lambda^n \in^(x^n)$ by lemma 6.1.9. Thus $|\lambda^n| \le \|x^n\|$ from what we've just shown. It follows that $r(x) \le \|x^n\|^{1/n}$ for every n, completing the proof.

Theorem 6.1.11: Spectrum Is Compact

Let $x \in A$ be an element of a complex unital Banach algebra A. Then the spectrum of x, $^(x)$ is a compact nonempty subset of $\mathbb{C}$, and the spectral radius of x is the limit of the convergence sequence $\|A^n\|^{1/n}$

***Proof*:**
Analysis now p.131, really cool proof that I should take the time to write down!

A cool consequence of this is that we get a classical result of Representation theory known as Artin-Wedderburn for algebraically closed algebras:

Corollary 6.1.12: Algebraically Closed Division Algebra $\mathbb{C}$

Let A be a unital Banach algebra over $\mathbb{C}$. Then if A is a division ring, $A = \mathbb{C}$

***Proof*:**
For each $x \in A$, there is some $\lambda \in sp(x)$ by the above theorem. Thus, $\lambda I - x \notin \mathrm{GL}(A)$, hence $x = \lambda I$.

6.2 Continuous Functional Calculus

Let $T \in B(\mathbb{H})$ and consider $\sigma(T)$. Consider $B(\sigma(T))$, which is a $*$-algebra where $*$ is complex onjugation. We may consider this algebra with the supremum norm. In this section, we'll be looking itno various sub $*$-algebra of $B(\sigma(T))$, for example the polynomials functions on $\sigma(T)$ or the continuous functions on $\sigma(T)$

Definition 6.2.1: Functional Calculus

Let $T \in B(\mathbb{H})$. Then a *functinoal calculus* of T is a continuous $*$-homomorphism from $*$-algebra of functions $(\sigma(T) \to \mathbb{C})$ to $B(\mathbb{H})$ which sents the constant function 1 to the idneitty operator,a and sends the identity function $x \mapsto x$ to T

Since it's a $*$-homomorphism, we have

$$\varphi(\overline{f}) = \varphi(f)^*$$

since we have the supremum norm and the $*$-homomorphism is continuous, $f_n \to f$ in the supremum norm then $f_n(T) \to f(T)$ in the norm topology of $B(\mathbb{H})$.

(more here, this is really cool!)